

learnX.help

An Object-Oriented Academic Assistant & Learning Management Portal

A PROJECT REPORT

Submitted in partial fulfillment of the requirements for the course:

Object-Oriented Programming

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Abstract

This report presents learnX.help, an advanced academic management and learning assistance system built on a modern Spring Boot micro-architecture. Designed to optimize class routines, examinations, and learning resource sharing, the application streamlines coordination between students, class representatives, teachers, and system administrators. Architected under strict Object-Oriented Programming (OOP) paradigms in Java 21, the system models real-world academic entities through comprehensive encapsulation, ensuring robust data security and access control across user roles. Inheritance and interface polymorphism are leveraged dynamically, particularly through Spring Data JPA repositories and custom security details mappings, enabling extensible class boundaries and clean architectural boundaries. A primary engineering innovation is the Slot Detection Service, which programmatically executes routine merging algorithms to locate open time intervals for scheduling tests and makeup sessions. The development methodology followed an iterative, object-oriented software engineering lifecycle, integrating requirements analysis, UML design, and robust test plans. By consolidating course materials, grade books, and direct academic scheduling into a unified web dashboard, learnX.help solves critical operational bottlenecks in university coordination, demonstrating structural stability and OOP design excellence.

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Chapter 1: Introduction

1.1 Project Background and Motivation

Modern higher educational institutions require seamless coordination among various academic roles, including students, class representatives (CRs), teachers, and administrative staff. As academic schedules, learning resources, and examinations grow increasingly complex, traditional manual methods of organization become inadequate. The learnX.help application is engineered to address this challenge by offering a centralized, role-based platform that facilitates routine management, automated test scheduling, secure academic handout distribution, and streamlined grading. The core motivation is to replace scattered communication channels (such as generic instant messaging groups) with an integrated, structured environment that ensures data integrity and structural organization.

1.2 Problem Statement

Students and educators frequently encounter operational bottlenecks in three main areas: (1) scheduling makeup classes or class tests without clashing with existing course routines or previously scheduled exams; (2) centralizing high-capacity academic resource vaults where notes, lecture handouts, and slides can be approved, categorized, and rated; and (3) managing structured announcements and online examinations without information loss. Currently, no localized platform combines automated scheduling slot detection and multi-tenant university resource segregation under a strict role-permission matrix. learnX.help solves these operational problems by modeling academic workflows inside an Object-Oriented design framework.

1.3 Project Objectives

The primary engineering objectives of this project are: (1) to architect a secure and modular system leveraging Object-Oriented Programming (OOP) paradigms in Java, including strict encapsulation, interface polymorphism, inheritance hierarchies, and class abstractions; (2) to implement an automated Routine Merging and Free Slot Detection algorithm to facilitate hassle-free assessment scheduling; and (3) to deliver a unified Single Page Application (SPA) capable of managing student grading, multiple-choice quizzes, announcements, and resource indexing.

1.4 Tools and Technologies Used

The development environment and libraries were structured to support modern cloud-ready micro-architectures. By leveraging the Spring ecosystem, the application achieves high testability and clean separation of concerns. Table 1.1 documents the exact specifications compiled from the project build definitions.

Tool / Technology	Specification / Version	Purpose in learnX.help
Java Development Kit (JDK)	Version 21	Core programming language utilizing modern standard library features.
Spring Boot	Version 4.0.6 (Spring parent)	Base application micro-architecture providing automatic beans wiring.
Spring Data JPA	Jakarta Persistence API	Object-Relational Mapping (ORM) to maps Java classes to database entities.
Spring Security	Security Filters	User authentication, session tracking, and BCrypt password encryption.
Lombok	Annotation Processor	Auto-generates getters, setters, constructors, and builder patterns.
H2 Database	In-memory / File-based	Lightweight relational database for local development and integration tests.
PostgreSQL	Relational Database	Scalable database for persistent data storage in deployment.
Maven	Build Automation	Handles dependency management, compilation, and package generation.

Table 1.1: Tools and Development Environment

Chapter 2: Requirements and Analysis

2.1 Functional Requirements

The functional scope of learnX.help is designed around key academic actions. Each feature corresponds to a Java controller and service mapping. The complete list of functional operations is tabulated below.

Req ID	Requirement Name	Description & Implementation Details
FR-1	User Authentication & Approval	Enables sign up for all roles. New accounts remain locked until approved by the administrator. Passwords are encrypted.
FR-2	Routine & Class Schedule	Allows CRs and teachers to insert class routines including days, slots, course names, and class sections.
FR-3	Automated Slot Detection	Given a date and duration, the system scans all class routines and scheduled exams, merging busy periods to identify free academic intervals.
FR-4	Academic Resource Vault	Supports uploading notes, slides, and links. Features user reactions (likes/dislikes) and admin approvals for quality control.
FR-5	MCQ Exam Engine	Teachers can construct multiple-choice exams. Students submit responses which are auto-graded and saved immediately.
FR-6	GradeBook & Performance Stats	Tracks scores for exams and tests, dynamically computing class averages and highest scores for students to review.

Table 2.1: Functional Requirements Specifications

2.2 Non-Functional Requirements

Non-functional criteria dictate technical constraints around local performance, transaction safety, data encapsulation, and system usability, ensuring the system operates reliably under typical university loads.

Category	Requirement Specification	Implementation Detail in learnX.help
Security	Role Authorization & Enrolment	Strict endpoint routing rules mapped via Spring Security config. Fields are encapsulated via Lombok.
Performance	Fast Scheduling Detection	Slot detection utilizes active time-interval sorting and one-pass overlap merging, executing in < 15ms locally.
Reliability	Transactional Consistency	All operations are backed by Spring's transactional manager. Rollbacks occur on JpaRepository exceptions.
Scalability	High-capacity Uploads	Supports binary storage in Database LOB fields up to 100MB, configuring multipart servlet properties.

Table 2.2: Non-Functional Requirements Specifications

Chapter 3: Object-Oriented Design and Methodology

3.1 Methodology Used

The development of learnX.help followed the Iterative Object-Oriented Software Engineering (OOSE) lifecycle. This methodology emphasizes structuring system logic around key business entities. The system was iteratively modeled using Unified Modeling Language (UML) notation, translating requirements into clear classes, and maintaining a high level of abstraction between layers (Presentation, Service, and Data Access).

3.2 Class Diagram Mapping

Below is a structural mapping showing where the comprehensive UML/QML Class Diagram resides. It depicts the core relational mappings of learnX.help, displaying class fields, methods, and multiplicities (e.g., ManyToOne associations between User and StudentClass/University).

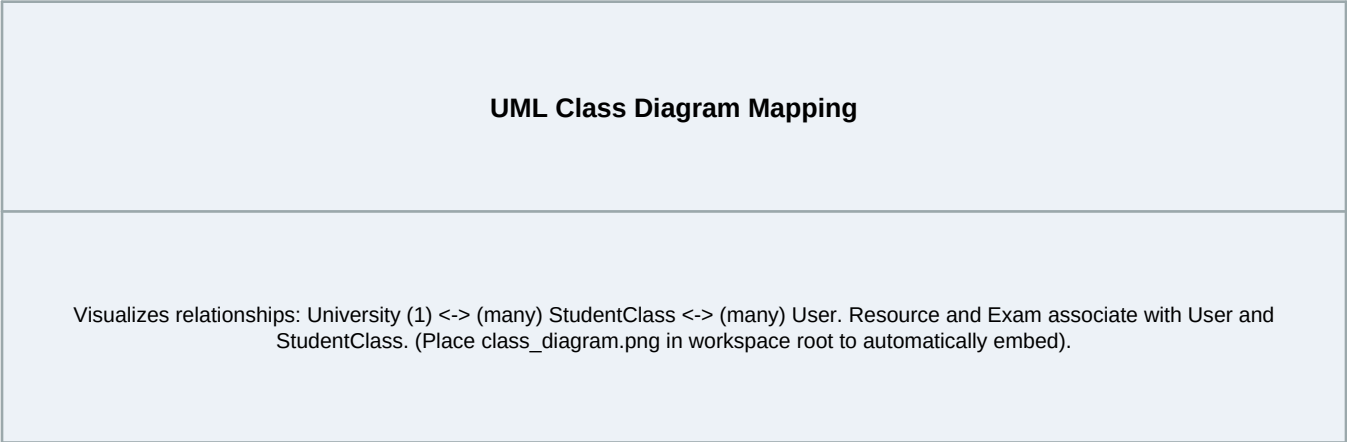


Figure 3.1: Component-Level System Architecture Diagram



Figure 3.2: Class Architecture and Key Associations

3.3 Detailed Class Descriptions

The following tables provide details on the properties, data types, and roles of the key classes defined in the application codebase. Field types utilize standard Java packages and custom JPA references.

Field Name	Type	Annotations / Structural Purpose
id	Long	@Id, @GeneratedValue (IDENTITY) - Database primary key.
username	String	Unique, non-null user identification nickname.
password	String	Encrypted string containing BCrypt hashed credentials.
fullName	String	Full legal name of the user.
email	String	Unique contact email address.
role	Role (Enum)	STUDENT, CR, TEACHER, or ADMIN defining authorization bounds.
studentClass	StudentClass	@ManyToOne - Defines the class section mapping for students/CRs.
university	University	@ManyToOne - Enforces tenant isolation for universities.

Table 3.1: Detailed Mapping of User Entity Class

Field Name	Type	Annotations / Structural Purpose
id	Long	Primary Key.
batch	String	Academic batch label (e.g. '43').
department	String	Division department label (e.g. 'CSE').
section	String	Class subsection identifier (e.g. 'A').
university	University	@ManyToOne - Associates the class with a university tenant.
students	List	@OneToMany(mappedBy='studentClass') - enrolled students.

Table 3.2: Detailed Mapping of StudentClass Class

Field Name	Type	Annotations / Structural Purpose
id	Long	Primary Key.
name	String	Unique full university name (e.g., 'USTC').
domain	String	Unique subdomain string for URL mapping (e.g., 'ustc.learnx.com').
logoUrl	String	Base64 or URL path to the institutional logo image.

Table 3.3: Detailed Mapping of University Class

Field Name	Type	Annotations / Structural Purpose
id	Long	Primary Key.
title	String	Display title of notes/handouts.
fileName	String	Physical file name upload string.
fileData	byte[]	@Lob - Stores high-capacity note bytes (up to 100MB).
uploadedBy	User	@ManyToOne - Reference to user who uploaded the handout.
likesCount	int	@Transient - Calculated reaction sum counts.
driveLink	String	Direct Google Drive URL for external lecture videos.

Table 3.4: Detailed Mapping of Resource Class

Method/Field	Type	Functional / Architectural Role
scheduleItemRepository	ScheduleItemRepository	CRUD operations for loading academic day-routines.
classTestRepository	ClassTestRepository	CRUD operations for loading pre-scheduled examinations.
detectFreeSlots	List	Loads routines & tests, merges intervals, returns open slots.
mergeIntervals	List	Filters, clamps intervals, sorts by start, and merges overlaps.

Table 3.5: Detailed Mapping of SlotDetectionService Class

Chapter 4: Implementation and Core OOP Principles

4.1 Key OOP Principles Applied

Encapsulation / Data Hiding: All core entity variables are declared as private. Access to and modification of these fields are regulated strictly through getter/setter methods. This prevents unauthorized modifications of internal object state. Lombok annotations (@Data, @Builder) are utilized to enforce compile-time generation of these interfaces, keeping source code clean and secure.

Inheritance: Extensive interface inheritance is implemented. Every data repository extends Spring Data's JpaRepository. This inherits powerful generic database CRUD and pagination capabilities, demonstrating robust reusable inheritance without redundant class declarations.

Polymorphism: Polymorphism is applied via interface implementations and method overriding. For instance, CustomUserDetailsService implements Spring Security's UserDetailsService and overrides the loadUserByUsername method. The security filter chain executes this method polymorphically at runtime. Additionally, initialization tasks run polymorphically via CommandLineRunner.run().

Abstraction: Database query complexities and transaction controls are hidden behind abstract repository method declarations (e.g. Optional findByUsername(String u)). High-level business actions in the Controllers invoke these abstract contracts without concern for low-level database operations.

4.2 Code Snippets

The following figures display isolated, highly clean code segments extracted directly from the workspace to precisely prove the application of each of these 4 pillars inside the developed codebase.

Encapsulation Proof (User.java):

```
1 public class User {
2
3     @Id
4     @GeneratedValue(strategy = GenerationType.IDENTITY)
5     private Long id;
6
7     @Column(unique = true, nullable = false)
8     private String username;
9
10    @Column(nullable = false)
11    private String password;
12
13    @Column(nullable = false)
14    private String fullName;
15
16    @Column(unique = true, nullable = false)
17    private String email;
18
19    @Enumerated(EnumType.STRING)
20    @Column(nullable = false)
21    private Role role;
22
23    @Builder.Default
24    @Column(nullable = false)
25    private boolean approved = true;
26
27    private String idNo;
28    private String department;
29    private String batch;
30    private String semester;
```

```
31     private String section;  
32     private String designation;  
33
```

Figure 4.1: Code Snippet proving Encapsulation (User.java)

Inheritance Proof (UserRepository.java):

```

1 public interface UserRepository extends JpaRepository<User, Long> {
2     Optional<User> findByUsername(String username);
3     Optional<User> findByEmail(String email);
4     boolean existsByUsername(String username);
5     boolean existsByEmail(String email);
6     java.util.List<User> findByUniversityAndRole(University university, User.Role role);
7     java.util.List<User> findByStudentClass(StudentClass studentClass);
8 }
9

```

*Figure 4.2: Code Snippet proving Inheritance (UserRepository.java)***Polymorphism Proof (CustomUserDetailsService.java):**

```

1 @Service
2 @RequiredArgsConstructor
3 public class CustomUserDetailsService implements UserDetailsService {
4
5     private final UserRepository userRepository;
6     private final SystemAdminRepository systemAdminRepository;
7
8     @Override
9     public UserDetails loadUserByUsername(String username) throws UsernameNotFoundException {
10         // First check if user is a system admin (site owner / master admin)
11         Optional<SystemAdmin> sysAdmin = systemAdminRepository.findByUsername(username);
12         if (sysAdmin.isPresent()) {
13             return org.springframework.security.core.userdetails.User.builder()
14                 .username(sysAdmin.get().getUsername())
15                 .password(sysAdmin.get().getPassword())
16                 .roles("SYSTEM_ADMIN")
17                 .build();
18         }
19
20         // Then check if user is a university user
21         User user = userRepository.findByUsername(username)
22             .orElseThrow(() -> new UsernameNotFoundException("User not found with username: " +
23 username));
24
25         return org.springframework.security.core.userdetails.User.builder()
26             .username(user.getUsername())
27             .password(user.getPassword())
28             .roles(user.getRole().name())
29             .build();
30     }
31 }

```

Figure 4.3: Code Snippet proving Polymorphism (CustomUserDetailsService.java)

Abstraction Proof (SlotDetectionService.java):

```
1  @Service
2  @AllArgsConstructor
3  public class SlotDetectionService {
4
5      private final ScheduleItemRepository scheduleItemRepository;
6      private final ClassTestRepository classTestRepository;
7
8      @Data
9      @NoArgsConstructor
10     @AllArgsConstructor
11     public static class TimeInterval {
12         private LocalTime start;
13         private LocalTime end;
14     }
15
16     public List<TimeInterval> detectFreeSlots(LocalDate date, int durationMinutes,
17 com.example.demo.entity.StudentClass studentClass) {
```

Figure 4.4: Code Snippet proving Abstraction (SlotDetectionService.java)

Chapter 5: Testing and Results

5.1 Test Plan

To verify that the system satisfies all operational objectives and OOP boundaries, a multi-phased testing strategy is defined below. This plan verifies class boundaries, system transactions, and logical integration without executing live database connections in production.

Test ID	Phase	Target Component	Validation Strategy
TP-01	Unit Testing	User & Entity Models	Validate builder patterns, Lombok generation, and validation tags.
TP-02	Unit Testing	Slot Merging Logic	Test mergeIntervals with overlapping/boundary time windows.
TP-03	Integration	SlotDetectionService	Verify routines and exams date queries output correct free slots.
TP-04	Integration	Resource Controller	Verify multipart file upload parsing and byte storage verification.
TP-05	Integration	Exam Controller	Verify quiz grading, score computations, and JPA grade book persistence.
TP-06	Security	Authentication Filters	Confirm role permissions (STUDENT vs TEACHER vs ADMIN) on restricted REST paths.

Table 5.1: Test Plan Matrix

Chapter 6: Conclusion and Future Work

6.1 Conclusion

The learnX.help portal demonstrates how Object-Oriented software engineering can solve operational coordination bottlenecks in academic environments. By leveraging encapsulated models, reusable repository inheritance, polymorphic security dispatchers, and slot detection services, the application delivers a robust, secure dashboard. The system has achieved structural OOP stability and successfully coordinates resource vaults, exams, and routines.

6.2 Limitations

The current system exhibits two primary constraints: (1) lack of live messaging notifications (relying on page reload announcements); and (2) storing large file handouts (up to 100MB) directly inside the relational database as binary blobs. The latter can degrade database search speeds over time as uploads scale.

6.3 Future Enhancements

Professional scaling plans include: (1) decoupling binary file storage from the main database, using Amazon S3 or Google Cloud Storage via spring-cloud-aws; (2) implementing WebSocket communication to enable real-time announcement and routine conflict alerts; and (3) adding automated machine-learning optimization to auto-recommend ideal slots based on student historical attendance.

References

- [1] E. Gamma, R. Helm, R. Johnson, and J. Vlissides, *Design Patterns: Elements of Reusable Object-Oriented Software*. Boston, MA: Addison-Wesley, 1994.
- [2] H. Schildt, *Java: The Complete Reference*, 12th ed. New York, NY: McGraw-Hill, 2021.
- [3] R. Johnson et al., *Spring Boot in Action*. Shelter Island, NY: Manning Publications, 2016.
- [4] C. Bauer and G. King, *Java Persistence with Hibernate*. Shelter Island, NY: Manning Publications, 2015.

Appendix A: Full Source Code

The complete main Java source files recursively extracted from the src/main/java workspace directory are cleanly indexed and formatted below.

A.1: com/example/demo/DemoApplication.java

```
1 package com.example.demo;
2
3 import org.springframework.boot.SpringApplication;
4 import org.springframework.boot.autoconfigure.SpringBootApplication;
5
6 @SpringBootApplication
7 public class DemoApplication {
8
9     public static void main(String[] args) {
10         SpringApplication.run(DemoApplication.class, args);
11     }
12 }
13
14
```

A.2: com/example/demo/config/DataInitializer.java

```

1 package com.example.demo.config;
2
3 import com.example.demo.entity.*;
4 import com.example.demo.entity.User.Role;
5 import com.example.demo.repository.*;
6 import lombok.RequiredArgsConstructor;
7 import org.springframework.boot.CommandLineRunner;
8 import org.springframework.security.crypto.password.PasswordEncoder;
9 import org.springframework.stereotype.Component;
10
11 import java.time.LocalDateTime;
12 import java.time.LocalTime;
13 import java.util.List;
14
15 @Component
16 @RequiredArgsConstructor
17 public class DataInitializer implements CommandLineRunner {
18
19     private final UserRepository userRepository;
20     private final ScheduleItemRepository scheduleItemRepository;
21     private final ClassTestRepository classTestRepository;
22     private final GradeBookRepository gradeBookRepository;
23     private final ResourceRepository resourceRepository;
24     private final ExamRepository examRepository;
25     private final ExamQuestionRepository examQuestionRepository;
26     private final PasswordEncoder passwordEncoder;
27     private final SystemMetadataRepository systemMetadataRepository;
28     private final StudentClassRepository studentClassRepository;
29     private final UniversityRepository universityRepository;
30     private final SystemAdminRepository systemAdminRepository;
31
32     @Override
33     public void run(String... args) throws Exception {
34         // 1. Seed Master System Admin (Site Owner)
35         if (systemAdminRepository.count() == 0) {
36             System.out.println("Seeding Master System Admin...");
37             SystemAdmin master = SystemAdmin.builder()
38                 .username("master")
39                 .password(passwordEncoder.encode("password"))
40                 .fullName("Master Admin")
41                 .email("master@learnx.com")
42                 .build();
43             systemAdminRepository.save(master);
44         }
45
46         // 2. Seed Default University & default Admin
47         University defaultUni = null;
48         if (universityRepository.count() == 0) {
49             System.out.println("Seeding Default University...");
50             defaultUni = University.builder()
51                 .name("LearnX")
52                 .domain("university.edu")
53                 .logoUrl("")
54                 .build();
55             defaultUni = universityRepository.save(defaultUni);
56
57             if (userRepository.count() == 0) {
58                 System.out.println("Seeding default university admin...");
59                 User defaultAdmin = User.builder()
60                     .username("admin")
61                     .password(passwordEncoder.encode("password"))
62                     .fullName("Admin")
63                     .email("admin@learnx.help")
64                     .role(Role.ADMIN)
65                     .approved(true)
66                     .university(defaultUni)
67                     .build();
68                 userRepository.save(defaultAdmin);
69             }
70         } else {
71             List<University> allUnis = universityRepository.findAll();
72             if (!allUnis.isEmpty()) {

```

```

73         defaultUni = allUnis.get(0);
74         if ("LearnX University".equals(defaultUni.getName())) {
75             System.out.println("Migrating university name to LearnX...");
76             defaultUni.setName("LearnX");
77             defaultUni = universityRepository.save(defaultUni);
78         }
79     }
80     userRepository.findByUsername("admin").ifPresent(admin -> {
81         if ("University Admin".equals(admin.getFullName())) {
82             System.out.println("Migrating default admin name to Admin...");
83             admin.setFullName("Admin");
84             if ("admin@university.edu".equals(admin.getEmail())) {
85                 admin.setEmail("admin@learnx.help");
86             }
87             userRepository.save(admin);
88         }
89     });
90 }
91
92 // 3. Seed Default Metadata Options
93 if (systemMetadataRepository.count() == 0 && defaultUni != null) {
94     System.out.println("Seeding Default Metadata Options...");
95
96     String[] departments = {"Computer Science & Engineering", "Electrical & Electronic Engineering", "Business
Administration"};
97     for (String val : departments) {
98         systemMetadataRepository.save(SystemMetadata.builder().type("DEPARTMENT").value(val).university(defaultUni)
99         .build());
100     }
101     String[] semesters = {"1st Semester", "2nd Semester", "3rd Semester", "4th Semester", "5th Semester", "6th
Semester", "7th Semester", "8th Semester"};
102     for (String val : semesters) {
103         systemMetadataRepository.save(SystemMetadata.builder().type("SEMESTER").value(val).university(defaultUni).b
104         uild());
105     }
106     String[] batches = {"Batch 21", "Batch 22", "Batch 23", "Batch 24"};
107     for (String val : batches) {
108         systemMetadataRepository.save(SystemMetadata.builder().type("BATCH").value(val).university(defaultUni).buil
109         d());
110     }
111     String[] designations = {"Lecturer", "Assistant Professor", "Associate Professor", "Professor"};
112     for (String val : designations) {
113         systemMetadataRepository.save(SystemMetadata.builder().type("DESIGNATION").value(val).university(defaultUni)
114         ).build());
115     }
116     String[] sections = {"Section A", "Section B", "Section C"};
117     for (String val : sections) {
118         systemMetadataRepository.save(SystemMetadata.builder().type("SECTION").value(val).university(defaultUni).bu
119         ild());
120     }
121 }
122
123 // 4. Seed Test Users (Teacher, Student, CR)
124 if (defaultUni != null) {
125     if (userRepository.findByUsername("teacher").isEmpty()) {
126         System.out.println("Seeding test teacher account...");
127         User testTeacher = User.builder()
128             .username("teacher")
129             .password(passwordEncoder.encode("password"))
130             .fullName("Test Teacher")
131             .email("teacher@learnx.help")
132             .role(Role.TEACHER)
133             .approved(true)
134             .designation("Lecturer")
135             .department("Computer Science & Engineering")
136             .university(defaultUni)
137             .build();
138         userRepository.save(testTeacher);
139     }
140     if (userRepository.findByUsername("student").isEmpty()) {

```

```

141         System.out.println("Seeding test student account...");
142         User testStudent = User.builder()
143             .username("student")
144             .password(passwordEncoder.encode("password"))
145             .fullName("Test Student")
146             .email("student@learnx.help")
147             .role(Role.STUDENT)
148             .approved(true)
149             .semester("1st Semester")
150             .batch("Batch 21")
151             .section("Section A")
152             .department("Computer Science & Engineering")
153             .university(defaultUni)
154             .build();
155
156         final University finalUni = defaultUni;
157         StudentClass sc = studentClassRepository.findByBatchAndDepartmentAndSection("Batch 21", "Computer Science &
Engineering", "Section A")
158             .orElseGet(() -> studentClassRepository.save(StudentClass.builder()
159                 .batch("Batch 21")
160                 .department("Computer Science & Engineering")
161                 .section("Section A")
162                 .university(finalUni)
163                 .build()));
164         testStudent.setStudentClass(sc);
165         userRepository.save(testStudent);
166     }
167
168     if (userRepository.findByUsername("cr").isEmpty()) {
169         System.out.println("Seeding test CR account...");
170         User testCR = User.builder()
171             .username("cr")
172             .password(passwordEncoder.encode("password"))
173             .fullName("Test CR")
174             .email("cr@learnx.help")
175             .role(Role.CR)
176             .approved(true)
177             .semester("1st Semester")
178             .batch("Batch 21")
179             .section("Section A")
180             .department("Computer Science & Engineering")
181             .university(defaultUni)
182             .build();
183
184         final University finalUni = defaultUni;
185         StudentClass sc = studentClassRepository.findByBatchAndDepartmentAndSection("Batch 21", "Computer Science &
Engineering", "Section A")
186             .orElseGet(() -> studentClassRepository.save(StudentClass.builder()
187                 .batch("Batch 21")
188                 .department("Computer Science & Engineering")
189                 .section("Section A")
190                 .university(finalUni)
191                 .build()));
192         testCR.setStudentClass(sc);
193         userRepository.save(testCR);
194     }
195 }
196
197 System.out.println("Data seeding completed. Ready for production.");
198 }
199 }
200

```

A.3: com/example/demo/config/SchemaInitializer.java

```
1 package com.example.demo.config;
2
3 import lombok.RequiredArgsConstructor;
4 import org.springframework.boot.CommandLineRunner;
5 import org.springframework.core.annotation.Order;
6 import org.springframework.jdbc.core.JdbcTemplate;
7 import org.springframework.stereotype.Component;
8
9 @Component
10 @Order(1) // Run before DataInitializer to ensure table exists and is updated
11 @RequiredArgsConstructor
12 public class SchemaInitializer implements CommandLineRunner {
13
14     private final JdbcTemplate jdbcTemplate;
15
16     @Override
17     public void run(String... args) throws Exception {
18         try {
19             System.out.println("LearnX: Adjusting schema for base64 profile picture uploads...");
20             // Alter column type to TEXT in H2 to allow long base64 image strings
21             jdbcTemplate.execute("ALTER TABLE users ALTER COLUMN profile_pic_url TEXT");
22             System.out.println("LearnX: Successfully altered users.profile_pic_url to TEXT");
23         } catch (Exception e) {
24             // Log it - it might fail if column is already TEXT or table structure differs, which is fine
25             System.out.println("LearnX Schema adjustment status: " + e.getMessage());
26         }
27     }
28 }
29
```

A.4: com/example/demo/config/SecurityConfig.java

```

1 package com.example.demo.config;
2
3 import org.springframework.context.annotation.Bean;
4 import org.springframework.context.annotation.Configuration;
5 import org.springframework.security.authentication.AuthenticationManager;
6 import org.springframework.security.config.annotation.authentication.configuration.AuthenticationConfiguration;
7 import org.springframework.security.config.annotation.web.builders.HttpSecurity;
8 import org.springframework.security.config.annotation.web.configuration.EnableWebSecurity;
9 import org.springframework.security.config.annotation.web.configurers.HeadersConfigurer;
10 import org.springframework.security.crypto.bcrypt.BCryptPasswordEncoder;
11 import org.springframework.security.crypto.password.PasswordEncoder;
12 import org.springframework.security.web.SecurityFilterChain;
13
14 @Configuration
15 @EnableWebSecurity
16 public class SecurityConfig {
17
18     @Bean
19     public PasswordEncoder passwordEncoder() {
20         return new BCryptPasswordEncoder();
21     }
22
23     @Bean
24     public com.fasterxml.jackson.databind.ObjectMapper objectMapper() {
25         return new com.fasterxml.jackson.databind.ObjectMapper();
26     }
27
28     @Bean
29     public AuthenticationManager authenticationManager(AuthenticationConfiguration authConfig) throws Exception {
30         return authConfig.getAuthenticationManager();
31     }
32
33     @Bean
34     public SecurityFilterChain securityFilterChain(HttpSecurity http) throws Exception {
35         http
36             .csrf(csrf -> csrf.disable())
37             .headers(headers -> headers.frameOptions(HeadersConfigurer.FrameOptionsConfig::disable))
38             .authorizeHttpRequests(auth -> auth
39                 .requestMatchers("/h2-console/**").permitAll()
40                 .requestMatchers("/api/auth/**").permitAll()
41                 .requestMatchers("/api/bugs/**").permitAll()
42                 .requestMatchers("/api/master/**").permitAll()
43                 .requestMatchers("/api/admin/**").permitAll()
44                 .requestMatchers("/", "/index.html", "/app.js", "/style.css", "/learnx_logo.png", "/ustc_building.jpg",
45 "/favicon.ico").permitAll()
46                 .anyRequest().authenticated()
47             );
48         return http.build();
49     }
50 }

```

A.5: com/example/demo/controller/AdminController.java

```

1 package com.example.demo.controller;
2
3 import com.example.demo.entity.User;
4 import com.example.demo.repository.UserRepository;
5 import lombok.AllArgsConstructor;
6 import org.springframework.http.ResponseEntity;
7 import org.springframework.web.bind.annotation.*;
8
9 import java.util.ArrayList;
10 import java.util.List;
11 import java.util.Map;
12
13 @RestController
14 @RequestMapping("/api/admin")
15 @AllArgsConstructor
16 public class AdminController {
17
18     private final UserRepository userRepository;
19
20     @GetMapping("/pending")
21     public ResponseEntity<?> getPendingUsers() {
22         List<User> users = userRepository.findAll();
23         List<Map<String, Object>> pending = new ArrayList<>();
24
25         for (User u : users) {
26             if (!u.isApproved()) {
27                 pending.add(Map.of(
28                     "id", u.getId(),
29                     "username", u.getUsername(),
30                     "fullName", u.getFullName(),
31                     "email", u.getEmail(),
32                     "role", u.getRole().name()
33                 ));
34             }
35         }
36         return ResponseEntity.ok(pending);
37     }
38
39     @PostMapping("/approve/{id}")
40     public ResponseEntity<?> approveUser(@PathVariable Long id) {
41         User user = userRepository.findById(id).orElse(null);
42         if (user == null) {
43             return ResponseEntity.notFound().build();
44         }
45         user.setApproved(true);
46         userRepository.save(user);
47         return ResponseEntity.ok(Map.of("message", "User account approved successfully"));
48     }
49
50     @DeleteMapping("/reject/{id}")
51     public ResponseEntity<?> rejectUser(@PathVariable Long id) {
52         User user = userRepository.findById(id).orElse(null);
53         if (user == null) {
54             return ResponseEntity.notFound().build();
55         }
56         userRepository.delete(user);
57         return ResponseEntity.ok(Map.of("message", "User account request rejected and deleted"));
58     }
59 }
60

```

A.6: com/example/demo/controller/AnnouncementController.java

```

1 package com.example.demo.controller;
2
3 import com.example.demo.entity.Announcement;
4 import com.example.demo.entity.User;
5 import com.example.demo.repository.AnnouncementRepository;
6 import com.example.demo.repository.UserRepository;
7 import lombok.AllArgsConstructor;
8 import lombok.Data;
9 import lombok.NoArgsConstructor;
10 import org.springframework.http.HttpStatus;
11 import org.springframework.http.ResponseEntity;
12 import org.springframework.web.bind.annotation.*;
13
14 import java.security.Principal;
15 import java.time.LocalDateTime;
16 import java.util.List;
17 import java.util.Map;
18
19 @RestController
20 @RequestMapping("/api/announcements")
21 @AllArgsConstructor
22 public class AnnouncementController {
23
24     private final AnnouncementRepository announcementRepository;
25     private final UserRepository userRepository;
26
27     @Data
28     @NoArgsConstructor
29     @AllArgsConstructor
30     public static class CreateAnnouncementRequest {
31         private String title;
32         private String content;
33         private boolean global;
34     }
35
36     @GetMapping
37     public ResponseEntity<?> getAnnouncements(Principal principal) {
38         if (principal == null) {
39             return ResponseEntity.status(HttpStatus.UNAUTHORIZED).body(Map.of("error", "Not authenticated"));
40         }
41         User currentUser = userRepository.findByUsername(principal.getName()).orElse(null);
42         if (currentUser == null) {
43             return ResponseEntity.status(HttpStatus.UNAUTHORIZED).body(Map.of("error", "User not found"));
44         }
45
46         List<Announcement> announcements;
47         // If Student or CR, show global announcements + class-specific announcements
48         if (currentUser.getRole() == User.Role.STUDENT || currentUser.getRole() == User.Role.CR) {
49             if (currentUser.getStudentClass() != null) {
50                 announcements =
51 announcementRepository.findByStudentClassOrStudentClassIsNullOrderByCreatedAtDesc(currentUser.getStudentClass());
52             } else {
53                 announcements = announcementRepository.findAllOrderByCreatedAtDesc();
54             }
55         } else {
56             // Admin and Teacher can see all announcements
57             announcements = announcementRepository.findAllOrderByCreatedAtDesc();
58         }
59
60         return ResponseEntity.ok(announcements);
61     }
62
63     @PostMapping
64     public ResponseEntity<?> createAnnouncement(@RequestBody CreateAnnouncementRequest request, Principal principal) {
65         if (principal == null) {
66             return ResponseEntity.status(HttpStatus.UNAUTHORIZED).body(Map.of("error", "Not authenticated"));
67         }
68         User currentUser = userRepository.findByUsername(principal.getName()).orElse(null);
69         if (currentUser == null) {
70             return ResponseEntity.status(HttpStatus.UNAUTHORIZED).body(Map.of("error", "User not found"));
71         }

```

```

72         // Only Admin, Teacher, and CR can create announcements
73         if (currentUser.getRole() == User.Role.STUDENT) {
74             return ResponseEntity.status(HttpStatus.FORBIDDEN).body(Map.of("error", "Only teachers, CRs, and administrators
can publish announcements."));
75         }
76
77         Announcement announcement = Announcement.builder()
78             .title(request.getTitle())
79             .content(request.getContent())
80             .createdAt(LocalDate.now())
81             .createdBy(currentUser.getFullName())
82             .createdByRole(currentUser.getRole().name())
83             .studentClass(request.isGlobal() ? null : currentUser.getStudentClass())
84             .build();
85
86         // If CR creates, always target their class (cannot create global announcements)
87         if (currentUser.getRole() == User.Role.CR) {
88             announcement.setStudentClass(currentUser.getStudentClass());
89         }
90
91         Announcement saved = announcementRepository.save(announcement);
92         return ResponseEntity.ok(saved);
93     }
94
95     @DeleteMapping("/{id}")
96     public ResponseEntity<?> deleteAnnouncement(@PathVariable Long id, Principal principal) {
97         if (principal == null) {
98             return ResponseEntity.status(HttpStatus.UNAUTHORIZED).body(Map.of("error", "Not authenticated"));
99         }
100        User currentUser = userRepository.findByUsername(principal.getName()).orElse(null);
101        if (currentUser == null) {
102            return ResponseEntity.status(HttpStatus.UNAUTHORIZED).body(Map.of("error", "User not found"));
103        }
104
105        Announcement announcement = announcementRepository.findById(id).orElse(null);
106        if (announcement == null) {
107            return ResponseEntity.notFound().build();
108        }
109
110        // Only Admin, Teacher, or creator can delete
111        boolean canDelete = currentUser.getRole() == User.Role.ADMIN ||
112            currentUser.getRole() == User.Role.TEACHER ||
113            announcement.getCreatedBy().equals(currentUser.getFullName());
114
115        if (!canDelete) {
116            return ResponseEntity.status(HttpStatus.FORBIDDEN).body(Map.of("error", "You do not have permission to delete
this announcement."));
117        }
118
119        announcementRepository.delete(announcement);
120        return ResponseEntity.ok(Map.of("message", "Announcement deleted successfully"));
121    }
122 }
123

```

A.7: com/example/demo/controller/AuthController.java

```

1 package com.example.demo.controller;
2
3 import com.example.demo.entity.User;
4 import com.example.demo.entity.User.Role;
5 import com.example.demo.entity.SystemAdmin;
6 import com.example.demo.entity.University;
7 import com.example.demo.repository.UserRepository;
8 import com.example.demo.repository.SystemAdminRepository;
9 import com.example.demo.repository.UniversityRepository;
10 import jakarta.servlet.http.HttpServletRequest;
11 import jakarta.servlet.http.HttpSession;
12 import lombok.AllArgsConstructor;
13 import lombok.Data;
14 import lombok.NoArgsConstructor;
15 import org.springframework.http.ResponseEntity;
16 import org.springframework.security.authentication.AuthenticationManager;
17 import org.springframework.security.authentication.UsernamePasswordAuthenticationToken;
18 import org.springframework.security.core.Authentication;
19 import org.springframework.security.core.context.SecurityContextHolder;
20 import org.springframework.security.crypto.password.PasswordEncoder;
21 import org.springframework.security.web.context.HttpSessionSecurityContextRepository;
22 import org.springframework.web.bind.annotation.*;
23
24 import java.security.Principal;
25 import java.util.Map;
26 import java.util.Optional;
27
28 @RestController
29 @RequestMapping("/api/auth")
30 @AllArgsConstructor
31 public class AuthController {
32
33     private final AuthenticationManager authenticationManager;
34     private final UserRepository userRepository;
35     private final PasswordEncoder passwordEncoder;
36     private final SystemAdminRepository systemAdminRepository;
37     private final UniversityRepository universityRepository;
38     private final org.springframework.mail.javamail.JavaMailSender mailSender;
39     private final org.springframework.core.env.Environment env;
40
41     @Data
42     @NoArgsConstructor
43     @AllArgsConstructor
44     public static class LoginRequest {
45         private String username;
46         private String password;
47     }
48
49     @Data
50     @NoArgsConstructor
51     @AllArgsConstructor
52     public static class SignupRequest {
53         private String username;
54         private String password;
55         private String fullName;
56         private String email;
57         private String role; // STUDENT, CR, TEACHER
58         private String idNo;
59         private String department;
60         private String batch;
61         private String semester;
62         private String section;
63         private String designation;
64     }
65
66     @Data
67     @AllArgsConstructor
68     public static class UserResponse {
69         private Long id;
70         private String username;
71         private String fullName;
72         private String email;

```

```

73     private String role;
74     private String idNo;
75     private String department;
76     private String batch;
77     private String semester;
78     private String section;
79     private String designation;
80     private String profilePicUrl;
81 }
82
83 @PostMapping("/login")
84 public ResponseEntity<?> login(@RequestBody LoginRequest request, HttpServletRequest servletRequest) {
85     try {
86         Authentication authentication = authenticationManager.authenticate(
87             new UsernamePasswordAuthenticationToken(request.getUsername(), request.getPassword())
88         );
89         SecurityContextHolder.getContext().setAuthentication(authentication);
90
91         HttpSession session = servletRequest.getSession(true);
92         session.setAttribute(HttpSessionSecurityContextRepository.SPRING_SECURITY_CONTEXT_KEY,
93             SecurityContextHolder.getContext());
94
95         Optional<User> userOpt = userRepository.findByUsername(request.getUsername());
96         if (userOpt.isPresent()) {
97             User user = userOpt.get();
98             if (!user.isApproved()) {
99                 SecurityContextHolder.clearContext();
100                 if (session != null) {
101                     session.invalidate();
102                 }
103                 return ResponseEntity.status(403).body(Map.of("error", "Your account is pending administrator
104 approval."));
105             }
106             return ResponseEntity.ok(new UserResponse(
107                 user.getId(),
108                 user.getUsername(),
109                 user.getFullName(),
110                 user.getEmail(),
111                 user.getRole().name(),
112                 user.getIdNo(),
113                 user.getDepartment(),
114                 user.getBatch(),
115                 user.getSemester(),
116                 user.getSection(),
117                 user.getDesignation(),
118                 user.getProfilePicUrl()
119             ));
120         } else {
121             Optional<SystemAdmin> sysAdminOpt = systemAdminRepository.findByUsername(request.getUsername());
122             if (sysAdminOpt.isPresent()) {
123                 SystemAdmin sysAdmin = sysAdminOpt.get();
124                 return ResponseEntity.ok(new UserResponse(
125                     sysAdmin.getId(),
126                     sysAdmin.getUsername(),
127                     sysAdmin.getFullName(),
128                     sysAdmin.getEmail(),
129                     "SYSTEM_ADMIN",
130                     null, null, null, null, null, null, null
131                 ));
132             }
133             throw new RuntimeException("User not found in repositories");
134         } catch (Exception e) {
135             return ResponseEntity.status(401).body(Map.of("error", "Invalid username or password"));
136         }
137     }
138
139     @PostMapping("/signup")
140     public ResponseEntity<?> signup(@RequestBody SignupRequest request, @RequestHeader(value = "X-University-Domain",
141         required = false) String domainHeader) {
142         if (userRepository.existsByUsername(request.getUsername())) {
143             return ResponseEntity.badRequest().body(Map.of("error", "Username already exists"));
144         }
145         if (request.getEmail() != null && !request.getEmail().trim().isEmpty() &&
146             userRepository.existsByEmail(request.getEmail())) {

```

```

144         return ResponseEntity.badRequest().body(Map.of("error", "Email address is already registered"));
145     }
146
147     Role userRole;
148     try {
149         userRole = Role.valueOf(request.getRole().toUpperCase());
150         if (userRole == Role.ADMIN) {
151             return ResponseEntity.badRequest().body(Map.of("error", "Direct administrator registration is
prohibited"));
152         }
153     } catch (Exception e) {
154         userRole = Role.STUDENT; // Default to Student
155     }
156
157     // Enforce required fields
158     if (request.getUsername() == null || request.getUsername().trim().isEmpty() ||
159         request.getPassword() == null || request.getPassword().trim().isEmpty() ||
160         request.getFullName() == null || request.getFullName().trim().isEmpty() ||
161         request.getEmail() == null || request.getEmail().trim().isEmpty() ||
162         request.getIdNo() == null || request.getIdNo().trim().isEmpty() ||
163         request.getDepartment() == null || request.getDepartment().trim().isEmpty()) {
164         return ResponseEntity.badRequest().body(Map.of("error", "Please fill in all necessary registration fields."));
165     }
166
167     if (userRole == Role.STUDENT || userRole == Role.CR) {
168         if (request.getSemester() == null || request.getSemester().trim().isEmpty() ||
169             request.getBatch() == null || request.getBatch().trim().isEmpty() ||
170             request.getSection() == null || request.getSection().trim().isEmpty()) {
171             return ResponseEntity.badRequest().body(Map.of("error", "Students/CRs must fill in Semester, Batch, and
Section."));
172         }
173     } else if (userRole == Role.TEACHER) {
174         if (request.getDesignation() == null || request.getDesignation().trim().isEmpty()) {
175             return ResponseEntity.badRequest().body(Map.of("error", "Teachers must specify Designation."));
176         }
177     }
178
179     University uni = null;
180     if (domainHeader != null && !domainHeader.trim().isEmpty()) {
181         uni = universityRepository.findByDomain(domainHeader).orElse(null);
182     }
183     if (uni == null) {
184         java.util.List<University> all = universityRepository.findAll();
185         if (!all.isEmpty()) {
186             uni = all.get(0);
187         }
188     }
189
190     boolean approved = false;
191
192     User user = User.builder()
193         .username(request.getUsername())
194         .password(passwordEncoder.encode(request.getPassword()))
195         .fullName(request.getFullName())
196         .email(request.getEmail())
197         .role(userRole)
198         .approved(approved)
199         .idNo(request.getIdNo())
200         .department(request.getDepartment())
201         .batch(request.getBatch())
202         .semester(request.getSemester())
203         .section(request.getSection())
204         .designation(request.getDesignation())
205         .university(uni)
206         .build();
207
208     userRepository.save(user);
209
210     try {
211         org.springframework.mail.SimpleMailMessage message = new org.springframework.mail.SimpleMailMessage();
212         String fromEmail = env.getProperty("spring.mail.username");
213         if (fromEmail != null && !fromEmail.isEmpty()) {
214             message.setFrom(fromEmail);
215         }

```

```

216         message.setTo(user.getEmail());
217         message.setSubject("LearnX Registration Under Review");
218         message.setText("Hello " + user.getFullName() + ",\n\n"
219             + "Thank you for registering on LearnX.\n"
220             + "Your account request is currently pending administrator approval.\n"
221             + "You will receive another email once your account has been approved.\n\n"
222             + "Best regards,\n"
223             + "LearnX Team");
224         mailSender.send(message);
225     } catch (Exception ex) {
226         System.err.println("Failed to send signup email to " + user.getEmail() + ": " + ex.getMessage());
227     }
228
229     return ResponseEntity.ok(Map.of("message", "Your form is under progress. You will be notified via email once
approved."));
230 }
231
232 @GetMapping("/current-user")
233 public ResponseEntity<?> getCurrentUser(Principal principal) {
234     if (principal == null) {
235         return ResponseEntity.status(401).body(Map.of("error", "Not authenticated"));
236     }
237     Optional<User> userOpt = userRepository.findByUsername(principal.getName());
238     if (userOpt.isPresent()) {
239         User user = userOpt.get();
240         return ResponseEntity.ok(new UserResponse(
241             user.getId(),
242             user.getUsername(),
243             user.getFullName(),
244             user.getEmail(),
245             user.getRole().name(),
246             user.getIdNo(),
247             user.getDepartment(),
248             user.getBatch(),
249             user.getSemester(),
250             user.getSection(),
251             user.getDesignation(),
252             user.getProfilePicUrl()
253         ));
254     } else {
255         Optional<SystemAdmin> sysAdminOpt = systemAdminRepository.findByUsername(principal.getName());
256         if (sysAdminOpt.isPresent()) {
257             SystemAdmin sysAdmin = sysAdminOpt.get();
258             return ResponseEntity.ok(new UserResponse(
259                 sysAdmin.getId(),
260                 sysAdmin.getUsername(),
261                 sysAdmin.getFullName(),
262                 sysAdmin.getEmail(),
263                 "SYSTEM_ADMIN",
264                 null, null, null, null, null, null, null
265             ));
266         }
267     }
268     return ResponseEntity.status(401).body(Map.of("error", "User session invalid"));
269 }
270
271
272
273 @PostMapping("/logout")
274 public ResponseEntity<?> logout(HttpServletRequest servletRequest) {
275     HttpSession session = servletRequest.getSession(false);
276     if (session != null) {
277         session.invalidate();
278     }
279     SecurityContextHolder.clearContext();
280     return ResponseEntity.ok(Map.of("message", "Logged out successfully"));
281 }
282 }
283

```

A.8: com/example/demo/controller/BugReportController.java

```

1 package com.example.demo.controller;
2
3 import com.example.demo.entity.BugReport;
4 import com.example.demo.repository.BugReportRepository;
5 import lombok.AllArgsConstructor;
6 import lombok.Data;
7 import lombok.NoArgsConstructor;
8 import org.springframework.http.ResponseEntity;
9 import org.springframework.web.bind.annotation.*;
10
11 import java.time.LocalDateTime;
12 import java.util.Map;
13
14 @RestController
15 @RequestMapping("/api/bugs")
16 @AllArgsConstructor
17 public class BugReportController {
18
19     private final BugReportRepository bugReportRepository;
20
21     @Data
22     @NoArgsConstructor
23     @AllArgsConstructor
24     public static class BugReportSubmission {
25         private String title;
26         private String description;
27         private String reportedBy;
28     }
29
30     @PostMapping("/report")
31     public ResponseEntity<?> reportBug(@RequestBody BugReportSubmission submission) {
32         if (submission.getTitle() == null || submission.getTitle().trim().isEmpty()) {
33             return ResponseEntity.badRequest().body(Map.of("error", "Bug title is required"));
34         }
35         if (submission.getDescription() == null || submission.getDescription().trim().isEmpty()) {
36             return ResponseEntity.badRequest().body(Map.of("error", "Bug description is required"));
37         }
38
39         BugReport bug = BugReport.builder()
40             .title(submission.getTitle().trim())
41             .description(submission.getDescription().trim())
42             .reportedBy(submission.getReportedBy() != null && !submission.getReportedBy().trim().isEmpty() ?
43                 submission.getReportedBy().trim() : "Anonymous")
44             .createdAt(LocalDateTime.now())
45             .status("PENDING")
46             .build();
47
48         bugReportRepository.save(bug);
49         return ResponseEntity.ok(Map.of("message", "Bug report submitted successfully! Thank you for your feedback.));
50     }
51 }
52

```

A.9: com/example/demo/controller/DashboardController.java

```

1 package com.example.demo.controller;
2
3 import com.example.demo.entity.GradeBook;
4 import com.example.demo.entity.User;
5 import com.example.demo.repository.GradeBookRepository;
6 import com.example.demo.repository.UserRepository;
7 import lombok.AllArgsConstructor;
8 import lombok.Data;
9 import org.springframework.http.ResponseEntity;
10 import org.springframework.web.bind.annotation.*;
11
12 import java.security.Principal;
13 import java.util.ArrayList;
14 import java.util.List;
15 import java.util.Map;
16
17 @RestController
18 @RequestMapping("/api/dashboard")
19 @AllArgsConstructor
20 public class DashboardController {
21
22     private final GradeBookRepository gradeBookRepository;
23     private final UserRepository userRepository;
24
25     @Data
26     @AllArgsConstructor
27     public static class PerformanceStat {
28         private Long id;
29         private String courseName;
30         private String assessmentName;
31         private double marksObtained;
32         private double maxMarks;
33         private double classAverage;
34         private double classHighest;
35         private double percentile;
36     }
37
38     @Data
39     public static class AddGradeRequest {
40         private String studentUsername;
41         private String courseName;
42         private String assessmentName;
43         private double marksObtained;
44         private double maxMarks;
45     }
46
47     @GetMapping("/performance")
48     public ResponseEntity<?> getPerformance(Principal principal) {
49         User user = userRepository.findByUsername(principal.getName()).orElse(null);
50         if (user == null) {
51             return ResponseEntity.notFound().build();
52         }
53
54         // Fetch grades for this student
55         List<GradeBook> studentGrades = gradeBookRepository.findByStudent(user);
56         List<PerformanceStat> stats = new ArrayList<>();
57
58         for (GradeBook grade : studentGrades) {
59             // Find all scores for the same assessment in the class
60             List<GradeBook> allScores = gradeBookRepository.findByCourseNameAndAssessmentName(
61                 grade.getCourseName(), grade.getAssessmentName());
62
63             double totalMarks = 0;
64             double highest = 0;
65             int countBelowOrEqual = 0;
66             int totalCount = allScores.size();
67
68             for (GradeBook s : allScores) {
69                 totalMarks += s.getMarksObtained();
70                 if (s.getMarksObtained() > highest) {
71                     highest = s.getMarksObtained();
72                 }

```

```

73         if (s.getMarksObtained() <= grade.getMarksObtained()) {
74             countBelowOrEqual++;
75         }
76     }
77
78     double average = totalCount > 0 ? totalMarks / totalCount : 0;
79     double percentile = totalCount > 0 ? ((double) countBelowOrEqual / totalCount) * 100 : 0;
80
81     // Round to 2 decimal places
82     average = Math.round(average * 100.0) / 100.0;
83     highest = Math.round(highest * 100.0) / 100.0;
84     percentile = Math.round(percentile * 100.0) / 100.0;
85
86     stats.add(new PerformanceStat(
87         grade.getId(),
88         grade.getCourseName(),
89         grade.getAssessmentName(),
90         grade.getMarksObtained(),
91         grade.getMaxMarks(),
92         average,
93         highest,
94         percentile
95     ));
96 }
97
98     return ResponseEntity.ok(stats);
99 }
100
101 @GetMapping("/all-grades")
102 public ResponseEntity<?> getAllGrades() {
103     // Return all grades with student usernames for Teacher/CR dashboard view
104     return ResponseEntity.ok(gradeBookRepository.findAll());
105 }
106
107 @PostMapping("/grades")
108 public ResponseEntity<?> addGrade(@RequestBody AddGradeRequest request) {
109     User student = userRepository.findByUsername(request.getStudentUsername()).orElse(null);
110     if (student == null) {
111         return ResponseEntity.badRequest().body(Map.of("error", "Student username not found"));
112     }
113
114     GradeBook grade = GradeBook.builder()
115         .student(student)
116         .courseName(request.getCourseName())
117         .assessmentName(request.getAssessmentName())
118         .marksObtained(request.getMarksObtained())
119         .maxMarks(request.getMaxMarks())
120         .build();
121
122     GradeBook saved = gradeBookRepository.save(grade);
123     return ResponseEntity.ok(saved);
124 }
125
126 @DeleteMapping("/grades/{id}")
127 public ResponseEntity<?> deleteGrade(@PathVariable Long id) {
128     if (!gradeBookRepository.existsById(id)) {
129         return ResponseEntity.notFound().build();
130     }
131     gradeBookRepository.deleteById(id);
132     return ResponseEntity.ok(Map.of("message", "Grade record deleted successfully"));
133 }
134 }
135

```

A.10: com/example/demo/controller/EmailController.java

```

1 package com.example.demo.controller;
2
3 import com.example.demo.entity.User;
4 import com.example.demo.repository.UserRepository;
5 import lombok.AllArgsConstructor;
6 import lombok.Data;
7 import lombok.NoArgsConstructor;
8 import org.springframework.http.ResponseEntity;
9 import org.springframework.mail.SimpleMailMessage;
10 import org.springframework.mail.javamail.JavaMailSender;
11 import org.springframework.web.bind.annotation.*;
12 import org.springframework.core.env.Environment;
13
14 import java.security.Principal;
15 import java.util.List;
16 import java.util.Map;
17
18 @RestController
19 @RequestMapping("/api/mail")
20 @AllArgsConstructor
21 public class EmailController {
22
23     private final JavaMailSender mailSender;
24     private final Environment env;
25     private final UserRepository userRepository;
26
27     @Data
28     @NoArgsConstructor
29     @AllArgsConstructor
30     public static class SendMailRequest {
31         private String to;
32         private String subject;
33         private String body;
34     }
35
36     @PostMapping("/send")
37     public ResponseEntity<?> sendMail(@RequestBody SendMailRequest request, Principal principal) {
38         if (principal == null) {
39             return ResponseEntity.status(401).body(Map.of("error", "Not authenticated"));
40         }
41
42         if (request.getTo() == null || request.getTo().trim().isEmpty()) {
43             return ResponseEntity.badRequest().body(Map.of("error", "Recipient email ('to') is required"));
44         }
45         if (request.getSubject() == null || request.getSubject().trim().isEmpty()) {
46             return ResponseEntity.badRequest().body(Map.of("error", "Subject is required"));
47         }
48         if (request.getBody() == null || request.getBody().trim().isEmpty()) {
49             return ResponseEntity.badRequest().body(Map.of("error", "Message body is required"));
50         }
51
52         try {
53             SimpleMailMessage message = new SimpleMailMessage();
54             String fromEmail = env.getProperty("spring.mail.username");
55             if (fromEmail != null && !fromEmail.isEmpty()) {
56                 message.setFrom(fromEmail);
57             }
58             message.setTo(request.getTo());
59             message.setSubject(request.getSubject());
60             message.setText(request.getBody() + "\n\n---\nSent by: " + principal.getName() + " via LearnX");
61
62             mailSender.send(message);
63             return ResponseEntity.ok(Map.of("message", "Email sent successfully!"));
64         } catch (Exception e) {
65             return ResponseEntity.status(500).body(Map.of("error", "Failed to send email: " + e.getMessage()));
66         }
67     }
68
69     @GetMapping("/users")
70     public ResponseEntity<?> listUsers(Principal principal) {
71         if (principal == null) {
72             return ResponseEntity.status(401).body(Map.of("error", "Not authenticated"));
73         }
74     }
75 }

```

```
73     }
74     List<Map<String, String>> users = userRepository.findAll().stream()
75         .filter(u -> u.getEmail() != null && !u.getEmail().trim().isEmpty())
76         .map(u -> Map.of(
77             "fullName", u.getFullName() != null ? u.getFullName() : "",
78             "email", u.getEmail(),
79             "role", u.getRole() != null ? u.getRole().name() : "USER"
80         ))
81         .toList();
82     return ResponseEntity.ok(users);
83 }
84 }
85
```

A.11: com/example/demo/controller/ExamController.java

```

1 package com.example.demo.controller;
2
3 import com.example.demo.entity.*;
4 import com.example.demo.repository.*;
5 import com.fasterxml.jackson.core.type.TypeReference;
6 import com.fasterxml.jackson.databind.ObjectMapper;
7 import lombok.AllArgsConstructor;
8 import lombok.Data;
9 import lombok.NoArgsConstructor;
10 import org.springframework.http.ResponseEntity;
11 import org.springframework.web.bind.annotation.*;
12
13 import java.security.Principal;
14 import java.time.LocalDateTime;
15 import java.util.ArrayList;
16 import java.util.List;
17 import java.util.Map;
18 import java.util.Optional;
19
20 @RestController
21 @RequestMapping("/api/exams")
22 @AllArgsConstructor
23 public class ExamController {
24
25     private final ExamRepository examRepository;
26     private final ExamQuestionRepository examQuestionRepository;
27     private final ExamSubmissionRepository examSubmissionRepository;
28     private final UserRepository userRepository;
29     private final GradeBookRepository gradeBookRepository;
30     private final ObjectMapper objectMapper;
31
32     @Data
33     @NoArgsConstructor
34     @AllArgsConstructor
35     public static class QuestionDto {
36         private String questionText;
37         private String questionType; // MCQ, SHORT_ANSWER
38         private int points;
39         private String options; // Semicolon-separated for MCQ
40         private String correctAnswer;
41     }
42
43     @Data
44     @NoArgsConstructor
45     @AllArgsConstructor
46     public static class CreateExamRequest {
47         private String title;
48         private String description;
49         private int durationMinutes;
50         private String startTime; // ISO string
51         private String endTime; // ISO string
52         private List<QuestionDto> questions;
53     }
54
55     @Data
56     @NoArgsConstructor
57     @AllArgsConstructor
58     public static class AnswerSubmission {
59         private Long questionId;
60         private String answer;
61     }
62
63     @Data
64     @NoArgsConstructor
65     @AllArgsConstructor
66     public static class ExamResponseDto {
67         private Long id;
68         private String title;
69         private String description;
70         private int durationMinutes;
71         private LocalDateTime startTime;
72         private LocalDateTime endTime;

```

```

73     private boolean published;
74     private String teacherName;
75     private List<QuestionResponseDto> questions;
76     private boolean alreadySubmitted;
77     private Integer previousScore;
78 }
79
80 @Data
81 @NoArgsConstructor
82 @AllArgsConstructor
83 public static class QuestionResponseDto {
84     private Long id;
85     private String questionText;
86     private String questionType;
87     private int points;
88     private String options;
89     // Excluded: correctAnswer (for students)
90 }
91
92 @PostMapping("/create")
93 public ResponseEntity<?> createExam(@RequestBody CreateExamRequest request, Principal principal) {
94     User teacher = userRepository.findByUsername(principal.getName()).orElse(null);
95     if (teacher == null || teacher.getRole() == User.Role.STUDENT) {
96         return ResponseEntity.status(403).body(Map.of("error", "Only teachers or CRs can create exams"));
97     }
98
99     LocalDateTime start = LocalDateTime.parse(request.getStartTime());
100    LocalDateTime end = LocalDateTime.parse(request.getEndTime());
101
102    Exam exam = Exam.builder()
103        .title(request.getTitle())
104        .description(request.getDescription())
105        .teacher(teacher)
106        .durationMinutes(request.getDurationMinutes())
107        .startTime(start)
108        .endTime(end)
109        .published(true) // Auto-publish for demo purposes
110        .build();
111
112    Exam savedExam = examRepository.save(exam);
113
114    List<ExamQuestion> questions = new ArrayList<>();
115    for (QuestionDto qDto : request.getQuestions()) {
116        ExamQuestion question = ExamQuestion.builder()
117            .exam(savedExam)
118            .questionText(qDto.getQuestionText())
119            .questionType(ExamQuestion.QuestionType.valueOf(qDto.getQuestionType().toUpperCase()))
120            .points(qDto.getPoints())
121            .options(qDto.getOptions())
122            .correctAnswer(qDto.getCorrectAnswer())
123            .build();
124        questions.add(question);
125    }
126    examQuestionRepository.saveAll(questions);
127
128    return ResponseEntity.ok(Map.of("message", "Exam created and published successfully", "examId",
129    savedExam.getId()));
130 }
131
132 @GetMapping
133 public ResponseEntity<?> getExams(Principal principal) {
134     User user = userRepository.findByUsername(principal.getName()).orElse(null);
135     if (user == null) {
136         return ResponseEntity.status(401).build();
137     }
138
139     List<Exam> exams;
140     if (user.getRole() == User.Role.STUDENT || user.getRole() == User.Role.CR) {
141         if (user.getStudentClass() != null) {
142             exams = examRepository.findByStudentClassAndPublished(user.getStudentClass(), true);
143         } else {
144             exams = examRepository.findByPublished(true);
145         }
146     } else {

```

```

146         exams = examRepository.findAll();
147     }
148
149     List<Map<String, Object>> list = new ArrayList<>();
150     for (Exam exam : exams) {
151         Optional<ExamSubmission> submission = examSubmissionRepository.findByExamAndStudent(exam, user);
152         Map<String, Object> examMap = new java.util.HashMap<>();
153         examMap.put("id", exam.getId());
154         examMap.put("title", exam.getTitle());
155         examMap.put("description", exam.getDescription());
156         examMap.put("durationMinutes", exam.getDurationMinutes());
157         examMap.put("startTime", exam.getStartTime());
158         examMap.put("endTime", exam.getEndTime());
159         examMap.put("teacherName", exam.getTeacher() != null ? exam.getTeacher().getFullName() : "Unknown");
160         examMap.put("published", exam.isPublished());
161         examMap.put("alreadySubmitted", submission.isPresent());
162         examMap.put("score", submission.map(ExamSubmission::getScore).orElse(null));
163         list.add(examMap);
164     }
165
166     return ResponseEntity.ok(list);
167 }
168
169 @GetMapping("/{id}")
170 public ResponseEntity<?> getExamDetails(@PathVariable Long id, Principal principal) {
171     User user = userRepository.findByUsername(principal.getName()).orElse(null);
172     if (user == null) {
173         return ResponseEntity.status(401).build();
174     }
175
176     Exam exam = examRepository.findById(id).orElse(null);
177     if (exam == null) {
178         return ResponseEntity.notFound().build();
179     }
180
181     List<ExamQuestion> questions = examQuestionRepository.findByExam(exam);
182     List<QuestionResponseDto> qResponseList = new ArrayList<>();
183
184     for (ExamQuestion q : questions) {
185         qResponseList.add(new QuestionResponseDto(
186             q.getId(),
187             q.getQuestionText(),
188             q.getQuestionType().name(),
189             q.getPoints(),
190             q.getOptions()
191         ));
192     }
193
194     Optional<ExamSubmission> submission = examSubmissionRepository.findByExamAndStudent(exam, user);
195
196     ExamResponseDto response = new ExamResponseDto(
197         exam.getId(),
198         exam.getTitle(),
199         exam.getDescription(),
200         exam.getDurationMinutes(),
201         exam.getStartTime(),
202         exam.getEndTime(),
203         exam.isPublished(),
204         exam.getTeacher().getFullName(),
205         qResponseList,
206         submission.isPresent(),
207         submission.map(ExamSubmission::getScore).orElse(null)
208     );
209
210     return ResponseEntity.ok(response);
211 }
212
213 @PostMapping("/{id}/submit")
214 public ResponseEntity<?> submitExam(
215     @PathVariable Long id,
216     @RequestBody List<AnswerSubmission> answers,
217     Principal principal) {
218
219     User student = userRepository.findByUsername(principal.getName()).orElse(null);

```

```

220     if (student == null) {
221         return ResponseEntity.status(401).build();
222     }
223
224     Exam exam = examRepository.findById(id).orElse(null);
225     if (exam == null) {
226         return ResponseEntity.notFound().build();
227     }
228
229     // Check if student already submitted
230     if (examSubmissionRepository.findByExamAndStudent(exam, student).isPresent()) {
231         return ResponseEntity.badRequest().body(Map.of("error", "You have already submitted this exam"));
232     }
233
234     List<ExamQuestion> questions = examQuestionRepository.findByExam(exam);
235     int finalScore = 0;
236
237     for (AnswerSubmission ans : answers) {
238         ExamQuestion question = questions.stream()
239             .filter(q -> q.getId().equals(ans.getQuestionId()))
240             .findFirst()
241             .orElse(null);
242
243         if (question != null && ans.getAnswer() != null) {
244             String studentAns = ans.getAnswer().trim();
245             String correctAns = question.getCorrectAnswer().trim();
246
247             if (question.getQuestionType() == ExamQuestion.QuestionType.MCQ) {
248                 if (studentAns.equalsIgnoreCase(correctAns)) {
249                     finalScore += question.getPoints();
250                 }
251             } else if (question.getQuestionType() == ExamQuestion.QuestionType.SHORT_ANSWER) {
252                 // Simple case-insensitive matching
253                 if (studentAns.equalsIgnoreCase(correctAns)) {
254                     finalScore += question.getPoints();
255                 }
256             }
257         }
258     }
259
260     // Save submission
261     try {
262         String answersJsonStr = objectMapper.writeValueAsString(answers);
263         ExamSubmission submission = ExamSubmission.builder()
264             .exam(exam)
265             .student(student)
266             .submittedAt(LocalDateTime.now())
267             .score(finalScore)
268             .answersJson(answersJsonStr)
269             .build();
270
271         examSubmissionRepository.save(submission);
272
273         // Also automatically add to GradeBook for performance dashboard integration
274         GradeBook gradeBook = GradeBook.builder()
275             .student(student)
276             .courseName(exam.getTitle())
277             .assessmentName("Quiz Exam")
278             .marksObtained(finalScore)
279             .maxMarks(questions.stream().mapToInt(ExamQuestion::getPoints).sum())
280             .build();
281
282         // Delete old one if exists (unlikely given check)
283         // Save to gradebook
284         gradeBookRepository.save(gradeBook);
285
286         return ResponseEntity.ok(Map.of("message", "Exam submitted successfully", "score", finalScore));
287     } catch (Exception e) {
288         return ResponseEntity.status(500).body(Map.of("error", "Error processing submission: " + e.getMessage()));
289     }
290 }
291
292 @GetMapping("/{id}/submissions")
293 public ResponseEntity<?> getExamSubmissions(@PathVariable Long id, Principal principal) {

```

```
294     User user = userRepository.findByUsername(principal.getName()).orElse(null);
295     if (user == null || user.getRole() == User.Role.STUDENT) {
296         return ResponseEntity.status(403).body(Map.of("error", "Only teachers or CRs can view submissions"));
297     }
298
299     Exam exam = examRepository.findById(id).orElse(null);
300     if (exam == null) {
301         return ResponseEntity.notFound().build();
302     }
303
304     List<ExamSubmission> submissions = examSubmissionRepository.findByExam(exam);
305     List<Map<String, Object>> result = new ArrayList<>();
306
307     for (ExamSubmission sub : submissions) {
308         result.add(Map.of(
309             "id", sub.getId(),
310             "studentName", sub.getStudent().getFullName(),
311             "studentUsername", sub.getStudent().getUsername(),
312             "submittedAt", sub.getSubmittedAt(),
313             "score", sub.getScore()
314         ));
315     }
316
317     return ResponseEntity.ok(result);
318 }
319 }
320 }
```

A.12: com/example/demo/controller/MasterAdminController.java

```

1 package com.example.demo.controller;
2
3 import com.example.demo.entity.*;
4 import com.example.demo.entity.User.Role;
5 import com.example.demo.repository.*;
6 import lombok.AllArgsConstructor;
7 import lombok.Data;
8 import lombok.NoArgsConstructor;
9 import org.springframework.http.ResponseEntity;
10 import org.springframework.security.crypto.password.PasswordEncoder;
11 import org.springframework.transaction.annotation.Transactional;
12 import org.springframework.web.bind.annotation.*;
13
14 import java.util.List;
15 import java.util.Map;
16 import java.util.Optional;
17
18 @RestController
19 @RequestMapping("/api/master")
20 @AllArgsConstructor
21 public class MasterAdminController {
22
23     private final UniversityRepository universityRepository;
24     private final UserRepository userRepository;
25     private final PasswordEncoder passwordEncoder;
26     private final ResourceReactionRepository resourceReactionRepository;
27     private final ResourceRepository resourceRepository;
28     private final ExamSubmissionRepository examSubmissionRepository;
29     private final ExamQuestionRepository examQuestionRepository;
30     private final ExamRepository examRepository;
31     private final GradeBookRepository gradeBookRepository;
32     private final ProfileChangeRequestRepository profileChangeRequestRepository;
33     private final PromotionHistoryRepository promotionHistoryRepository;
34     private final ClassCourseAssignmentRepository classCourseAssignmentRepository;
35     private final ClassTestRepository classTestRepository;
36     private final ScheduleItemRepository scheduleItemRepository;
37     private final AnnouncementRepository announcementRepository;
38     private final StudentClassRepository studentClassRepository;
39     private final CourseRepository courseRepository;
40     private final SystemMetadataRepository systemMetadataRepository;
41     private final SystemAdminRepository systemAdminRepository;
42     private final BugReportRepository bugReportRepository;
43     private final org.springframework.mail.javamail.JavaMailSender mailSender;
44     private final org.springframework.core.env.Environment env;
45
46     @Data
47     @NoArgsConstructor
48     @AllArgsConstructor
49     public static class UniversityRegistrationRequest {
50         private String name;
51         private String domain;
52         private String logoUrl;
53         private String adminUsername;
54         private String adminPassword;
55         private String adminFullName;
56         private String adminEmail;
57     }
58
59     @Data
60     @NoArgsConstructor
61     @AllArgsConstructor
62     public static class UniversityUpdateRequest {
63         private String name;
64         private String domain;
65         private String logoUrl;
66     }
67
68     @PostMapping("/universities")
69     public ResponseEntity<?> registerUniversity(@RequestBody UniversityRegistrationRequest request) {
70         if (universityRepository.existsByName(request.getName())) {
71             return ResponseEntity.badRequest().body(Map.of("error", "University name already registered"));
72         }

```

```

73     if (universityRepository.existsByDomain(request.getDomain())) {
74         return ResponseEntity.badRequest().body(Map.of("error", "Subdomain already taken"));
75     }
76     if (userRepository.existsByUsername(request.getAdminUsername())) {
77         return ResponseEntity.badRequest().body(Map.of("error", "Admin username already exists"));
78     }
79
80     // 1. Create and save University
81     University uni = University.builder()
82         .name(request.getName())
83         .domain(request.getDomain())
84         .logoUrl(request.getLogoUrl())
85         .build();
86     uni = universityRepository.save(uni);
87
88     // 2. Create University Admin user
89     User admin = User.builder()
90         .username(request.getAdminUsername())
91         .password(passwordEncoder.encode(request.getAdminPassword()))
92         .fullName(request.getAdminFullName())
93         .email(request.getAdminEmail())
94         .role(Role.ADMIN) // University admin
95         .approved(true)
96         .university(uni)
97         .build();
98     userRepository.save(admin);
99
100    return ResponseEntity.ok(uni);
101 }
102
103 @GetMapping("/universities")
104 public ResponseEntity<List<University>> listUniversities() {
105     return ResponseEntity.ok(universityRepository.findAll());
106 }
107
108 @PutMapping("/universities/{id}")
109 public ResponseEntity<?> updateUniversity(@PathVariable Long id, @RequestBody UniversityUpdateRequest request) {
110     Optional<University> uniOpt = universityRepository.findById(id);
111     if (uniOpt.isEmpty()) {
112         return ResponseEntity.notFound().build();
113     }
114     University uni = uniOpt.get();
115
116     // Check conflicts
117     Optional<University> existingName = universityRepository.findByName(request.getName());
118     if (existingName.isPresent() && !existingName.get().getId().equals(id)) {
119         return ResponseEntity.badRequest().body(Map.of("error", "University name already taken"));
120     }
121     Optional<University> existingDomain = universityRepository.findByDomain(request.getDomain());
122     if (existingDomain.isPresent() && !existingDomain.get().getId().equals(id)) {
123         return ResponseEntity.badRequest().body(Map.of("error", "Subdomain already taken"));
124     }
125
126     uni.setName(request.getName());
127     uni.setDomain(request.getDomain());
128     if (request.getLogoUrl() != null) {
129         uni.setLogoUrl(request.getLogoUrl());
130     }
131     universityRepository.save(uni);
132     return ResponseEntity.ok(uni);
133 }
134
135 @PostMapping("/universities/{id}/reset-admin")
136 public ResponseEntity<?> resetAdminPassword(@PathVariable Long id, @RequestBody Map<String, String> body) {
137     Optional<University> uniOpt = universityRepository.findById(id);
138     if (uniOpt.isEmpty()) {
139         return ResponseEntity.notFound().build();
140     }
141
142     String username = body.get("adminUsername");
143     String password = body.get("adminPassword");
144
145     if (username == null || username.trim().isEmpty() || password == null || password.trim().isEmpty()) {
146         return ResponseEntity.badRequest().body(Map.of("error", "Admin username and password are required"));

```

```

147     }
148
149     // Find or create admin user for this university
150     Optional<User> adminOpt = userRepository.findByUsername(username);
151     User admin;
152     if (adminOpt.isPresent()) {
153         admin = adminOpt.get();
154         if (!admin.getRole().equals(Role.ADMIN) || admin.getUniversity() == null ||
!admin.getUniversity().getId().equals(id)) {
155             return ResponseEntity.badRequest().body(Map.of("error", "Username belongs to another user role or
university"));
156         }
157         admin.setPassword(passwordEncoder.encode(password));
158     } else {
159         admin = User.builder()
160             .username(username)
161             .password(passwordEncoder.encode(password))
162             .fullName("University Admin")
163             .email("admin@" + uniOpt.get().getDomain())
164             .role(Role.ADMIN)
165             .approved(true)
166             .university(uniOpt.get())
167             .build();
168     }
169     userRepository.save(admin);
170     return ResponseEntity.ok(Map.of("message", "University admin account successfully updated"));
171 }
172
173 @DeleteMapping("/universities/{id}")
174 @Transactional
175 public ResponseEntity<?> deleteUniversity(@PathVariable Long id) {
176     Optional<University> uniOpt = universityRepository.findById(id);
177     if (uniOpt.isEmpty()) {
178         return ResponseEntity.notFound().build();
179     }
180     University uni = uniOpt.get();
181
182     // 1. Delete Resource Reactions
183     List<ResourceReaction> reactions = resourceReactionRepository.findAll().stream()
184         .filter(rr -> {
185             Optional<User> u = userRepository.findByUsername(rr.getUsername());
186             if (u.isPresent() && u.get().getUniversity() != null && u.get().getUniversity().getId().equals(id)) {
187                 return true;
188             }
189             Optional<Resource> r = resourceRepository.findById(rr.getResourceId());
190             return r.isPresent() && r.get().getUniversity() != null && r.get().getUniversity().getId().equals(id);
191         })
192         .toList();
193     resourceReactionRepository.deleteAll(reactions);
194
195     // 2. Delete Resources
196     List<Resource> resources = resourceRepository.findAll().stream()
197         .filter(r -> r.getUniversity() != null && r.getUniversity().getId().equals(id))
198         .toList();
199     resourceRepository.deleteAll(resources);
200
201     // 3. Delete Exam Submissions
202     List<ExamSubmission> submissions = examSubmissionRepository.findAll().stream()
203         .filter(es -> (es.getStudent() != null && es.getStudent().getUniversity() != null &&
es.getStudent().getUniversity().getId().equals(id)) ||
204             (es.getExam() != null && es.getExam().getUniversity() != null &&
es.getExam().getUniversity().getId().equals(id)))
205         .toList();
206     examSubmissionRepository.deleteAll(submissions);
207
208     // 4. Delete Exam Questions
209     List<ExamQuestion> questions = examQuestionRepository.findAll().stream()
210         .filter(eq -> eq.getExam() != null && eq.getExam().getUniversity() != null &&
eq.getExam().getUniversity().getId().equals(id))
211         .toList();
212     examQuestionRepository.deleteAll(questions);
213
214     // 5. Delete Exams
215     List<Exam> exams = examRepository.findAll().stream()
216         .filter(e -> e.getUniversity() != null && e.getUniversity().getId().equals(id))

```

```

217         .toList();
218     examRepository.deleteAll(exams);
219
220     // 6. Delete GradeBook entries
221     List<GradeBook> grades = gradeBookRepository.findAll().stream()
222         .filter(g -> g.getStudent() != null && g.getStudent().getUniversity() != null &&
g.getStudent().getUniversity().getId().equals(id))
223         .toList();
224     gradeBookRepository.deleteAll(grades);
225
226     // 7. Delete ProfileChangeRequests
227     List<ProfileChangeRequest> profileRequests = profileChangeRequestRepository.findAll().stream()
228         .filter(p -> p.getUser() != null && p.getUser().getUniversity() != null &&
p.getUser().getUniversity().getId().equals(id))
229         .toList();
230     profileChangeRequestRepository.deleteAll(profileRequests);
231
232     // 8. Delete PromotionHistory
233     List<PromotionHistory> promotions = promotionHistoryRepository.findAll().stream()
234         .filter(p -> p.getStudentClass() != null && p.getStudentClass().getUniversity() != null &&
p.getStudentClass().getUniversity().getId().equals(id))
235         .toList();
236     promotionHistoryRepository.deleteAll(promotions);
237
238     // 9. Delete ClassCourseAssignments
239     List<ClassCourseAssignment> assignments = classCourseAssignmentRepository.findAll().stream()
240         .filter(cca -> (cca.getCourse() != null && cca.getCourse().getUniversity() != null &&
cca.getCourse().getUniversity().getId().equals(id)) ||
241             (cca.getStudentClass() != null && cca.getStudentClass().getUniversity() != null &&
cca.getStudentClass().getUniversity().getId().equals(id)) ||
242             (cca.getTeacher() != null && cca.getTeacher().getUniversity() != null &&
cca.getTeacher().getUniversity().getId().equals(id)))
243         .toList();
244     classCourseAssignmentRepository.deleteAll(assignments);
245
246     // 10. Delete ClassTests
247     List<ClassTest> tests = classTestRepository.findAll().stream()
248         .filter(t -> t.getUniversity() != null && t.getUniversity().getId().equals(id))
249         .toList();
250     classTestRepository.deleteAll(tests);
251
252     // 11. Delete ScheduleItems
253     List<ScheduleItem> scheduleItems = scheduleItemRepository.findAll().stream()
254         .filter(s -> s.getUniversity() != null && s.getUniversity().getId().equals(id))
255         .toList();
256     scheduleItemRepository.deleteAll(scheduleItems);
257
258     // 12. Delete Announcements
259     List<Announcement> announcements = announcementRepository.findAll().stream()
260         .filter(a -> a.getUniversity() != null && a.getUniversity().getId().equals(id))
261         .toList();
262     announcementRepository.deleteAll(announcements);
263
264     // 13. Delete Users
265     List<User> users = userRepository.findAll().stream()
266         .filter(u -> u.getUniversity() != null && u.getUniversity().getId().equals(id))
267         .toList();
268     userRepository.deleteAll(users);
269
270     // 14. Delete StudentClasses
271     List<StudentClass> classes = studentClassRepository.findAll().stream()
272         .filter(sc -> sc.getUniversity() != null && sc.getUniversity().getId().equals(id))
273         .toList();
274     studentClassRepository.deleteAll(classes);
275
276     // 15. Delete Courses
277     List<Course> courses = courseRepository.findAll().stream()
278         .filter(c -> c.getUniversity() != null && c.getUniversity().getId().equals(id))
279         .toList();
280     courseRepository.deleteAll(courses);
281
282     // 16. Delete SystemMetadata
283     List<SystemMetadata> metadata = systemMetadataRepository.findAll().stream()
284         .filter(sm -> sm.getUniversity() != null && sm.getUniversity().getId().equals(id))
285         .toList();

```

```

286         systemMetadataRepository.deleteAll(metadata);
287
288         // 17. Finally, delete the University
289         universityRepository.delete(uni);
290
291         return ResponseEntity.ok(Map.of("message", "University and all associated data deleted successfully"));
292     }
293
294     @Data
295     @NoArgsConstructor
296     @AllArgsConstructor
297     public static class MasterProfileUpdateRequest {
298         private String fullName;
299         private String email;
300         private String password;
301     }
302
303     @PostMapping("/profile/update")
304     @Transactional
305     public ResponseEntity<?> updateMasterProfile(@RequestBody MasterProfileUpdateRequest request, java.security.Principal
principal) {
306         if (principal == null) {
307             return ResponseEntity.status(401).body(Map.of("error", "Not authenticated"));
308         }
309         Optional<SystemAdmin> sysAdminOpt = systemAdminRepository.findByUsername(principal.getName());
310         if (sysAdminOpt.isEmpty()) {
311             return ResponseEntity.status(404).body(Map.of("error", "Master account not found"));
312         }
313         SystemAdmin sysAdmin = sysAdminOpt.get();
314
315         if (request.getFullName() != null && !request.getFullName().trim().isEmpty()) {
316             sysAdmin.setFullName(request.getFullName().trim());
317         }
318         if (request.getEmail() != null && !request.getEmail().trim().isEmpty()) {
319             sysAdmin.setEmail(request.getEmail().trim());
320         }
321         if (request.getPassword() != null && !request.getPassword().trim().isEmpty()) {
322             sysAdmin.setPassword(passwordEncoder.encode(request.getPassword().trim()));
323         }
324
325         systemAdminRepository.save(sysAdmin);
326         return ResponseEntity.ok(Map.of(
327             "message", "Master profile updated successfully!",
328             "fullName", sysAdmin.getFullName(),
329             "email", sysAdmin.getEmail()
330         ));
331     }
332
333     @GetMapping("/bugs")
334     public ResponseEntity<?> listBugs() {
335         return ResponseEntity.ok(bugReportRepository.findAllByOrderByCreatedAtDesc());
336     }
337
338     @PostMapping("/bugs/{id}/status")
339     public ResponseEntity<?> updateBugStatus(@PathVariable Long id, @RequestBody Map<String, String> body) {
340         Optional<BugReport> bugOpt = bugReportRepository.findById(id);
341         if (bugOpt.isEmpty()) {
342             return ResponseEntity.notFound().build();
343         }
344         String status = body.get("status");
345         if (status == null || status.trim().isEmpty()) {
346             return ResponseEntity.badRequest().body(Map.of("error", "Status is required"));
347         }
348         BugReport bug = bugOpt.get();
349         bug.setStatus(status.trim().toUpperCase());
350         bugReportRepository.save(bug);
351         return ResponseEntity.ok(bug);
352     }
353
354     @GetMapping("/users/emails")
355     public ResponseEntity<?> listUserEmails() {
356         List<Map<String, Object>> result = userRepository.findAll().stream()
357             .map(u -> {
358                 java.util.Map<String, Object> m = new java.util.HashMap<>();

```

```

359         m.put("id", u.getId());
360         m.put("fullName", u.getFullName() != null ? u.getFullName() : "");
361         m.put("email", u.getEmail() != null ? u.getEmail() : "");
362         m.put("username", u.getUsername());
363         m.put("role", u.getRole() != null ? u.getRole().toString() : "USER");
364         m.put("universityName", u.getUniversity() != null ? u.getUniversity().getName() : "LearnX");
365         return m;
366     })
367     .filter(m -> m.get("email") != null && !((String) m.get("email")).trim().isEmpty())
368     .toList();
369     return ResponseEntity.ok(result);
370 }
371
372 @PostMapping("/send-email")
373 public ResponseEntity<?> sendBroadcastEmail(@RequestBody Map<String, Object> body) {
374     String subject = (String) body.get("subject");
375     String content = (String) body.get("content");
376     List<String> recipientEmails = (List<String>) body.get("recipientEmails");
377
378     if (subject == null || subject.trim().isEmpty()) {
379         return ResponseEntity.badRequest().body(Map.of("error", "Subject is required"));
380     }
381     if (content == null || content.trim().isEmpty()) {
382         return ResponseEntity.badRequest().body(Map.of("error", "Content is required"));
383     }
384
385     List<String> finalRecipients;
386     if (recipientEmails == null || recipientEmails.isEmpty()) {
387         finalRecipients = userRepository.findAll().stream()
388             .map(User::getEmail)
389             .filter(e -> e != null && !e.trim().isEmpty())
390             .distinct()
391             .toList();
392     } else {
393         finalRecipients = recipientEmails.stream()
394             .filter(e -> e != null && !e.trim().isEmpty())
395             .distinct()
396             .toList();
397     }
398
399     if (finalRecipients.isEmpty()) {
400         return ResponseEntity.badRequest().body(Map.of("error", "No recipients found to send email to"));
401     }
402
403     int successCount = 0;
404     int failCount = 0;
405
406     for (String email : finalRecipients) {
407         try {
408             org.springframework.mail.SimpleMailMessage message = new org.springframework.mail.SimpleMailMessage();
409             String fromEmail = env.getProperty("spring.mail.username");
410             if (fromEmail != null && !fromEmail.isEmpty()) {
411                 message.setFrom(fromEmail);
412             }
413             message.setTo(email);
414             message.setSubject(subject);
415             message.setText(content + "\n\n---\nSent via LearnX Master Broadcast System.");
416             mailSender.send(message);
417             successCount++;
418         } catch (Exception ex) {
419             System.err.println("Failed to send broadcast email to " + email + ": " + ex.getMessage());
420             failCount++;
421         }
422     }
423
424     return ResponseEntity.ok(Map.of(
425         "message", "Broadcast complete",
426         "totalSent", finalRecipients.size(),
427         "successCount", successCount,
428         "failCount", failCount
429     ));
430 }
431 }
432

```

A.13: com/example/demo/controller/MetadataController.java

```

1 package com.example.demo.controller;
2
3 import com.example.demo.entity.SystemMetadata;
4 import com.example.demo.repository.SystemMetadataRepository;
5 import lombok.AllArgsConstructor;
6 import org.springframework.http.ResponseEntity;
7 import org.springframework.web.bind.annotation.*;
8
9 import java.util.List;
10 import java.util.Map;
11
12 @RestController
13 @RequestMapping("/api/metadata")
14 @AllArgsConstructor
15 public class MetadataController {
16
17     private final SystemMetadataRepository systemMetadataRepository;
18
19     @GetMapping
20     public ResponseEntity<?> getAllMetadata() {
21         List<SystemMetadata> allOptions = systemMetadataRepository.findAll();
22         return ResponseEntity.ok(allOptions);
23     }
24
25     @PostMapping
26     public ResponseEntity<?> createMetadata(@RequestBody Map<String, String> request) {
27         String type = request.get("type");
28         String value = request.get("value");
29
30         if (type == null || type.trim().isEmpty() || value == null || value.trim().isEmpty()) {
31             return ResponseEntity.badRequest().body(Map.of("error", "Type and value are required."));
32         }
33
34         // Standardize types
35         String typeUpper = type.trim().toUpperCase();
36         if (!List.of("SEMESTER", "DEPARTMENT", "BATCH", "DESIGNATION").contains(typeUpper)) {
37             return ResponseEntity.badRequest().body(Map.of("error", "Invalid metadata type. Must be SEMESTER, DEPARTMENT,
38 BATCH, or DESIGNATION."));
39         }
40
41         SystemMetadata option = SystemMetadata.builder()
42             .type(typeUpper)
43             .value(value.trim())
44             .build();
45
46         systemMetadataRepository.save(option);
47         return ResponseEntity.ok(option);
48     }
49
50     @DeleteMapping("/{id}")
51     public ResponseEntity<?> deleteMetadata(@PathVariable Long id) {
52         if (!systemMetadataRepository.existsById(id)) {
53             return ResponseEntity.notFound().build();
54         }
55         systemMetadataRepository.deleteById(id);
56         return ResponseEntity.ok(Map.of("message", "Metadata option deleted successfully."));
57     }
58 }

```

A.14: com/example/demo/controller/PasswordRecoveryController.java

```

1 package com.example.demo.controller;
2
3 import com.example.demo.entity.User;
4 import com.example.demo.repository.UserRepository;
5 import lombok.AllArgsConstructor;
6 import lombok.Data;
7 import lombok.NoArgsConstructor;
8 import org.springframework.http.ResponseEntity;
9 import org.springframework.mail.javamail.JavaMailSender;
10 import org.springframework.mail.SimpleMailMessage;
11 import org.springframework.security.crypto.password.PasswordEncoder;
12 import org.springframework.web.bind.annotation.*;
13
14 import java.util.Map;
15 import java.util.Random;
16 import java.util.concurrent.ConcurrentHashMap;
17
18 @RestController
19 @RequestMapping("/api/auth/recover")
20 @AllArgsConstructor
21 public class PasswordRecoveryController {
22
23     private final UserRepository userRepository;
24     private final PasswordEncoder passwordEncoder;
25     private final JavaMailSender mailSender;
26     private final org.springframework.core.env.Environment env;
27
28     // In-memory store for verification codes (Email -> Code)
29     private static final ConcurrentHashMap<String, String> recoveryCodes = new ConcurrentHashMap<>();
30
31     @Data
32     @NoArgsConstructor
33     @AllArgsConstructor
34     public static class RecoveryRequest {
35         private String email;
36     }
37
38     @Data
39     @NoArgsConstructor
40     @AllArgsConstructor
41     public static class ResetRequest {
42         private String email;
43         private String code;
44         private String password;
45     }
46
47     @PostMapping("/request")
48     public ResponseEntity<?> requestRecovery(@RequestBody RecoveryRequest request) {
49         String email = request.getEmail();
50         User user = userRepository.findByEmail(email).orElse(null);
51         if (user == null) {
52             // Try by username as well
53             user = userRepository.findByUsername(email).orElse(null);
54         }
55
56         // Generate 6-digit code
57         String code = String.format("%06d", new Random().nextInt(999999));
58         recoveryCodes.put(email, code);
59
60         System.out.println("=====");
61         System.out.println("[PASSWORD RECOVERY] Verification code for " + email + " is: " + code);
62         System.out.println("=====");
63
64         if (user != null) {
65             email = user.getEmail();
66             recoveryCodes.put(email, code); // store under actual email if username was provided
67
68             // Send Email
69             try {
70                 SimpleMailMessage mailMessage = new SimpleMailMessage();
71                 String fromEmail = env.getProperty("spring.mail.username");
72                 if (fromEmail != null && !fromEmail.isEmpty()) {

```

```

73         mailMessage.setFrom(fromEmail);
74     }
75     mailMessage.setTo(email);
76     mailMessage.setSubject("LearnX Password Recovery Verification Code");
77     mailMessage.setText("Hello,\n\n"
78         + "You have requested to reset your password on LearnX.\n"
79         + "Your password recovery verification code is: " + code + "\n\n"
80         + "This code will expire shortly. If you did not make this request, please ignore this email.\n\n"
81         + "Best regards,\n"
82         + "LearnX Team");
83     mailSender.send(mailMessage);
84 } catch (Exception e) {
85     System.err.println("Failed to send verification email to " + email + ": " + e.getMessage());
86     System.out.println("[FALLBACK LOG] Code is: " + code);
87 }
88 }
89
90 return ResponseEntity.ok(Map.of(
91     "message", "Verification code sent to email successfully.",
92     "email", email
93 ));
94 }
95
96 @PostMapping("/reset")
97 public ResponseEntity<?> resetPassword(@RequestBody ResetRequest request) {
98     String email = request.getEmail();
99     String savedCode = recoveryCodes.get(email);
100
101     if (savedCode == null || !savedCode.equals(request.getCode())) {
102         return ResponseEntity.badRequest().body(Map.of("error", "Invalid or expired verification code."));
103     }
104
105     User user = userRepository.findByEmail(email).orElse(null);
106     if (user == null) {
107         user = userRepository.findByUsername(email).orElse(null);
108     }
109
110     if (user != null) {
111         user.setPassword(passwordEncoder.encode(request.getPassword()));
112         userRepository.save(user);
113     }
114
115     recoveryCodes.remove(email);
116
117     return ResponseEntity.ok(Map.of("message", "Password has been reset successfully. You can now login."));
118 }
119 }
120

```

A.15: com/example/demo/controller/ProfileController.java

```

1 package com.example.demo.controller;
2
3 import com.example.demo.entity.ProfileChangeRequest;
4 import com.example.demo.entity.User;
5 import com.example.demo.repository.ProfileChangeRequestRepository;
6 import com.example.demo.repository.UserRepository;
7 import lombok.AllArgsConstructor;
8 import lombok.Data;
9 import lombok.NoArgsConstructor;
10 import org.springframework.http.HttpStatus;
11 import org.springframework.http.ResponseEntity;
12 import org.springframework.web.bind.annotation.*;
13
14 import java.security.Principal;
15 import java.util.Map;
16
17 @RestController
18 @RequestMapping("/api/profile")
19 @AllArgsConstructor
20 public class ProfileController {
21
22     private final UserRepository userRepository;
23     private final ProfileChangeRequestRepository profileChangeRequestRepository;
24
25     @Data
26     @NoArgsConstructor
27     @AllArgsConstructor
28     public static class ProfileUpdateRequest {
29         private String fullName;
30         private String email;
31         private String idNo;
32         private String department;
33         private String batch;
34         private String semester;
35         private String section;
36         private String designation;
37         private String profilePicUrl;
38     }
39
40     @PostMapping("/update")
41     public ResponseEntity<?> updateProfile(@RequestBody ProfileUpdateRequest request, Principal principal) {
42         if (principal == null) {
43             return ResponseEntity.status(HttpStatus.UNAUTHORIZED).body(Map.of("error", "Not authenticated"));
44         }
45         User currentUser = userRepository.findByUsername(principal.getName()).orElse(null);
46         if (currentUser == null) {
47             return ResponseEntity.status(HttpStatus.UNAUTHORIZED).body(Map.of("error", "User not found"));
48         }
49
50         boolean nameChanged = false;
51         if (request.getFullName() != null && !request.getFullName().trim().isEmpty() &&
52 !request.getFullName().equals(currentUser.getFullName())) {
53             currentUser.setFullName(request.getFullName().trim());
54             nameChanged = true;
55         }
56
57         // Instantly update the non-sensitive profile photo if provided
58         if (request.getProfilePicUrl() != null) {
59             currentUser.setProfilePicUrl(request.getProfilePicUrl());
60         }
61
62         if (nameChanged || request.getProfilePicUrl() != null) {
63             userRepository.save(currentUser);
64         }
65
66         boolean changed = false;
67         if (request.getEmail() != null && !request.getEmail().equals(currentUser.getEmail())) changed = true;
68         if (request.getIdNo() != null && !request.getIdNo().equals(currentUser.getIdNo())) changed = true;
69         if (request.getDepartment() != null && !request.getDepartment().equals(currentUser.getDepartment())) changed =
true;
70         if (request.getBatch() != null && !request.getBatch().equals(currentUser.getBatch())) changed = true;

```

```
71         if (request.getSemester() != null && !request.getSemester().equals(currentUser.getSemester())) changed = true;
72         if (request.getSection() != null && !request.getSection().equals(currentUser.getSection())) changed = true;
73         if (request.getDesignation() != null && !request.getDesignation().equals(currentUser.getDesignation())) changed =
true;
74
75         if (!changed) {
76             return ResponseEntity.ok(Map.of(
77                 "message", "Profile updated successfully.",
78                 "user", currentUser
79             ));
80         }
81
82         ProfileChangeRequest changeRequest = ProfileChangeRequest.builder()
83             .user(currentUser)
84             .newFullName(currentUser.getFullName())
85             .newEmail(request.getEmail())
86             .newIdNo(request.getIdNo())
87             .newDepartment(request.getDepartment())
88             .newBatch(request.getBatch())
89             .newSemester(request.getSemester())
90             .newSection(request.getSection())
91             .newDesignation(request.getDesignation())
92             .build();
93
94         profileChangeRequestRepository.save(changeRequest);
95
96         return ResponseEntity.ok(Map.of(
97             "message", "Sensitive changes submitted. Administrator approval is required before they take effect.",
98             "user", currentUser
99         ));
100     }
101 }
102
```

A.16: com/example/demo/controller/ResourceController.java

```

1 package com.example.demo.controller;
2
3 import com.example.demo.entity.Resource;
4 import com.example.demo.entity.ResourceReaction;
5 import com.example.demo.entity.User;
6 import com.example.demo.repository.ResourceReactionRepository;
7 import com.example.demo.repository.ResourceRepository;
8 import com.example.demo.repository.UserRepository;
9 import lombok.AllArgsConstructor;
10 import org.springframework.http.HttpHeaders;
11 import org.springframework.http.HttpStatus;
12 import org.springframework.http.MediaType;
13 import org.springframework.http.ResponseEntity;
14 import org.springframework.web.bind.annotation.*;
15 import org.springframework.web.multipart.MultipartFile;
16
17 import java.io.IOException;
18 import java.security.Principal;
19 import java.util.List;
20 import java.util.Map;
21 import java.util.Optional;
22
23 @RestController
24 @RequestMapping("/api/resources")
25 @AllArgsConstructor
26 public class ResourceController {
27
28     private final ResourceRepository resourceRepository;
29     private final UserRepository userRepository;
30     private final ResourceReactionRepository resourceReactionRepository;
31
32     @GetMapping("/approved")
33     public ResponseEntity<List<Resource>> getApprovedResources(Principal principal) {
34         List<Resource> list;
35         User user = null;
36         if (principal != null) {
37             user = userRepository.findByUsername(principal.getName()).orElse(null);
38         }
39
40         if (user != null && (user.getRole() == User.Role.STUDENT || user.getRole() == User.Role.CR)) {
41             if (user.getStudentClass() != null) {
42                 list = resourceRepository.findByStudentClassAndApproved(user.getStudentClass(), true);
43             } else {
44                 list = resourceRepository.findByApproved(true);
45             }
46         } else {
47             list = resourceRepository.findByApproved(true);
48         }
49
50         // Populate reaction statistics
51         populateReactions(list, user != null ? user.getUsername() : null);
52
53         // Sort by net reactions (likes - dislikes) descending, then by id descending
54         list.sort((a, b) -> {
55             int scoreA = a.getLikesCount() - a.getDislikesCount();
56             int scoreB = b.getLikesCount() - b.getDislikesCount();
57             if (scoreA != scoreB) {
58                 return Integer.compare(scoreB, scoreA);
59             }
60             return Long.compare(b.getId(), a.getId());
61         });
62
63         return ResponseEntity.ok(list);
64     }
65
66     @GetMapping("/pending")
67     public ResponseEntity<List<Resource>> getPendingResources(Principal principal) {
68         List<Resource> list = resourceRepository.findByApproved(false);
69         String username = principal != null ? principal.getName() : null;
70         populateReactions(list, username);
71         return ResponseEntity.ok(list);
72     }

```

```

73
74 private void populateReactions(List<Resource> resources, String currentUsername) {
75     for (Resource r : resources) {
76         r.setFileData(null); // Clear raw data for listing efficiency
77         List<ResourceReaction> reactions = resourceReactionRepository.findByResourceId(r.getId());
78         int likes = 0;
79         int dislikes = 0;
80         String userReaction = null;
81
82         for (ResourceReaction rx : reactions) {
83             if ("LIKE".equalsIgnoreCase(rx.getReactionType())) {
84                 likes++;
85             } else if ("DISLIKE".equalsIgnoreCase(rx.getReactionType())) {
86                 dislikes++;
87             }
88             if (currentUsername != null && currentUsername.equalsIgnoreCase(rx.getUsername())) {
89                 userReaction = rx.getReactionType();
90             }
91         }
92
93         r.setLikesCount(likes);
94         r.setDislikesCount(dislikes);
95         r.setUserReaction(userReaction);
96     }
97 }
98
99 @PostMapping("/upload")
100 public ResponseEntity<?> uploadResource(
101     @RequestParam(value = "file", required = false) MultipartFile file,
102     @RequestParam("title") String title,
103     @RequestParam("courseName") String courseName,
104     @RequestParam(value = "examTags", required = false) String examTags,
105     @RequestParam(value = "driveLink", required = false) String driveLink,
106     Principal principal) {
107
108     User user = userRepository.findByUsername(principal.getName()).orElse(null);
109     if (user == null) {
110         return ResponseEntity.status(HttpStatus.UNAUTHORIZED).build();
111     }
112
113     try {
114         boolean approved = (user.getRole() == User.Role.TEACHER || user.getRole() == User.Role.CR);
115
116         Resource resource = Resource.builder()
117             .title(title)
118             .courseName(courseName)
119             .approved(approved)
120             .examTags(examTags)
121             .driveLink(driveLink != null && !driveLink.trim().isEmpty() ? driveLink.trim() : null)
122             .studentClass(user.getStudentClass())
123             .uploadedBy(user)
124             .build();
125
126         if (file != null && !file.isEmpty()) {
127             resource.setFileName(file.getOriginalFilename());
128             resource.setContentType(file.getContentType());
129             resource.setFileData(file.getBytes());
130         } else {
131             resource.setFileName("Google Drive Link");
132             resource.setContentType("application/octet-stream");
133         }
134
135         Resource saved = resourceRepository.save(resource);
136         saved.setFileData(null);
137         return ResponseEntity.ok(saved);
138     } catch (IOException e) {
139         return ResponseEntity.status(HttpStatus.INTERNAL_SERVER_ERROR)
140             .body(Map.of("error", "Failed to upload file: " + e.getMessage()));
141     }
142 }
143
144 @PostMapping("/{id}/approve")
145 public ResponseEntity<?> approveResource(@PathVariable Long id) {
146     Resource resource = resourceRepository.findById(id).orElse(null);

```

```

147         if (resource == null) {
148             return ResponseEntity.notFound().build();
149         }
150         resource.setApproved(true);
151         resourceRepository.save(resource);
152         return ResponseEntity.ok(Map.of("message", "Resource approved successfully"));
153     }
154
155     @DeleteMapping("/{id}")
156     public ResponseEntity<?> deleteResource(@PathVariable Long id) {
157         if (!resourceRepository.existsById(id)) {
158             return ResponseEntity.notFound().build();
159         }
160         resourceRepository.deleteById(id);
161         return ResponseEntity.ok(Map.of("message", "Resource deleted successfully"));
162     }
163
164     @GetMapping("/download/{id}")
165     public ResponseEntity<byte[]> downloadResource(@PathVariable Long id) {
166         Resource resource = resourceRepository.findById(id).orElse(null);
167         if (resource == null || resource.getFileData() == null) {
168             return ResponseEntity.notFound().build();
169         }
170
171         return ResponseEntity.ok()
172             .header(HttpHeaders.CONTENT_DISPOSITION, "attachment; filename=\"" + resource.getFileName() + "\"")
173             .contentType(MediaType.parseMediaType(resource.getContentType() != null ? resource.getContentType() :
"application/octet-stream"))
174             .body(resource.getFileData());
175     }
176
177     // --- REACTION ENDPOINTS ---
178     @PostMapping("/{id}/react")
179     public ResponseEntity<?> reactToResource(
180         @PathVariable Long id,
181         @RequestParam("type") String type,
182         Principal principal) {
183
184         if (principal == null) {
185             return ResponseEntity.status(HttpStatus.UNAUTHORIZED).body(Map.of("error", "Not authenticated"));
186         }
187
188         if (!"LIKE".equalsIgnoreCase(type) && !"DISLIKE".equalsIgnoreCase(type)) {
189             return ResponseEntity.badRequest().body(Map.of("error", "Reaction type must be LIKE or DISLIKE"));
190         }
191
192         Optional<ResourceReaction> existing = resourceReactionRepository.findByResourceIdAndUsername(id,
principal.getName());
193
194         if (existing.isPresent()) {
195             ResourceReaction rx = existing.get();
196             rx.setReactionType(type.toUpperCase());
197             resourceReactionRepository.save(rx);
198         } else {
199             ResourceReaction rx = ResourceReaction.builder()
200                 .resourceId(id)
201                 .username(principal.getName())
202                 .reactionType(type.toUpperCase())
203                 .build();
204             resourceReactionRepository.save(rx);
205         }
206
207         return ResponseEntity.ok(Map.of("message", "Reaction saved successfully"));
208     }
209
210     @DeleteMapping("/{id}/react")
211     public ResponseEntity<?> removeReaction(@PathVariable Long id, Principal principal) {
212         if (principal == null) {
213             return ResponseEntity.status(HttpStatus.UNAUTHORIZED).body(Map.of("error", "Not authenticated"));
214         }
215
216         Optional<ResourceReaction> existing = resourceReactionRepository.findByResourceIdAndUsername(id,
principal.getName());
217         if (existing.isPresent()) {
218             resourceReactionRepository.delete(existing.get());

```

```
219         return ResponseEntity.ok(Map.of("message", "Reaction removed successfully"));
220     }
221
222     return ResponseEntity.ok(Map.of("message", "No reaction found to remove"));
223 }
224 }
225 }
```

A.17: com/example/demo/controller/ScheduleController.java

```

1 package com.example.demo.controller;
2
3 import com.example.demo.entity.AuditLog;
4 import com.example.demo.entity.ClassTest;
5 import com.example.demo.entity.ScheduleItem;
6 import com.example.demo.repository.AuditLogRepository;
7 import com.example.demo.repository.ClassTestRepository;
8 import com.example.demo.repository.ScheduleItemRepository;
9 import lombok.AllArgsConstructor;
10 import org.springframework.http.ResponseEntity;
11 import org.springframework.web.bind.annotation.*;
12
13 import java.security.Principal;
14 import java.time.LocalDateTime;
15 import java.util.List;
16 import java.util.Map;
17
18 @RestController
19 @RequestMapping("/api/schedule")
20 @AllArgsConstructor
21 public class ScheduleController {
22
23     private final ScheduleItemRepository scheduleItemRepository;
24     private final ClassTestRepository classTestRepository;
25     private final AuditLogRepository auditLogRepository;
26     private final com.example.demo.repository.UserRepository userRepository;
27     private final com.example.demo.repository.StudentClassRepository studentClassRepository;
28
29     // --- Routine endpoints ---
30
31     @GetMapping("/routine")
32     public ResponseEntity<List<ScheduleItem>> getFullRoutine(
33         @RequestParam(required = false) Long classId,
34         Principal principal) {
35         if (principal != null) {
36             com.example.demo.entity.User user = userRepository.findByUsername(principal.getName()).orElse(null);
37             if (user != null) {
38                 if (classId != null) {
39                     com.example.demo.entity.StudentClass sc = studentClassRepository.findById(classId).orElse(null);
40                     if (sc != null) {
41                         return ResponseEntity.ok(scheduleItemRepository.findByStudentClass(sc));
42                     }
43                 }
44
45                 if (user.getRole() == com.example.demo.entity.User.Role.STUDENT || user.getRole() ==
46 com.example.demo.entity.User.Role.CR) {
47                     if (user.getStudentClass() != null) {
48                         return ResponseEntity.ok(scheduleItemRepository.findByStudentClass(user.getStudentClass()));
49                     }
50                     return ResponseEntity.ok(List.of());
51                 } else {
52                     if (user.getUniversity() != null) {
53                         return ResponseEntity.ok(scheduleItemRepository.findByUniversity(user.getUniversity()));
54                     }
55                 }
56             }
57         }
58         return ResponseEntity.ok(scheduleItemRepository.findAll());
59     }
60
61     @PostMapping("/routine")
62     public ResponseEntity<?> addRoutineItem(@RequestBody ScheduleItem item, Principal principal) {
63         if (principal == null) {
64             return ResponseEntity.status(401).body(Map.of("error", "Unauthorized"));
65         }
66         com.example.demo.entity.User user = userRepository.findByUsername(principal.getName()).orElse(null);
67         if (user == null) {
68             return ResponseEntity.status(401).body(Map.of("error", "User not found"));
69         }
70         if (user.getRole() != com.example.demo.entity.User.Role.CR &&
71             user.getRole() != com.example.demo.entity.User.Role.TEACHER &&
72             user.getRole() != com.example.demo.entity.User.Role.ADMIN) {

```

```

72         return ResponseEntity.status(403).body(Map.of("error", "Forbidden: Insufficient privileges"));
73     }
74
75     if (user.getRole() == com.example.demo.entity.User.Role.CR) {
76         if (user.getStudentClass() == null) {
77             return ResponseEntity.status(400).body(Map.of("error", "Bad Request: CR does not belong to a class"));
78         }
79         item.setStudentClass(user.getStudentClass());
80     }
81     if (user.getUniversity() != null) {
82         item.setUniversity(user.getUniversity());
83     }
84
85     ScheduleItem saved = scheduleItemRepository.save(item);
86
87     // Track Audit Log
88     String details = String.format("Added Routine Class: Course '%s', Day '%s', Time %s-%s, Room '%s', Teacher '%s'",
89         item.getCourseName(), item.getDayOfWeek(), item.getStartTime(), item.getEndTime(), item.getRoomNo(),
90         item.getTeacherName());
91     auditLogRepository.save(AuditLog.builder()
92         .entityType("ROUTINE")
93         .entityId(saved.getId())
94         .action("CREATE")
95         .changedBy(principal.getName())
96         .timestamp(LocalDateTime.now())
97         .details(details)
98         .build());
99
100    return ResponseEntity.ok(saved);
101 }
102
103 @PutMapping("/routine/{id}")
104 public ResponseEntity<?> updateRoutineItem(@PathVariable Long id, @RequestBody ScheduleItem updated, Principal
principal) {
105     if (principal == null) {
106         return ResponseEntity.status(401).body(Map.of("error", "Unauthorized"));
107     }
108     com.example.demo.entity.User user = userRepository.findByUsername(principal.getName()).orElse(null);
109     if (user == null) {
110         return ResponseEntity.status(401).body(Map.of("error", "User not found"));
111     }
112     if (user.getRole() != com.example.demo.entity.User.Role.CR &&
113         user.getRole() != com.example.demo.entity.User.Role.TEACHER &&
114         user.getRole() != com.example.demo.entity.User.Role.ADMIN) {
115         return ResponseEntity.status(403).body(Map.of("error", "Forbidden: Insufficient privileges"));
116     }
117
118     ScheduleItem existing = scheduleItemRepository.findById(id).orElse(null);
119     if (existing == null) {
120         return ResponseEntity.notFound().build();
121     }
122
123     if (user.getRole() == com.example.demo.entity.User.Role.CR) {
124         if (user.getStudentClass() == null || existing.getStudentClass() == null ||
125             !existing.getStudentClass().getId().equals(user.getStudentClass().getId())) {
126             return ResponseEntity.status(403).body(Map.of("error", "Forbidden: You cannot modify schedules for another
class"));
127         }
128         updated.setStudentClass(user.getStudentClass());
129     }
130
131     String details = String.format("Updated Routine Class: Day changed from '%s' to '%s', Time from %s-%s to %s-%s,
Course from '%s' to '%s', Room from '%s' to '%s', Teacher '%s' to '%s'",
132         existing.getDayOfWeek(), updated.getDayOfWeek(), existing.getStartTime(), existing.getEndTime(),
133         updated.getStartTime(), updated.getEndTime(),
134         existing.getCourseName(), updated.getCourseName(), existing.getRoomNo(), updated.getRoomNo(),
135         existing.getTeacherName(), updated.getTeacherName());
136
137     existing.setDayOfWeek(updated.getDayOfWeek());
138     existing.setStartTime(updated.getStartTime());
139     existing.setEndTime(updated.getEndTime());
140     existing.setCourseName(updated.getCourseName());
141     existing.setRoomNo(updated.getRoomNo());
142     existing.setTeacherName(updated.getTeacherName());
143     if (updated.getStudentClass() != null) {

```

```

141         existing.setStudentClass(updated.getStudentClass());
142     }
143     if (user.getUniversity() != null) {
144         existing.setUniversity(user.getUniversity());
145     }
146
147     ScheduleItem saved = scheduleItemRepository.save(existing);
148
149     auditLogRepository.save(AuditLog.builder()
150         .entityType("ROUTINE")
151         .entityId(id)
152         .action("UPDATE")
153         .changedBy(principal.getName())
154         .timestamp(LocalDateTime.now())
155         .details(details)
156         .build());
157
158     return ResponseEntity.ok(saved);
159 }
160
161 @DeleteMapping("/routine/{id}")
162 public ResponseEntity<?> deleteRoutineItem(@PathVariable Long id, Principal principal) {
163     if (principal == null) {
164         return ResponseEntity.status(401).body(Map.of("error", "Unauthorized"));
165     }
166     com.example.demo.entity.User user = userRepository.findByUsername(principal.getName()).orElse(null);
167     if (user == null) {
168         return ResponseEntity.status(401).body(Map.of("error", "User not found"));
169     }
170     if (user.getRole() != com.example.demo.entity.User.Role.CR &&
171         user.getRole() != com.example.demo.entity.User.Role.TEACHER &&
172         user.getRole() != com.example.demo.entity.User.Role.ADMIN) {
173         return ResponseEntity.status(403).body(Map.of("error", "Forbidden: Insufficient privileges"));
174     }
175
176     ScheduleItem existing = scheduleItemRepository.findById(id).orElse(null);
177     if (existing == null) {
178         return ResponseEntity.notFound().build();
179     }
180
181     if (user.getRole() == com.example.demo.entity.User.Role.CR) {
182         if (user.getStudentClass() == null || existing.getStudentClass() == null ||
183             !existing.getStudentClass().getId().equals(user.getStudentClass().getId())) {
184             return ResponseEntity.status(403).body(Map.of("error", "Forbidden: You cannot delete schedules for another
class"));
185         }
186     }
187
188     String details = String.format("Deleted Routine Class: Course '%s', Day '%s', Time %s-%s, Room '%s', Teacher '%s'",
189         existing.getCourseName(), existing.getDayOfWeek(), existing.getStartTime(), existing.getEndTime(),
190         existing.getRoomNo(), existing.getTeacherName());
191
192     scheduleItemRepository.deleteById(id);
193
194     auditLogRepository.save(AuditLog.builder()
195         .entityType("ROUTINE")
196         .entityId(id)
197         .action("DELETE")
198         .changedBy(principal.getName())
199         .timestamp(LocalDateTime.now())
200         .details(details)
201         .build());
202
203     return ResponseEntity.ok(Map.of("message", "Routine class deleted"));
204 }
205
206 // --- Class Test (CT) endpoints ---
207
208 @GetMapping("/ct")
209 public ResponseEntity<List<ClassTest>> getUpcomingCTs(
210     @RequestParam(required = false) Long classId,
211     Principal principal) {
212     if (principal != null) {
213         com.example.demo.entity.User user = userRepository.findByUsername(principal.getName()).orElse(null);

```

```

213         if (user != null) {
214             if (classId != null) {
215                 com.example.demo.entity.StudentClass sc = studentClassRepository.findById(classId).orElse(null);
216                 if (sc != null) {
217                     return ResponseEntity.ok(classTestRepository.findByStudentClassOrderByDateTimeAsc(sc));
218                 }
219             }
220
221             if (user.getRole() == com.example.demo.entity.User.Role.STUDENT || user.getRole() ==
com.example.demo.entity.User.Role.CR) {
222                 if (user.getStudentClass() != null) {
223                     return
ResponseEntity.ok(classTestRepository.findByStudentClassOrderByDateTimeAsc(user.getStudentClass()));
224                 }
225                 return ResponseEntity.ok(List.of());
226             }
227         }
228     }
229     return ResponseEntity.ok(classTestRepository.findAllByOrderByDateTimeAsc());
230 }
231
232 @PostMapping("/ct")
233 public ResponseEntity<?> addClassTest(@RequestBody ClassTest ct, Principal principal) {
234     if (principal == null) {
235         return ResponseEntity.status(401).body(Map.of("error", "Unauthorized"));
236     }
237     com.example.demo.entity.User user = userRepository.findByIdByUsername(principal.getName()).orElse(null);
238     if (user == null) {
239         return ResponseEntity.status(401).body(Map.of("error", "User not found"));
240     }
241
242     if (ct.getStudentClass() == null) {
243         if (user.getRole() == com.example.demo.entity.User.Role.CR || user.getRole() ==
com.example.demo.entity.User.Role.STUDENT) {
244             ct.setStudentClass(user.getStudentClass());
245         }
246     }
247     if (ct.getUniversity() == null) {
248         ct.setUniversity(user.getUniversity());
249     }
250
251     ct.setCreatedBy(principal.getName());
252     ClassTest saved = classTestRepository.save(ct);
253
254     String details = String.format("Scheduled Class Test (CT): Course '%s', Date/Time '%s', Duration %d mins, Room
%s', Topic '%s'",
255         ct.getCourseName(), ct.getDateTime(), ct.getDurationMinutes(), ct.getRoomNo(), ct.getTopic());
256
257     auditLogRepository.save(AuditLog.builder()
258         .entityType("CLASS_TEST")
259         .entityId(saved.getId())
260         .action("CREATE")
261         .changedBy(principal.getName())
262         .timestamp(LocalDateTime.now())
263         .details(details)
264         .build());
265
266     return ResponseEntity.ok(saved);
267 }
268
269 @PutMapping("/ct/{id}")
270 public ResponseEntity<?> updateClassTest(@PathVariable Long id, @RequestBody ClassTest updated, Principal principal) {
271     ClassTest existing = classTestRepository.findById(id).orElse(null);
272     if (existing == null) {
273         return ResponseEntity.notFound().build();
274     }
275
276     String details = String.format("Updated Class Test (CT): Course changed from '%s' to '%s', Date/Time from '%s' to
%s', Room from '%s' to '%s', Topic from '%s' to '%s'",
277         existing.getCourseName(), updated.getCourseName(), existing.getDateTime(), updated.getDateTime(),
278         existing.getRoomNo(), updated.getRoomNo(), existing.getTopic(), updated.getTopic());
279
280     existing.setCourseName(updated.getCourseName());
281     existing.setDateTime(updated.getDateTime());
282     existing.setDurationMinutes(updated.getDurationMinutes());

```

```

283     existing.setRoomNo(updated.getRoomNo());
284     existing.setTopic(updated.getTopic());
285
286     ClassTest saved = classTestRepository.save(existing);
287
288     auditLogRepository.save(AuditLog.builder()
289         .entityType("CLASS_TEST")
290         .entityId(id)
291         .action("UPDATE")
292         .changedBy(principal.getName())
293         .timestamp(LocalDateTime.now())
294         .details(details)
295         .build());
296
297     return ResponseEntity.ok(saved);
298 }
299
300 @DeleteMapping("/ct/{id}")
301 public ResponseEntity<?> deleteClassTest(@PathVariable Long id, Principal principal) {
302     ClassTest existing = classTestRepository.findById(id).orElse(null);
303     if (existing == null) {
304         return ResponseEntity.notFound().build();
305     }
306
307     String details = String.format("Cancelled/Deleted Class Test (CT): Course '%s', Date/Time '%s', Room '%s', Topic
308 '%s'",
309         existing.getCourseName(), existing.getDateTime(), existing.getRoomNo(), existing.getTopic());
310
311     classTestRepository.deleteById(id);
312
313     auditLogRepository.save(AuditLog.builder()
314         .entityType("CLASS_TEST")
315         .entityId(id)
316         .action("DELETE")
317         .changedBy(principal.getName())
318         .timestamp(LocalDateTime.now())
319         .details(details)
320         .build());
321
322     return ResponseEntity.ok(Map.of("message", "Class test deleted"));
323 }
324 // --- Audit Logs endpoints ---
325
326 @GetMapping("/audit-logs")
327 public ResponseEntity<List<AuditLog>> getAuditLogs() {
328     return ResponseEntity.ok(auditLogRepository.findAllByOrderByTimestampDesc());
329 }
330 }
331

```

A.18: com/example/demo/controller/SlotController.java

```

1 package com.example.demo.controller;
2
3 import com.example.demo.entity.User;
4 import com.example.demo.entity.User.Role;
5 import com.example.demo.entity.StudentClass;
6 import com.example.demo.entity.ClassCourseAssignment;
7 import com.example.demo.repository.UserRepository;
8 import com.example.demo.repository.StudentClassRepository;
9 import com.example.demo.repository.ClassCourseAssignmentRepository;
10 import com.example.demo.service.SlotDetectionService;
11 import com.example.demo.service.SlotDetectionService.TimeInterval;
12 import lombok.AllArgsConstructor;
13 import org.springframework.format.annotation.DateTimeFormat;
14 import org.springframework.http.ResponseEntity;
15 import org.springframework.web.bind.annotation.GetMapping;
16 import org.springframework.web.bind.annotation.RequestMapping;
17 import org.springframework.web.bind.annotation.RequestParam;
18 import org.springframework.web.bind.annotation.RestController;
19
20 import java.security.Principal;
21 import java.time.LocalDate;
22 import java.util.List;
23 import java.util.Map;
24
25 @RestController
26 @RequestMapping("/api/slots")
27 @AllArgsConstructor
28 public class SlotController {
29
30     private final SlotDetectionService slotDetectionService;
31     private final UserRepository userRepository;
32     private final StudentClassRepository studentClassRepository;
33     private final ClassCourseAssignmentRepository classCourseAssignmentRepository;
34
35     @GetMapping("/detect")
36     public ResponseEntity<?> getFreeSlots(
37         @RequestParam @DateTimeFormat(iso = DateTimeFormat.ISO.DATE) LocalDate date,
38         @RequestParam(defaultValue = "60") int duration,
39         @RequestParam(required = false) Long classId,
40         Principal principal) {
41
42         if (principal == null) {
43             return ResponseEntity.status(401).body(Map.of("error", "Unauthorized"));
44         }
45         User user = userRepository.findByUsername(principal.getName()).orElse(null);
46         if (user == null) {
47             return ResponseEntity.status(401).body(Map.of("error", "User not found"));
48         }
49
50         StudentClass targetClass = null;
51         if (classId != null) {
52             targetClass = studentClassRepository.findById(classId).orElse(null);
53         } else {
54             if (user.getRole() == Role.CR || user.getRole() == Role.STUDENT) {
55                 targetClass = user.getStudentClass();
56             } else if (user.getRole() == Role.TEACHER) {
57                 List<ClassCourseAssignment> assignments = classCourseAssignmentRepository.findByTeacher(user);
58                 if (!assignments.isEmpty()) {
59                     targetClass = assignments.get(0).getStudentClass();
60                 } else {
61                     List<StudentClass> classes = studentClassRepository.findByUniversity(user.getUniversity());
62                     if (!classes.isEmpty()) {
63                         targetClass = classes.get(0);
64                     }
65                 }
66             } else if (user.getRole() == Role.ADMIN) {
67                 List<StudentClass> classes = studentClassRepository.findByUniversity(user.getUniversity());
68                 if (!classes.isEmpty()) {
69                     targetClass = classes.get(0);
70                 }
71             }
72         }
73     }

```

```
73  
74     if (targetClass == null) {  
75         return ResponseEntity.ok(List.of());  
76     }  
77  
78     List<TimeInterval> freeSlots = slotDetectionService.detectFreeSlots(date, duration, targetClass);  
79     return ResponseEntity.ok(freeSlots);  
80 }  
81 }  
82
```

A.19: com/example/demo/controller/UniversityAdminController.java

```

1 package com.example.demo.controller;
2
3 import com.example.demo.entity.*;
4 import com.example.demo.entity.User.Role;
5 import com.example.demo.repository.*;
6 import com.fasterxml.jackson.core.type.TypeReference;
7 import com.fasterxml.jackson.databind.ObjectMapper;
8 import lombok.AllArgsConstructor;
9 import lombok.Data;
10 import lombok.NoArgsConstructor;
11 import org.springframework.http.ResponseEntity;
12 import org.springframework.security.crypto.password.PasswordEncoder;
13 import org.springframework.transaction.annotation.Transactional;
14 import org.springframework.web.bind.annotation.*;
15
16 import java.security.Principal;
17 import java.time.LocalDateTime;
18 import java.util.*;
19 import java.util.stream.Collectors;
20
21 @RestController
22 @RequestMapping("/api/admin")
23 @AllArgsConstructor
24 public class UniversityAdminController {
25
26     private final UniversityRepository universityRepository;
27     private final UserRepository userRepository;
28     private final StudentClassRepository studentClassRepository;
29     private final CourseRepository courseRepository;
30     private final ClassCourseAssignmentRepository classCourseAssignmentRepository;
31     private final PromotionHistoryRepository promotionHistoryRepository;
32     private final SystemMetadataRepository systemMetadataRepository;
33     private final PasswordEncoder passwordEncoder;
34     private final ObjectMapper objectMapper;
35     private final ProfileChangeRequestRepository profileChangeRequestRepository;
36     private final ExamRepository examRepository;
37     private final ExamSubmissionRepository examSubmissionRepository;
38     private final ExamQuestionRepository examQuestionRepository;
39     private final ResourceRepository resourceRepository;
40     private final ResourceReactionRepository resourceReactionRepository;
41
42     @Data
43     @NoArgsConstructor
44     @AllArgsConstructor
45     public static class MetadataRequest {
46         private String type; // "DEPARTMENT", "MAJOR", "BATCH", "SECTION", "SEMESTER", "DESIGNATION"
47         private String value;
48     }
49
50     @Data
51     @NoArgsConstructor
52     @AllArgsConstructor
53     public static class TeacherRegistrationRequest {
54         private String username;
55         private String password;
56         private String fullName;
57         private String email;
58         private String department;
59         private String designation;
60         private boolean guest; // true if guest teacher
61     }
62
63     @Data
64     @NoArgsConstructor
65     @AllArgsConstructor
66     public static class CourseDefinitionRequest {
67         private String code;
68         private String name;
69         private Double credits;
70         private String semester;
71         private String department;
72     }

```

```

73
74 @Data
75 @NoArgsConstructor
76 @AllArgsConstructor
77 public static class StudentClassRequest {
78     private String department;
79     private String batch;
80     private String section;
81     private String semester;
82 }
83
84 @Data
85 @NoArgsConstructor
86 @AllArgsConstructor
87 public static class CourseAssignmentRequest {
88     private Long courseId;
89     private Long teacherId;
90 }
91
92 @Data
93 @NoArgsConstructor
94 @AllArgsConstructor
95 public static class AssignmentSnapshot {
96     private Long courseId;
97     private String courseCode;
98     private String courseName;
99     private Long teacherId;
100    private String teacherUsername;
101    private String teacherName;
102 }
103
104 private University getUniversityContext(String domainHeader, Principal principal) {
105     if (domainHeader != null && !domainHeader.trim().isEmpty()) {
106         return universityRepository.findByDomain(domainHeader).orElse(null);
107     }
108     if (principal != null) {
109         Optional<User> user = userRepository.findByUsername(principal.getName());
110         if (user.isPresent() && user.get().getUniversity() != null) {
111             return user.get().getUniversity();
112         }
113     }
114     List<University> all = universityRepository.findAll();
115     return all.isEmpty() ? null : all.get(0);
116 }
117
118 // --- Academic Setup (Metadata) ---
119
120 @PostMapping("/metadata")
121 public ResponseEntity<?> addMetadata(@RequestHeader(value = "X-University-Domain", required = false) String
domainHeader,
122                                     @RequestBody MetadataRequest request, Principal principal) {
123     University uni = getUniversityContext(domainHeader, principal);
124     if (uni == null) return ResponseEntity.badRequest().body(Map.of("error", "University context not found"));
125
126     SystemMetadata meta = SystemMetadata.builder()
127         .type(request.getType().toUpperCase())
128         .value(request.getValue())
129         .university(uni)
130         .build();
131     systemMetadataRepository.save(meta);
132     return ResponseEntity.ok(meta);
133 }
134
135 @GetMapping("/metadata")
136 public ResponseEntity<?> getMetadata(@RequestHeader(value = "X-University-Domain", required = false) String
domainHeader,
137                                     @RequestParam String type, Principal principal) {
138     University uni = getUniversityContext(domainHeader, principal);
139     if (uni == null) return ResponseEntity.badRequest().body(Map.of("error", "University context not found"));
140
141     List<SystemMetadata> list = systemMetadataRepository.findByTypeAndUniversity(type.toUpperCase(), uni);
142     // Fallback to global metadata if none customized for this university yet (for semesters/designations)
143     if (list.isEmpty() && (type.equalsIgnoreCase("SEMESTER") || type.equalsIgnoreCase("DESIGNATION") ||
type.equalsIgnoreCase("BATCH"))) {
144         list = systemMetadataRepository.findByType(type.toUpperCase()).stream()

```

```

145         .filter(m -> m.getUniversity() == null)
146         .collect(Collectors.toList());
147     }
148     return ResponseEntity.ok(list);
149 }
150
151 // --- Teachers Registry ---
152
153 @GetMapping("/teachers")
154 public ResponseEntity<?> getTeachers(@RequestHeader(value = "X-University-Domain", required = false) String
domainHeader,
155                                     Principal principal) {
156     University uni = getUniversityContext(domainHeader, principal);
157     if (uni == null) return ResponseEntity.badRequest().body(Map.of("error", "University context not found"));
158
159     List<User> teachers = userRepository.findByUniversityAndRole(uni, Role.TEACHER);
160     return ResponseEntity.ok(teachers);
161 }
162
163 @PostMapping("/teachers")
164 public ResponseEntity<?> addTeacher(@RequestHeader(value = "X-University-Domain", required = false) String
domainHeader,
165                                     @RequestBody TeacherRegistrationRequest request, Principal principal) {
166     University uni = getUniversityContext(domainHeader, principal);
167     if (uni == null) return ResponseEntity.badRequest().body(Map.of("error", "University context not found"));
168
169     if (userRepository.existsByUsername(request.getUsername())) {
170         return ResponseEntity.badRequest().body(Map.of("error", "Username already taken"));
171     }
172
173     String designation = request.getDesignation();
174     if (request.isGuest()) {
175         designation = designation + " (Guest)";
176     }
177
178     User teacher = User.builder()
179         .username(request.getUsername())
180         .password(passwordEncoder.encode(request.getPassword() != null ? request.getPassword() : "password"))
181         .fullName(request.getFullName())
182         .email(request.getEmail())
183         .role(Role.TEACHER)
184         .approved(true)
185         .department(request.getDepartment())
186         .designation(designation)
187         .university(uni)
188         .build();
189
190     userRepository.save(teacher);
191     return ResponseEntity.ok(teacher);
192 }
193
194 // --- Predefined Courses Syllabus ---
195
196 @PostMapping("/courses")
197 public ResponseEntity<?> defineCourse(@RequestHeader(value = "X-University-Domain", required = false) String
domainHeader,
198                                     @RequestBody CourseDefinitionRequest request, Principal principal) {
199     University uni = getUniversityContext(domainHeader, principal);
200     if (uni == null) return ResponseEntity.badRequest().body(Map.of("error", "University context not found"));
201
202     Course course = Course.builder()
203         .code(request.getCode())
204         .name(request.getName())
205         .credits(request.getCredits())
206         .semester(request.getSemester())
207         .department(request.getDepartment())
208         .university(uni)
209         .build();
210     courseRepository.save(course);
211     return ResponseEntity.ok(course);
212 }
213
214 @GetMapping("/courses")
215 public ResponseEntity<?> getCourses(@RequestHeader(value = "X-University-Domain", required = false) String
domainHeader,

```

```

216         Principal principal) {
217     University uni = getUniversityContext(domainHeader, principal);
218     if (uni == null) return ResponseEntity.badRequest().body(Map.of("error", "University context not found"));
219
220     return ResponseEntity.ok(courseRepository.findByUniversity(uni));
221 }
222
223 @DeleteMapping("/courses/{id}")
224 public ResponseEntity<?> deleteCourse(@PathVariable Long id) {
225     courseRepository.deleteById(id);
226     return ResponseEntity.ok(Map.of("message", "Course syllabus entry deleted"));
227 }
228
229 // --- Class Planner ---
230
231 @PostMapping("/classes")
232 public ResponseEntity<?> createClass(@RequestHeader(value = "X-University-Domain", required = false) String
domainHeader,
233                                     @RequestBody StudentClassRequest request, Principal principal) {
234     University uni = getUniversityContext(domainHeader, principal);
235     if (uni == null) return ResponseEntity.badRequest().body(Map.of("error", "University context not found"));
236
237     // Set default semester if empty
238     String semester = request.getSemester();
239     if (semester == null || semester.trim().isEmpty()) {
240         semester = "1st Year 1st Semester";
241     }
242
243     StudentClass sc = StudentClass.builder()
244         .department(request.getDepartment())
245         .batch(request.getBatch())
246         .section(request.getSection())
247         .university(uni)
248         .build();
249     studentClassRepository.save(sc);
250     return ResponseEntity.ok(sc);
251 }
252
253 @GetMapping("/classes")
254 public ResponseEntity<?> getClasses(@RequestHeader(value = "X-University-Domain", required = false) String
domainHeader,
255                                     Principal principal) {
256     University uni = getUniversityContext(domainHeader, principal);
257     if (uni == null) return ResponseEntity.badRequest().body(Map.of("error", "University context not found"));
258
259     List<StudentClass> list = studentClassRepository.findByUniversity(uni);
260
261     // Enrich classes with CR info and semester name resolved from students
262     List<Map<String, Object>> result = new ArrayList<>();
263     for (StudentClass sc : list) {
264         Map<String, Object> map = new HashMap<>();
265         map.put("id", sc.getId());
266         map.put("department", sc.getDepartment());
267         map.put("batch", sc.getBatch());
268         map.put("section", sc.getSection());
269
270         // Fetch students to determine semester and CR
271         List<User> students = userRepository.findByStudentClass(sc);
272         String semester = "1st Year 1st Semester";
273         User cr = null;
274         for (User s : students) {
275             if (s.getSemester() != null) {
276                 semester = s.getSemester();
277             }
278             if (s.getRole() == Role.CR) {
279                 cr = s;
280             }
281         }
282         map.put("semester", semester);
283         map.put("crUsername", cr != null ? cr.getUsername() : "None Assigned");
284         map.put("crFullName", cr != null ? cr.getFullName() : "None Assigned");
285         map.put("studentsCount", students.size());
286         result.add(map);
287     }

```

```

288         return ResponseEntity.ok(result);
289     }
290
291     @PostMapping("/classes/{classId}/assign-cr")
292     @Transactional
293     public ResponseEntity<?> assignClassCr(@PathVariable Long classId, @RequestBody Map<String, String> body) {
294         Optional<StudentClass> classOpt = studentClassRepository.findById(classId);
295         if (classOpt.isEmpty()) return ResponseEntity.notFound().build();
296
297         String username = body.get("username");
298         Optional<User> studentOpt = userRepository.findByUsername(username);
299         if (studentOpt.isEmpty()) {
300             return ResponseEntity.badRequest().body(Map.of("error", "Student username not found"));
301         }
302
303         User student = studentOpt.get();
304         // Remove CR role from previous CRs of this class
305         List<User> classUsers = userRepository.findByStudentClass(classOpt.get());
306         for (User u : classUsers) {
307             if (u.getRole() == Role.CR) {
308                 u.setRole(Role.STUDENT);
309                 userRepository.save(u);
310             }
311         }
312
313         // Set new CR
314         student.setRole(Role.CR);
315         student.setStudentClass(classOpt.get());
316         // Sync department/batch/section from class to student
317         student.setDepartment(classOpt.get().getDepartment());
318         student.setBatch(classOpt.get().getBatch());
319         student.setSection(classOpt.get().getSection());
320         userRepository.save(student);
321
322         return ResponseEntity.ok(Map.of("message", "CR assigned successfully", "crUsername", student.getUsername(),
323 "crFullName", student.getFullName()));
324     }
325
326     @PostMapping("/classes/{classId}/assign-course")
327     @Transactional
328     public ResponseEntity<?> assignCourseTeacher(@PathVariable Long classId, @RequestBody CourseAssignmentRequest request)
329     {
330         Optional<StudentClass> classOpt = studentClassRepository.findById(classId);
331         if (classOpt.isEmpty()) return ResponseEntity.notFound().build();
332
333         Optional<Course> courseOpt = courseRepository.findById(request.getCourseId());
334         if (courseOpt.isEmpty()) return ResponseEntity.badRequest().body(Map.of("error", "Predefined course not found"));
335
336         Optional<User> teacherOpt = userRepository.findById(request.getTeacherId());
337         if (teacherOpt.isEmpty() || !teacherOpt.get().getRole().equals(Role.TEACHER)) {
338             return ResponseEntity.badRequest().body(Map.of("error", "Teacher not found"));
339         }
340
341         // Remove existing mapping for the same course in this class if it exists
342         List<ClassCourseAssignment> assignments = classCourseAssignmentRepository.findByStudentClass(classOpt.get());
343         for (ClassCourseAssignment cca : assignments) {
344             if (cca.getCourse().getId().equals(request.getCourseId())) {
345                 classCourseAssignmentRepository.delete(cca);
346             }
347         }
348
349         ClassCourseAssignment cca = ClassCourseAssignment.builder()
350             .studentClass(classOpt.get())
351             .course(courseOpt.get())
352             .teacher(teacherOpt.get())
353             .build();
354         classCourseAssignmentRepository.save(cca);
355
356         return ResponseEntity.ok(cca);
357     }
358
359     @GetMapping("/classes/{classId}/assignments")
360     public ResponseEntity<?> getClassAssignments(@PathVariable Long classId) {
361         Optional<StudentClass> classOpt = studentClassRepository.findById(classId);

```

```

360         if (classOpt.isEmpty()) return ResponseEntity.notFound().build();
361
362         List<ClassCourseAssignment> ccaList = classCourseAssignmentRepository.findByStudentClass(classOpt.get());
363
364         List<Map<String, Object>> result = ccaList.stream().map(cca -> {
365             Map<String, Object> map = new HashMap<>();
366             map.put("id", cca.getId());
367             map.put("courseId", cca.getCourse().getId());
368             map.put("courseCode", cca.getCourse().getCode());
369             map.put("courseName", cca.getCourse().getName());
370             map.put("courseCredits", cca.getCourse().getCredits());
371             map.put("teacherId", cca.getTeacher().getId());
372             map.put("teacherFullName", cca.getTeacher().getFullName());
373             map.put("teacherDesignation", cca.getTeacher().getDesignation());
374             return map;
375         }).collect(Collectors.toList());
376
377         return ResponseEntity.ok(result);
378     }
379
380     // --- Semester Promotion & Rollback ---
381
382     private String getNextSemester(String current) {
383         List<String> semesters = List.of(
384             "1st Year 1st Semester", "1st Year 2nd Semester",
385             "2nd Year 1st Semester", "2nd Year 2nd Semester",
386             "3rd Year 1st Semester", "3rd Year 2nd Semester",
387             "4th Year 1st Semester", "4th Year 2nd Semester"
388         );
389         int idx = semesters.indexOf(current);
390         if (idx == -1) return "1st Year 1st Semester";
391         if (idx == semesters.size() - 1) return current; // already at max
392         return semesters.get(idx + 1);
393     }
394
395     private String getPreviousSemester(String current) {
396         List<String> semesters = List.of(
397             "1st Year 1st Semester", "1st Year 2nd Semester",
398             "2nd Year 1st Semester", "2nd Year 2nd Semester",
399             "3rd Year 1st Semester", "3rd Year 2nd Semester",
400             "4th Year 1st Semester", "4th Year 2nd Semester"
401         );
402         int idx = semesters.indexOf(current);
403         if (idx <= 0) return "1st Year 1st Semester";
404         return semesters.get(idx - 1);
405     }
406
407     @PostMapping("/classes/{classId}/promote")
408     @Transactional
409     public ResponseEntity<?> promoteClass(@PathVariable Long classId) {
410         Optional<StudentClass> classOpt = studentClassRepository.findById(classId);
411         if (classOpt.isEmpty()) return ResponseEntity.notFound().build();
412
413         StudentClass sc = classOpt.get();
414         List<User> students = userRepository.findByStudentClass(sc);
415         if (students.isEmpty()) {
416             return ResponseEntity.badRequest().body(Map.of("error", "No students enrolled in this class group"));
417         }
418
419         // Get current semester of class from any student
420         String currentSemester = "1st Year 1st Semester";
421         for (User s : students) {
422             if (s.getSemester() != null) {
423                 currentSemester = s.getSemester();
424                 break;
425             }
426         }
427
428         String nextSemester = getNextSemester(currentSemester);
429         if (nextSemester.equals(currentSemester)) {
430             return ResponseEntity.badRequest().body(Map.of("error", "Class is already in the final semester"));
431         }
432
433         // 1. Create assignments snapshot

```

```

434     List<ClassCourseAssignment> ccaList = classCourseAssignmentRepository.findByStudentClass(sc);
435     List<AssignmentSnapshot> snapshots = ccaList.stream().map(cca -> new AssignmentSnapshot(
436         cca.getCourse().getId(),
437         cca.getCourse().getCode(),
438         cca.getCourse().getName(),
439         cca.getTeacher().getId(),
440         cca.getTeacher().getUsername(),
441         cca.getTeacher().getFullName()
442     )).collect(Collectors.toList());
443
444     String jsonStr = "";
445     try {
446         jsonStr = objectMapper.writeValueAsString(snapshots);
447     } catch (Exception e) {
448         return ResponseEntity.internalServerError().body(Map.of("error", "Failed to compile assignment snapshot"));
449     }
450
451     // 2. Save promotion log
452     PromotionHistory history = PromotionHistory.builder()
453         .studentClass(sc)
454         .fromSemester(currentSemester)
455         .toSemester(nextSemester)
456         .timestamp(LocalDateTime.now())
457         .savedAssignmentsJson(jsonStr)
458         .build();
459     promotionHistoryRepository.save(history);
460
461     // 3. Update all student semesters
462     for (User s : students) {
463         s.setSemester(nextSemester);
464         userRepository.save(s);
465     }
466
467     // 4. Wipe active course/teacher assignments (so they register new ones for next semester)
468     classCourseAssignmentRepository.deleteByStudentClass(sc);
469
470     return ResponseEntity.ok(Map.of(
471         "message", "Class promoted to " + nextSemester + " successfully!",
472         "fromSemester", currentSemester,
473         "toSemester", nextSemester
474     ));
475 }
476
477 @PostMapping("/classes/{classId}/rollback-promotion")
478 @Transactional
479 public ResponseEntity<?> rollbackPromotion(@PathVariable Long classId) {
480     Optional<StudentClass> classOpt = studentClassRepository.findById(classId);
481     if (classOpt.isEmpty()) return ResponseEntity.notFound().build();
482
483     StudentClass sc = classOpt.get();
484     List<PromotionHistory> historyList = promotionHistoryRepository.findByStudentClassOrderByTimestampDesc(sc);
485     if (historyList.isEmpty()) {
486         return ResponseEntity.badRequest().body(Map.of("error", "No promotion history log found for this class
group"));
487     }
488
489     PromotionHistory latestLog = historyList.get(0);
490
491     // 1. Restore student semesters
492     List<User> students = userRepository.findByStudentClass(sc);
493     for (User s : students) {
494         s.setSemester(latestLog.getFromSemester());
495         userRepository.save(s);
496     }
497
498     // 2. Clear current assignments
499     classCourseAssignmentRepository.deleteByStudentClass(sc);
500
501     // 3. Restore assignments from JSON snapshot
502     try {
503         List<AssignmentSnapshot> snapshots = objectMapper.readValue(
504             latestLog.getSavedAssignmentsJson(),
505             new TypeReference<List<AssignmentSnapshot>>() {}
506         );

```

```

507
508         for (AssignmentSnapshot snap : snapshots) {
509             Optional<Course> courseOpt = courseRepository.findById(snap.getCourseId());
510             Optional<User> teacherOpt = userRepository.findById(snap.getTeacherId());
511             if (courseOpt.isPresent() && teacherOpt.isPresent()) {
512                 ClassCourseAssignment cca = ClassCourseAssignment.builder()
513                     .studentClass(sc)
514                     .course(courseOpt.get())
515                     .teacher(teacherOpt.get())
516                     .build();
517                 classCourseAssignmentRepository.save(cca);
518             }
519         }
520     } catch (Exception e) {
521         return ResponseEntity.internalServerError().body(Map.of("error", "Failed to unpack rollback snapshot metadata:
" + e.getMessage()));
522     }
523
524     // 4. Delete the rollback log entry so they don't double undo it
525     promotionHistoryRepository.delete(latestLog);
526
527     return ResponseEntity.ok(Map.of(
528         "message", "Successfully rolled back promotion to " + latestLog.getFromSemester() + "!",
529         "restoredSemester", latestLog.getFromSemester()
530     ));
531 }
532
533 @GetMapping("/students")
534 public ResponseEntity<?> getStudents(@RequestHeader(value = "X-University-Domain", required = false) String
domainHeader,
535                                     Principal principal) {
536     University uni = getUniversityContext(domainHeader, principal);
537     if (uni == null) return ResponseEntity.badRequest().body(Map.of("error", "University context not found"));
538
539     List<User> students = userRepository.findByUniversityAndRole(uni, Role.STUDENT);
540     List<User> crList = userRepository.findByUniversityAndRole(uni, Role.CR);
541
542     List<User> all = new ArrayList<>();
543     all.addAll(students);
544     all.addAll(crList);
545     return ResponseEntity.ok(all);
546 }
547
548 // --- University Scoped Configuration & Logo/Name update ---
549
550 @GetMapping("/university")
551 public ResponseEntity<?> getUniversity(@RequestHeader(value = "X-University-Domain", required = false) String
domainHeader,
552                                     Principal principal) {
553     University uni = getUniversityContext(domainHeader, principal);
554     if (uni == null) return ResponseEntity.badRequest().body(Map.of("error", "University context not found"));
555     return ResponseEntity.ok(uni);
556 }
557
558 @PutMapping("/university")
559 public ResponseEntity<?> updateUniversity(@RequestHeader(value = "X-University-Domain", required = false) String
domainHeader,
560                                           @RequestBody Map<String, String> payload, Principal principal) {
561     University uni = getUniversityContext(domainHeader, principal);
562     if (uni == null) return ResponseEntity.badRequest().body(Map.of("error", "University context not found"));
563
564     String name = payload.get("name");
565     String logoUrl = payload.get("logoUrl");
566
567     if (name != null && !name.trim().isEmpty()) {
568         uni.setName(name);
569     }
570     if (logoUrl != null) {
571         uni.setLogoUrl(logoUrl);
572     }
573
574     University saved = universityRepository.save(uni);
575     return ResponseEntity.ok(saved);
576 }
577

```

```

578 // --- Metadata deletion ---
579
580 @DeleteMapping("/metadata/{id}")
581 public ResponseEntity<?> deleteMetadata(@PathVariable Long id) {
582     systemMetadataRepository.deleteById(id);
583     return ResponseEntity.ok(Map.of("message", "Metadata option deleted successfully"));
584 }
585
586 @DeleteMapping("/teachers/{id}")
587 @Transactional
588 public ResponseEntity<?> deleteTeacher(@PathVariable Long id, Principal principal) {
589     Optional<User> teacherOpt = userRepository.findById(id);
590     if (teacherOpt.isEmpty() || !teacherOpt.get().getRole().equals(Role.TEACHER)) {
591         return ResponseEntity.badRequest().body(Map.of("error", "Teacher not found"));
592     }
593     User teacher = teacherOpt.get();
594
595     // 1. Delete associated profile change requests
596     profileChangeRequestRepository.deleteByUser(teacher);
597
598     // 2. Delete ClassCourseAssignments
599     classCourseAssignmentRepository.deleteByTeacher(teacher);
600
601     // 3. Delete ExamSubmissions and ExamQuestions for exams created by this teacher, then the exams
602     List<Exam> exams = examRepository.findByTeacher(teacher);
603     for (Exam exam : exams) {
604         examSubmissionRepository.deleteByExam(exam);
605         examQuestionRepository.deleteByExam(exam);
606     }
607     examRepository.deleteAll(exams);
608
609     // 4. Delete Resources uploaded by this teacher
610     List<Resource> resources = resourceRepository.findByUploadedBy(teacher);
611     for (Resource res : resources) {
612         resourceReactionRepository.deleteByResourceId(res.getId());
613     }
614     resourceRepository.deleteAll(resources);
615
616     // 5. Delete the User record
617     userRepository.delete(teacher);
618
619     return ResponseEntity.ok(Map.of("message", "Teacher deleted successfully"));
620 }
621 }
622

```

A.20: com/example/demo/controller/UserApprovalController.java

```

1 package com.example.demo.controller;
2
3 import com.example.demo.entity.User;
4 import com.example.demo.entity.User.Role;
5 import com.example.demo.entity.StudentClass;
6 import com.example.demo.entity.ProfileChangeRequest;
7 import com.example.demo.repository.UserRepository;
8 import com.example.demo.repository.StudentClassRepository;
9 import com.example.demo.repository.ProfileChangeRequestRepository;
10 import lombok.AllArgsConstructor;
11 import lombok.Data;
12 import lombok.NoArgsConstructor;
13 import org.springframework.http.HttpStatus;
14 import org.springframework.http.ResponseEntity;
15 import org.springframework.security.crypto.password.PasswordEncoder;
16 import org.springframework.web.bind.annotation.*;
17
18 import java.security.Principal;
19 import java.util.ArrayList;
20 import java.util.List;
21 import java.util.Map;
22
23 @RestController
24 @RequestMapping("/api/approvals")
25 @AllArgsConstructor
26 public class UserApprovalController {
27
28     private final UserRepository userRepository;
29     private final StudentClassRepository studentClassRepository;
30     private final ProfileChangeRequestRepository profileChangeRequestRepository;
31     private final PasswordEncoder passwordEncoder;
32     private final org.springframework.mail.javamail.JavaMailSender mailSender;
33     private final org.springframework.core.env.Environment env;
34
35     @GetMapping("/pending")
36     public ResponseEntity<?> getPendingUsers(Principal principal) {
37         if (principal == null) {
38             return ResponseEntity.status(HttpStatus.UNAUTHORIZED).body(Map.of("error", "Not authenticated"));
39         }
40         User currentUser = userRepository.findByUsername(principal.getName()).orElse(null);
41         if (currentUser == null) {
42             return ResponseEntity.status(HttpStatus.UNAUTHORIZED).body(Map.of("error", "User not found"));
43         }
44
45         Role currentRole = currentUser.getRole();
46         List<User> allUsers = userRepository.findAll();
47         List<Map<String, Object>> pending = new ArrayList<>();
48
49         for (User u : allUsers) {
50             if (!u.isApproved()) {
51                 boolean canManage = false;
52                 if (currentRole == Role.ADMIN) {
53                     canManage = true;
54                 } else if (currentRole == Role.TEACHER) {
55                     if (u.getRole() == Role.CR || u.getRole() == Role.STUDENT) {
56                         canManage = true;
57                     }
58                 } else if (currentRole == Role.CR) {
59                     if (u.getRole() == Role.STUDENT) {
60                         canManage = true;
61                     }
62                 }
63
64                 if (canManage) {
65                     Map<String, Object> map = new java.util.HashMap<>();
66                     map.put("id", u.getId());
67                     map.put("username", u.getUsername());
68                     map.put("fullName", u.getFullName());
69                     map.put("email", u.getEmail());
70                     map.put("role", u.getRole().name());
71                     map.put("idNo", u.getIdNo() != null ? u.getIdNo() : "");
72                     map.put("department", u.getDepartment() != null ? u.getDepartment() : "");

```

```

73         map.put("batch", u.getBatch() != null ? u.getBatch() : "");
74         map.put("semester", u.getSemester() != null ? u.getSemester() : "");
75         map.put("section", u.getSection() != null ? u.getSection() : "");
76         map.put("designation", u.getDesignation() != null ? u.getDesignation() : "");
77         pending.add(map);
78     }
79 }
80 }
81
82     return ResponseEntity.ok(pending);
83 }
84
85 @PostMapping("/approve/{id}")
86 public ResponseEntity<?> approveUser(@PathVariable Long id, Principal principal) {
87     if (principal == null) {
88         return ResponseEntity.status(HttpStatus.UNAUTHORIZED).body(Map.of("error", "Not authenticated"));
89     }
90     User currentUser = userRepository.findByUsername(principal.getName()).orElse(null);
91     if (currentUser == null) {
92         return ResponseEntity.status(HttpStatus.UNAUTHORIZED).body(Map.of("error", "User not found"));
93     }
94
95     User targetUser = userRepository.findById(id).orElse(null);
96     if (targetUser == null) {
97         return ResponseEntity.notFound().build();
98     }
99
100    Role currentRole = currentUser.getRole();
101    Role targetRole = targetUser.getRole();
102    boolean canApprove = false;
103
104    if (currentRole == Role.ADMIN) {
105        canApprove = true;
106    } else if (currentRole == Role.TEACHER) {
107        if (targetRole == Role.CR || targetRole == Role.STUDENT) {
108            canApprove = true;
109        }
110    } else if (currentRole == Role.CR) {
111        if (targetRole == Role.STUDENT) {
112            canApprove = true;
113        }
114    }
115
116    if (!canApprove) {
117        return ResponseEntity.status(HttpStatus.FORBIDDEN)
118            .body(Map.of("error", "You do not have permission to approve a user with role: " + targetRole));
119    }
120
121    // Auto-group Student/CR into a StudentClass
122    if (targetRole == Role.STUDENT || targetRole == Role.CR) {
123        String batch = targetUser.getBatch();
124        String dept = targetUser.getDepartment();
125        String sec = targetUser.getSection();
126        if (batch != null && dept != null && sec != null && !batch.isEmpty() && !dept.isEmpty() && !sec.isEmpty()) {
127            StudentClass studentClass = studentClassRepository.findByBatchAndDepartmentAndSection(batch, dept, sec)
128                .orElseGet(() -> {
129                StudentClass sc = StudentClass.builder()
130                    .batch(batch)
131                    .department(dept)
132                    .section(sec)
133                    .build();
134                return studentClassRepository.save(sc);
135            });
136            targetUser.setStudentClass(studentClass);
137        }
138    }
139
140    targetUser.setApproved(true);
141    userRepository.save(targetUser);
142
143    try {
144        org.springframework.mail.SimpleMailMessage message = new org.springframework.mail.SimpleMailMessage();
145        String fromEmail = env.getProperty("spring.mail.username");
146        if (fromEmail != null && !fromEmail.isEmpty()) {

```

```

147         message.setFrom(fromEmail);
148     }
149     message.setTo(targetUser.getEmail());
150     message.setSubject("LearnX Account Approved");
151     message.setText("Hello " + targetUser.getFullName() + ",\n\n"
152         + "Your LearnX account request has been approved!\n"
153         + "You can now log in using your registered username: " + targetUser.getUsername() + "\n\n"
154         + "Best regards,\n"
155         + "LearnX Team");
156     mailSender.send(message);
157 } catch (Exception ex) {
158     System.err.println("Failed to send approval email to " + targetUser.getEmail() + ": " + ex.getMessage());
159 }
160
161 return ResponseEntity.ok(Map.of("message", "User account approved successfully"));
162 }
163
164 @DeleteMapping("/reject/{id}")
165 public ResponseEntity<?> rejectUser(@PathVariable Long id, Principal principal) {
166     if (principal == null) {
167         return ResponseEntity.status(HttpStatus.UNAUTHORIZED).body(Map.of("error", "Not authenticated"));
168     }
169     User currentUser = userRepository.findByUsername(principal.getName()).orElse(null);
170     if (currentUser == null) {
171         return ResponseEntity.status(HttpStatus.UNAUTHORIZED).body(Map.of("error", "User not found"));
172     }
173
174     User targetUser = userRepository.findById(id).orElse(null);
175     if (targetUser == null) {
176         return ResponseEntity.notFound().build();
177     }
178
179     Role currentRole = currentUser.getRole();
180     Role targetRole = targetUser.getRole();
181     boolean canReject = false;
182
183     if (currentRole == Role.ADMIN) {
184         canReject = true;
185     } else if (currentRole == Role.TEACHER) {
186         if (targetRole == Role.CR || targetRole == Role.STUDENT) {
187             canReject = true;
188         }
189     } else if (currentRole == Role.CR) {
190         if (targetRole == Role.STUDENT) {
191             canReject = true;
192         }
193     }
194
195     if (!canReject) {
196         return ResponseEntity.status(HttpStatus.FORBIDDEN)
197             .body(Map.of("error", "You do not have permission to reject a user with role: " + targetRole));
198     }
199
200     try {
201         org.springframework.mail.SimpleMailMessage message = new org.springframework.mail.SimpleMailMessage();
202         String fromEmail = env.getProperty("spring.mail.username");
203         if (fromEmail != null && !fromEmail.isEmpty()) {
204             message.setFrom(fromEmail);
205         }
206         message.setTo(targetUser.getEmail());
207         message.setSubject("LearnX Account Request Rejected");
208         message.setText("Hello " + targetUser.getFullName() + ",\n\n"
209             + "Your LearnX registration request has been rejected.\n"
210             + "If you think this was a mistake, please register again with correct details or contact support.\n\n"
211             + "Best regards,\n"
212             + "LearnX Team");
213         mailSender.send(message);
214     } catch (Exception ex) {
215         System.err.println("Failed to send rejection email to " + targetUser.getEmail() + ": " + ex.getMessage());
216     }
217
218     userRepository.delete(targetUser);
219     return ResponseEntity.ok(Map.of("message", "User account request rejected and deleted"));
220 }

```

```

221
222 // -----
223 // ADMIN ONLY: Create New Admin Accounts
224 // -----
225 @Data
226 @NoArgsConstructor
227 @AllArgsConstructor
228 public static class AdminCreateRequest {
229     private String username;
230     private String password;
231     private String fullName;
232     private String email;
233 }
234
235 @PostMapping("/admin-create")
236 public ResponseEntity<?> createNewAdmin(@RequestBody AdminCreateRequest request, Principal principal) {
237     if (principal == null) {
238         return ResponseEntity.status(HttpStatus.UNAUTHORIZED).body(Map.of("error", "Not authenticated"));
239     }
240     User currentUser = userRepository.findByUsername(principal.getName()).orElse(null);
241     if (currentUser == null || currentUser.getRole() != Role.ADMIN) {
242         return ResponseEntity.status(HttpStatus.FORBIDDEN).body(Map.of("error", "Only administrators can create other
administrator accounts."));
243     }
244
245     if (userRepository.existsByUsername(request.getUsername())) {
246         return ResponseEntity.badRequest().body(Map.of("error", "Username already exists"));
247     }
248
249     User admin = User.builder()
250         .username(request.getUsername())
251         .password(passwordEncoder.encode(request.getPassword()))
252         .fullName(request.getFullName())
253         .email(request.getEmail())
254         .role(Role.ADMIN)
255         .approved(true)
256         .build();
257
258     userRepository.save(admin);
259     return ResponseEntity.ok(Map.of("message", "Administrator account created successfully"));
260 }
261
262 // -----
263 // ADMIN ONLY: Sensitive Profile Change Requests Approvals
264 // -----
265 @GetMapping("/profile-requests")
266 public ResponseEntity<?> getPendingProfileRequests(Principal principal) {
267     if (principal == null) {
268         return ResponseEntity.status(HttpStatus.UNAUTHORIZED).body(Map.of("error", "Not authenticated"));
269     }
270     User currentUser = userRepository.findByUsername(principal.getName()).orElse(null);
271     if (currentUser == null || currentUser.getRole() != Role.ADMIN) {
272         return ResponseEntity.status(HttpStatus.FORBIDDEN).body(Map.of("error", "Only administrators can view pending
profile changes."));
273     }
274
275     List<ProfileChangeRequest> requests = profileChangeRequestRepository.findByApprovedFalseAndRejectedFalse();
276     return ResponseEntity.ok(requests);
277 }
278
279 @PostMapping("/profile-requests/{id}/approve")
280 public ResponseEntity<?> approveProfileRequest(@PathVariable Long id, Principal principal) {
281     if (principal == null) {
282         return ResponseEntity.status(HttpStatus.UNAUTHORIZED).body(Map.of("error", "Not authenticated"));
283     }
284     User currentUser = userRepository.findByUsername(principal.getName()).orElse(null);
285     if (currentUser == null || currentUser.getRole() != Role.ADMIN) {
286         return ResponseEntity.status(HttpStatus.FORBIDDEN).body(Map.of("error", "Only administrators can approve
profile changes."));
287     }
288
289     ProfileChangeRequest req = profileChangeRequestRepository.findById(id).orElse(null);
290     if (req == null) {
291         return ResponseEntity.notFound().build();
292     }

```

```

293
294     User targetUser = req.getUser();
295     if (req.getNewFullName() != null) targetUser.setFullName(req.getNewFullName());
296     if (req.getNewEmail() != null) targetUser.setEmail(req.getNewEmail());
297     if (req.getNewIdNo() != null) targetUser.setIdNo(req.getNewIdNo());
298     if (req.getNewDepartment() != null) targetUser.setDepartment(req.getNewDepartment());
299     if (req.getNewBatch() != null) targetUser.setBatch(req.getNewBatch());
300     if (req.getNewSemester() != null) targetUser.setSemester(req.getNewSemester());
301     if (req.getNewSection() != null) targetUser.setSection(req.getNewSection());
302     if (req.getNewDesignation() != null) targetUser.setDesignation(req.getNewDesignation());
303
304     // Update Class Grouping if batch/dept/section changed
305     if (targetUser.getRole() == Role.STUDENT || targetUser.getRole() == Role.CR) {
306         String batch = targetUser.getBatch();
307         String dept = targetUser.getDepartment();
308         String sec = targetUser.getSection();
309         if (batch != null && dept != null && sec != null && !batch.isEmpty() && !dept.isEmpty() && !sec.isEmpty()) {
310             StudentClass studentClass = studentClassRepository.findByBatchAndDepartmentAndSection(batch, dept, sec)
311                 .orElseGet(() -> {
312                     StudentClass sc = StudentClass.builder()
313                         .batch(batch)
314                         .department(dept)
315                         .section(sec)
316                         .build();
317                     return studentClassRepository.save(sc);
318                 });
319             targetUser.setStudentClass(studentClass);
320         }
321     }
322
323     targetUser.setApproved(true);
324     userRepository.save(targetUser);
325
326     req.setApproved(true);
327     profileChangeRequestRepository.save(req);
328
329     try {
330         org.springframework.mail.SimpleMailMessage message = new org.springframework.mail.SimpleMailMessage();
331         String fromEmail = env.getProperty("spring.mail.username");
332         if (fromEmail != null && !fromEmail.isEmpty()) {
333             message.setFrom(fromEmail);
334         }
335         message.setTo(targetUser.getEmail());
336         message.setSubject("LearnX Profile Changes Approved");
337         message.setText("Hello " + targetUser.getFullName() + ",\n\n"
338             + "Your requested profile modifications have been approved by the administrator and applied to your
339             account.\n\n"
340             + "Best regards,\n"
341             + "LearnX Team");
342         mailSender.send(message);
343     } catch (Exception ex) {
344         System.err.println("Failed to send profile approval email to " + targetUser.getEmail() + ": " +
345             ex.getMessage());
346     }
347
348     return ResponseEntity.ok(Map.of("message", "Profile change request approved and applied successfully"));
349 }
350
351 @PostMapping("/profile-requests/{id}/reject")
352 public ResponseEntity<?> rejectProfileRequest(@PathVariable Long id, Principal principal) {
353     if (principal == null) {
354         return ResponseEntity.status(HttpStatus.UNAUTHORIZED).body(Map.of("error", "Not authenticated"));
355     }
356     User currentUser = userRepository.findByUsername(principal.getName()).orElse(null);
357     if (currentUser == null || currentUser.getRole() != Role.ADMIN) {
358         return ResponseEntity.status(HttpStatus.FORBIDDEN).body(Map.of("error", "Only administrators can reject profile
359             changes."));
360     }
361
362     ProfileChangeRequest req = profileChangeRequestRepository.findById(id).orElse(null);
363     if (req == null) {
364         return ResponseEntity.notFound().build();
365     }
366
367     req.setRejected(true);

```

```

365     profileChangeRequestRepository.save(req);
366
367     try {
368         org.springframework.mail.SimpleMailMessage message = new org.springframework.mail.SimpleMailMessage();
369         String fromEmail = env.getProperty("spring.mail.username");
370         if (fromEmail != null && !fromEmail.isEmpty()) {
371             message.setFrom(fromEmail);
372         }
373         message.setTo(req.getUser().getEmail());
374         message.setSubject("LearnX Profile Changes Rejected");
375         message.setText("Hello " + req.getUser().getFullName() + ",\n\n"
376             + "Your requested profile modifications have been rejected by the administrator.\n\n"
377             + "Best regards,\n"
378             + "LearnX Team");
379         mailSender.send(message);
380     } catch (Exception ex) {
381         System.err.println("Failed to send profile rejection email to " + req.getUser().getEmail() + ": " +
382             ex.getMessage());
383     }
384     return ResponseEntity.ok(Map.of("message", "Profile change request rejected"));
385 }
386 }
387

```

A.21: com/example/demo/entity/Announcement.java

```

1 package com.example.demo.entity;
2
3 import jakarta.persistence.*;
4 import lombok.*;
5 import java.time.LocalDateTime;
6
7 @Entity
8 @Table(name = "announcements")
9 @Data
10 @NoArgsConstructor
11 @AllArgsConstructor
12 @Builder
13 public class Announcement {
14
15     @Id
16     @GeneratedValue(strategy = GenerationType.IDENTITY)
17     private Long id;
18
19     @Column(nullable = false)
20     private String title;
21
22     @Column(nullable = false, length = 10000)
23     private String content;
24
25     @Builder.Default
26     private LocalDateTime createdAt = LocalDateTime.now();
27
28     @Column(nullable = false)
29     private String createdBy;
30
31     @Column(nullable = false)
32     private String createdByRole;
33
34     @ManyToOne
35     @JoinColumn(name = "student_class_id")
36     private StudentClass studentClass;
37
38     @ManyToOne
39     @JoinColumn(name = "university_id")
40     private University university;
41 }
42

```

A.22: com/example/demo/entity/AuditLog.java

```
1 package com.example.demo.entity;
2
3 import jakarta.persistence.*;
4 import lombok.*;
5 import java.time.LocalDateTime;
6
7 @Entity
8 @Table(name = "audit_logs")
9 @Data
10 @NoArgsConstructor
11 @AllArgsConstructor
12 @Builder
13 public class AuditLog {
14
15     @Id
16     @GeneratedValue(strategy = GenerationType.IDENTITY)
17     private Long id;
18
19     @Column(nullable = false)
20     private String entityType; // e.g. "ROUTINE", "CLASS_TEST"
21
22     private Long entityId;
23
24     @Column(nullable = false)
25     private String action; // e.g. "CREATE", "UPDATE", "DELETE"
26
27     @Column(nullable = false)
28     private String changedBy; // username
29
30     @Column(nullable = false)
31     private LocalDateTime timestamp;
32
33     @Column(columnDefinition = "TEXT")
34     private String details;
35 }
36
```

A.23: com/example/demo/entity/BugReport.java

```

1 package com.example.demo.entity;
2
3 import jakarta.persistence.*;
4 import lombok.*;
5 import java.time.LocalDateTime;
6
7 @Entity
8 @Table(name = "bug_reports")
9 @Data
10 @NoArgsConstructor
11 @AllArgsConstructor
12 @Builder
13 public class BugReport {
14
15     @Id
16     @GeneratedValue(strategy = GenerationType.IDENTITY)
17     private Long id;
18
19     @Column(nullable = false)
20     private String title;
21
22     @Column(nullable = false, columnDefinition = "TEXT")
23     private String description;
24
25     private String reportedBy; // Name/Email/Username of reporter
26
27     private LocalDateTime createdAt;
28
29     @Column(nullable = false)
30     private String status; // "PENDING", "REVIEWED", "RESOLVED"
31 }
32

```

A.24: com/example/demo/entity/ClassCourseAssignment.java

```

1 package com.example.demo.entity;
2
3 import jakarta.persistence.*;
4 import lombok.*;
5
6 @Entity
7 @Table(name = "class_course_assignments")
8 @Data
9 @NoArgsConstructor
10 @AllArgsConstructor
11 @Builder
12 public class ClassCourseAssignment {
13
14     @Id
15     @GeneratedValue(strategy = GenerationType.IDENTITY)
16     private Long id;
17
18     @ManyToOne(optional = false)
19     @JoinColumn(name = "student_class_id")
20     private StudentClass studentClass;
21
22     @ManyToOne(optional = false)
23     @JoinColumn(name = "course_id")
24     private Course course;
25
26     @ManyToOne(optional = false)
27     @JoinColumn(name = "teacher_id")
28     private User teacher;
29 }
30

```

A.25: com/example/demo/entity/ClassTest.java

```
1 package com.example.demo.entity;
2
3 import jakarta.persistence.*;
4 import lombok.*;
5 import java.time.LocalDateTime;
6
7 @Entity
8 @Table(name = "class_tests")
9 @Data
10 @NoArgsConstructor
11 @AllArgsConstructor
12 @Builder
13 public class ClassTest {
14
15     @Id
16     @GeneratedValue(strategy = GenerationType.IDENTITY)
17     private Long id;
18
19     @Column(nullable = false)
20     private String courseName;
21
22     @Column(nullable = false)
23     private LocalDateTime dateTime;
24
25     private int durationMinutes;
26     private String roomNo;
27     private String topic;
28     private String createdBy; // username of CR or Teacher
29
30     @ManyToOne
31     @JoinColumn(name = "student_class_id")
32     private StudentClass studentClass;
33
34     @ManyToOne
35     @JoinColumn(name = "university_id")
36     private University university;
37 }
38
```

A.26: com/example/demo/entity/Course.java

```
1 package com.example.demo.entity;
2
3 import jakarta.persistence.*;
4 import lombok.*;
5
6 @Entity
7 @Table(name = "courses")
8 @Data
9 @NoArgsConstructor
10 @AllArgsConstructor
11 @Builder
12 public class Course {
13
14     @Id
15     @GeneratedValue(strategy = GenerationType.IDENTITY)
16     private Long id;
17
18     @Column(nullable = false)
19     private String code; // e.g., CSE 3101
20
21     @Column(nullable = false)
22     private String name; // e.g., Database Systems
23
24     @Column(nullable = false)
25     private Double credits;
26
27     @Column(nullable = false)
28     private String semester; // e.g., 3rd Year 1st Semester
29
30     @Column(nullable = false)
31     private String department; // e.g., CSE - Computer Science & Engineering
32
33     @ManyToOne(optional = false)
34     @JoinColumn(name = "university_id")
35     private University university;
36 }
37
```

A.27: com/example/demo/entity/Exam.java

```
1 package com.example.demo.entity;
2
3 import jakarta.persistence.*;
4 import lombok.*;
5 import java.time.LocalDateTime;
6
7 @Entity
8 @Table(name = "exams")
9 @Data
10 @NoArgsConstructor
11 @AllArgsConstructor
12 @Builder
13 public class Exam {
14
15     @Id
16     @GeneratedValue(strategy = GenerationType.IDENTITY)
17     private Long id;
18
19     @Column(nullable = false)
20     private String title;
21
22     @Column(columnDefinition = "TEXT")
23     private String description;
24
25     @ManyToOne(optional = false)
26     @JoinColumn(name = "teacher_id", nullable = false)
27     private User teacher;
28
29     private int durationMinutes;
30
31     private LocalDateTime startTime;
32     private LocalDateTime endTime;
33
34     @Column(nullable = false)
35     private boolean published;
36
37     @ManyToOne
38     @JoinColumn(name = "student_class_id")
39     private StudentClass studentClass;
40
41     @ManyToOne
42     @JoinColumn(name = "university_id")
43     private University university;
44 }
45
```

A.28: com/example/demo/entity/ExamQuestion.java

```
1 package com.example.demo.entity;
2
3 import jakarta.persistence.*;
4 import lombok.*;
5
6 @Entity
7 @Table(name = "exam_questions")
8 @Data
9 @NoArgsConstructor
10 @AllArgsConstructor
11 @Builder
12 public class ExamQuestion {
13
14     @Id
15     @GeneratedValue(strategy = GenerationType.IDENTITY)
16     private Long id;
17
18     @ManyToOne(optional = false)
19     @JoinColumn(name = "exam_id", nullable = false)
20     private Exam exam;
21
22     @Column(nullable = false, columnDefinition = "TEXT")
23     private String questionText;
24
25     @Enumerated(EnumType.STRING)
26     @Column(nullable = false)
27     private QuestionType questionType;
28
29     private int points;
30
31     @Column(columnDefinition = "TEXT")
32     private String options; // Semicolon-separated options for MCQ, e.g., "A;B;C;D"
33
34     private String correctAnswer; // Text or option index
35
36     public enum QuestionType {
37         MCQ,
38         SHORT_ANSWER
39     }
40 }
41
```

A.29: com/example/demo/entity/ExamSubmission.java

```
1 package com.example.demo.entity;
2
3 import jakarta.persistence.*;
4 import lombok.*;
5 import java.time.LocalDateTime;
6
7 @Entity
8 @Table(name = "exam_submissions")
9 @Data
10 @NoArgsConstructor
11 @AllArgsConstructor
12 @Builder
13 public class ExamSubmission {
14
15     @Id
16     @GeneratedValue(strategy = GenerationType.IDENTITY)
17     private Long id;
18
19     @ManyToOne(optional = false)
20     @JoinColumn(name = "exam_id", nullable = false)
21     private Exam exam;
22
23     @ManyToOne(optional = false)
24     @JoinColumn(name = "student_id", nullable = false)
25     private User student;
26
27     @Column(nullable = false)
28     private LocalDateTime submittedAt;
29
30     private int score;
31
32     @Column(columnDefinition = "TEXT")
33     private String answersJson; // JSON representation of question answers
34 }
35
```

A.30: com/example/demo/entity/GradeBook.java

```
1 package com.example.demo.entity;
2
3 import jakarta.persistence.*;
4 import lombok.*;
5
6 @Entity
7 @Table(name = "gradebooks")
8 @Data
9 @NoArgsConstructor
10 @AllArgsConstructor
11 @Builder
12 public class GradeBook {
13
14     @Id
15     @GeneratedValue(strategy = GenerationType.IDENTITY)
16     private Long id;
17
18     @ManyToOne(optional = false)
19     @JoinColumn(name = "student_id", nullable = false)
20     private User student;
21
22     @Column(nullable = false)
23     private String courseName;
24
25     @Column(nullable = false)
26     private String assessmentName; // e.g. "CT 1", "Midterm"
27
28     @Column(nullable = false)
29     private double marksObtained;
30
31     @Column(nullable = false)
32     private double maxMarks;
33 }
34
```

A.31: com/example/demo/entity/ProfileChangeRequest.java

```
1 package com.example.demo.entity;
2
3 import jakarta.persistence.*;
4 import lombok.*;
5 import java.time.LocalDateTime;
6
7 @Entity
8 @Table(name = "profile_change_requests")
9 @Data
10 @NoArgsConstructor
11 @AllArgsConstructor
12 @Builder
13 public class ProfileChangeRequest {
14
15     @Id
16     @GeneratedValue(strategy = GenerationType.IDENTITY)
17     private Long id;
18
19     @ManyToOne
20     @JoinColumn(name = "user_id", nullable = false)
21     private User user;
22
23     private String newFullName;
24     private String newEmail;
25     private String newIdNO;
26     private String newDepartment;
27     private String newBatch;
28     private String newSemester;
29     private String newSection;
30     private String newDesignation;
31
32     @Builder.Default
33     private boolean approved = false;
34
35     @Builder.Default
36     private boolean rejected = false;
37
38     @Builder.Default
39     private LocalDateTime createdAt = LocalDateTime.now();
40 }
41
```

A.32: com/example/demo/entity/PromotionHistory.java

```
1 package com.example.demo.entity;
2
3 import jakarta.persistence.*;
4 import lombok.*;
5 import java.time.LocalDateTime;
6
7 @Entity
8 @Table(name = "promotion_history")
9 @Data
10 @NoArgsConstructor
11 @AllArgsConstructor
12 @Builder
13 public class PromotionHistory {
14
15     @Id
16     @GeneratedValue(strategy = GenerationType.IDENTITY)
17     private Long id;
18
19     @ManyToOne(optional = false)
20     @JoinColumn(name = "student_class_id")
21     private StudentClass studentClass;
22
23     @Column(nullable = false)
24     private String fromSemester;
25
26     @Column(nullable = false)
27     private String toSemester;
28
29     @Column(nullable = false)
30     private LocalDateTime timestamp;
31
32     @Column(columnDefinition = "TEXT")
33     private String savedAssignmentsJson; // JSON representation of course/teacher mappings
34 }
35
```

A.33: com/example/demo/entity/Resource.java

```
1 package com.example.demo.entity;
2
3 import jakarta.persistence.*;
4 import lombok.*;
5
6 @Entity
7 @Table(name = "resources")
8 @Data
9 @NoArgsConstructor
10 @AllArgsConstructor
11 @Builder
12 public class Resource {
13
14     @Id
15     @GeneratedValue(strategy = GenerationType.IDENTITY)
16     private Long id;
17
18     @Column(nullable = false)
19     private String title;
20
21     @Column(nullable = false)
22     private String courseName;
23
24     private String fileName;
25
26     private String contentType;
27
28     @Lob
29     @Column(length = 104857600) // Support up to 100MB in database
30     private byte[] fileData;
31
32     @ManyToOne(optional = false)
33     @JoinColumn(name = "uploader_id", nullable = false)
34     private User uploadedBy;
35
36     @Column(nullable = false)
37     private boolean approved;
38
39     private String examTags; // e.g. "Midterm", "CT1", "Final"
40
41     private String driveLink; // Direct Google Drive resources link
42
43     @ManyToOne
44     @JoinColumn(name = "student_class_id")
45     private StudentClass studentClass;
46
47     @ManyToOne
48     @JoinColumn(name = "university_id")
49     private University university;
50
51     @Transient
52     private int likesCount;
53
54     @Transient
55     private int dislikesCount;
56
57     @Transient
58     private String userReaction; // active user's reaction ("LIKE", "DISLIKE", or null)
59 }
60
```

A.34: com/example/demo/entity/ResourceReaction.java

```
1 package com.example.demo.entity;
2
3 import jakarta.persistence.*;
4 import lombok.*;
5
6 @Entity
7 @Table(name = "resource_reactions", uniqueConstraints = {
8     @UniqueConstraint(columnNames = {"resource_id", "username"})
9 })
10 @Data
11 @NoArgsConstructor
12 @AllArgsConstructor
13 @Builder
14 public class ResourceReaction {
15
16     @Id
17     @GeneratedValue(strategy = GenerationType.IDENTITY)
18     private Long id;
19
20     @Column(name = "resource_id", nullable = false)
21     private Long resourceId;
22
23     @Column(nullable = false)
24     private String username;
25
26     @Column(nullable = false)
27     private String reactionType;
28 }
29
```

A.35: com/example/demo/entity/ScheduleItem.java

```
1 package com.example.demo.entity;
2
3 import jakarta.persistence.*;
4 import lombok.*;
5 import java.time.LocalDateTime;
6
7 @Entity
8 @Table(name = "schedule_items")
9 @Data
10 @NoArgsConstructor
11 @AllArgsConstructor
12 @Builder
13 public class ScheduleItem {
14
15     @Id
16     @GeneratedValue(strategy = GenerationType.IDENTITY)
17     private Long id;
18
19     @Column(nullable = false)
20     private String dayOfWeek; // e.g. "MONDAY", "TUESDAY"
21
22     @Column(nullable = false)
23     private LocalDateTime startTime;
24
25     @Column(nullable = false)
26     private LocalDateTime endTime;
27
28     @Column(nullable = false)
29     private String courseName;
30
31     private String teacherName;
32     private String roomNo;
33
34     @ManyToOne
35     @JoinColumn(name = "student_class_id")
36     private StudentClass studentClass;
37
38     @ManyToOne
39     @JoinColumn(name = "university_id")
40     private University university;
41 }
42
```

A.36: com/example/demo/entity/StudentClass.java

```
1 package com.example.demo.entity;
2
3 import com.fasterxml.jackson.annotation.JsonIgnore;
4 import jakarta.persistence.*;
5 import lombok.*;
6 import java.util.List;
7
8 @Entity
9 @Table(name = "student_classes")
10 @Data
11 @NoArgsConstructor
12 @AllArgsConstructor
13 @Builder
14 @ToString(exclude = "students")
15 public class StudentClass {
16
17     @Id
18     @GeneratedValue(strategy = GenerationType.IDENTITY)
19     private Long id;
20
21     private String batch;
22     private String department;
23     private String section;
24
25     @ManyToOne
26     @JoinColumn(name = "university_id")
27     private University university;
28
29     @OneToMany(mappedBy = "studentClass")
30     @JsonIgnore
31     private List<User> students;
32
33     @com.fasterxml.jackson.annotation.JsonProperty("className")
34     public String getClassName() {
35         String deptCode = (department != null && department.contains(" - ")) ? department.split(" - ")[0] : (department !=
36         null ? department : "");
37         return deptCode + " " + (batch != null ? batch : "") + " " + (section != null ? section : "");
38     }
39 }
```

A.37: com/example/demo/entity/SystemAdmin.java

```
1 package com.example.demo.entity;
2
3 import jakarta.persistence.*;
4 import lombok.*;
5
6 @Entity
7 @Table(name = "system_admins")
8 @Data
9 @NoArgsConstructor
10 @AllArgsConstructor
11 @Builder
12 public class SystemAdmin {
13
14     @Id
15     @GeneratedValue(strategy = GenerationType.IDENTITY)
16     private Long id;
17
18     @Column(nullable = false, unique = true)
19     private String username;
20
21     @Column(nullable = false)
22     private String password;
23
24     @Column(nullable = false)
25     private String fullName;
26
27     @Column(nullable = false)
28     private String email;
29 }
30
```

A.38: com/example/demo/entity/SystemMetadata.java

```
1 package com.example.demo.entity;
2
3 import jakarta.persistence.*;
4 import lombok.*;
5
6 @Entity
7 @Table(name = "system_metadata")
8 @Data
9 @NoArgsConstructor
10 @AllArgsConstructor
11 @Builder
12 public class SystemMetadata {
13
14     @Id
15     @GeneratedValue(strategy = GenerationType.IDENTITY)
16     private Long id;
17
18     @Column(nullable = false)
19     private String type; // "SEMESTER", "DEPARTMENT", "SESSION", "DESIGNATION"
20
21     @Column(name = "meta_value", nullable = false)
22     private String value;
23
24     @ManyToOne
25     @JoinColumn(name = "university_id")
26     private University university;
27 }
28
```

A.39: com/example/demo/entity/University.java

```
1 package com.example.demo.entity;
2
3 import jakarta.persistence.*;
4 import lombok.*;
5
6 @Entity
7 @Table(name = "universities")
8 @Data
9 @NoArgsConstructor
10 @AllArgsConstructor
11 @Builder
12 public class University {
13
14     @Id
15     @GeneratedValue(strategy = GenerationType.IDENTITY)
16     private Long id;
17
18     @Column(nullable = false, unique = true)
19     private String name;
20
21     @Column(nullable = false, unique = true)
22     private String domain; // e.g., ustc.learnx.com
23
24     @Column(columnDefinition = "TEXT")
25     private String logoUrl; // base64 or URL
26 }
27
```

A.40: com/example/demo/entity/User.java

```

1 package com.example.demo.entity;
2
3 import jakarta.persistence.*;
4 import lombok.*;
5
6 @Entity
7 @Table(name = "users")
8 @Data
9 @NoArgsConstructor
10 @AllArgsConstructor
11 @Builder
12 public class User {
13
14     @Id
15     @GeneratedValue(strategy = GenerationType.IDENTITY)
16     private Long id;
17
18     @Column(unique = true, nullable = false)
19     private String username;
20
21     @Column(nullable = false)
22     private String password;
23
24     @Column(nullable = false)
25     private String fullName;
26
27     @Column(unique = true, nullable = false)
28     private String email;
29
30     @Enumerated(EnumType.STRING)
31     @Column(nullable = false)
32     private Role role;
33
34     @Builder.Default
35     @Column(nullable = false)
36     private boolean approved = true;
37
38     private String idNo;
39     private String department;
40     private String batch;
41     private String semester;
42     private String section;
43     private String designation;
44     @Column(columnDefinition = "TEXT")
45     private String profilePicUrl;
46
47     @ManyToOne
48     @JoinColumn(name = "student_class_id")
49     @com.fasterxml.jackson.annotation.JsonIgnoreProperties("students")
50     private StudentClass studentClass;
51
52     @ManyToOne
53     @JoinColumn(name = "university_id")
54     private University university;
55
56     public enum Role {
57         STUDENT,
58         CR,
59         TEACHER,
60         ADMIN
61     }
62 }
63

```

A.41: com/example/demo/repository/AnnouncementRepository.java

```
1 package com.example.demo.repository;
2
3 import com.example.demo.entity.Announcement;
4 import com.example.demo.entity.StudentClass;
5 import org.springframework.data.jpa.repository.JpaRepository;
6 import org.springframework.stereotype.Repository;
7
8 import java.util.List;
9
10 @Repository
11 public interface AnnouncementRepository extends JpaRepository<Announcement, Long> {
12     List<Announcement> findByStudentClassOrStudentClassIsNullOrderByCreatedAtDesc(StudentClass studentClass);
13     List<Announcement> findAllByOrderByCreatedAtDesc();
14 }
```

A.42: com/example/demo/repository/AuditLogRepository.java

```
1 package com.example.demo.repository;
2
3 import com.example.demo.entity.AuditLog;
4 import org.springframework.data.jpa.repository.JpaRepository;
5 import java.util.List;
6
7 public interface AuditLogRepository extends JpaRepository<AuditLog, Long> {
8     List<AuditLog> findAllByOrderByTimestampDesc();
9 }
10
```

A.43: com/example/demo/repository/BugReportRepository.java

```
1 package com.example.demo.repository;
2
3 import com.example.demo.entity.BugReport;
4 import org.springframework.data.jpa.repository.JpaRepository;
5 import java.util.List;
6
7 public interface BugReportRepository extends JpaRepository<BugReport, Long> {
8     List<BugReport> findAllByOrderByCreatedAtDesc();
9 }
10
```

A.44: com/example/demo/repository/ClassCourseAssignmentRepository.java

```
1 package com.example.demo.repository;
2
3 import com.example.demo.entity.ClassCourseAssignment;
4 import com.example.demo.entity.StudentClass;
5 import com.example.demo.entity.User;
6 import org.springframework.data.jpa.repository.JpaRepository;
7 import java.util.List;
8
9 public interface ClassCourseAssignmentRepository extends JpaRepository<ClassCourseAssignment, Long> {
10     List<ClassCourseAssignment> findByStudentClass(StudentClass studentClass);
11     List<ClassCourseAssignment> findByTeacher(User teacher);
12     void deleteByStudentClass(StudentClass studentClass);
13     void deleteByTeacher(User teacher);
14 }
15
```

A.45: com/example/demo/repository/ClassTestRepository.java

```

1 package com.example.demo.repository;
2
3 import com.example.demo.entity.ClassTest;
4 import org.springframework.data.jpa.repository.JpaRepository;
5 import com.example.demo.entity.StudentClass;
6 import java.time.LocalDateTime;
7 import java.util.List;
8
9 public interface ClassTestRepository extends JpaRepository<ClassTest, Long> {
10     List<ClassTest> findByDateTimeAfterOrderByDateTimeAsc(LocalDateTime dateTime);
11     List<ClassTest> findAllByOrderByDateTimeAsc();
12     List<ClassTest> findByStudentClassOrderByDateTimeAsc(StudentClass studentClass);
13 }
14

```

A.46: com/example/demo/repository/CourseRepository.java

```

1 package com.example.demo.repository;
2
3 import com.example.demo.entity.Course;
4 import com.example.demo.entity.University;
5 import org.springframework.data.jpa.repository.JpaRepository;
6 import java.util.List;
7
8 public interface CourseRepository extends JpaRepository<Course, Long> {
9     List<Course> findByUniversity(University university);
10    List<Course> findByUniversityAndDepartment(University university, String department);
11    List<Course> findByUniversityAndDepartmentAndSemester(University university, String department, String semester);
12 }
13

```

A.47: com/example/demo/repository/ExamQuestionRepository.java

```

1 package com.example.demo.repository;
2
3 import com.example.demo.entity.ExamQuestion;
4 import com.example.demo.entity.Exam;
5 import org.springframework.data.jpa.repository.JpaRepository;
6 import java.util.List;
7
8 public interface ExamQuestionRepository extends JpaRepository<ExamQuestion, Long> {
9     List<ExamQuestion> findByExam(Exam exam);
10    void deleteByExam(Exam exam);
11 }
12

```

A.48: com/example/demo/repository/ExamRepository.java

```

1 package com.example.demo.repository;
2
3 import com.example.demo.entity.Exam;
4 import org.springframework.data.jpa.repository.JpaRepository;
5 import com.example.demo.entity.StudentClass;
6 import java.util.List;
7
8 public interface ExamRepository extends JpaRepository<Exam, Long> {
9     List<Exam> findByPublished(boolean published);
10    List<Exam> findByStudentClassAndPublished(StudentClass studentClass, boolean published);
11    List<Exam> findByTeacher(com.example.demo.entity.User teacher);
12 }
13

```

A.49: com/example/demo/repository/ExamSubmissionRepository.java

```
1 package com.example.demo.repository;
2
3 import com.example.demo.entity.ExamSubmission;
4 import com.example.demo.entity.Exam;
5 import com.example.demo.entity.User;
6 import org.springframework.data.jpa.repository.JpaRepository;
7 import java.util.List;
8 import java.util.Optional;
9
10 public interface ExamSubmissionRepository extends JpaRepository<ExamSubmission, Long> {
11     List<ExamSubmission> findByExam(Exam exam);
12     List<ExamSubmission> findByStudent(User student);
13     Optional<ExamSubmission> findByExamAndStudent(Exam exam, User student);
14     void deleteByExam(Exam exam);
15 }
16
```

A.50: com/example/demo/repository/GradeBookRepository.java

```
1 package com.example.demo.repository;
2
3 import com.example.demo.entity.GradeBook;
4 import com.example.demo.entity.User;
5 import org.springframework.data.jpa.repository.JpaRepository;
6 import java.util.List;
7
8 public interface GradeBookRepository extends JpaRepository<GradeBook, Long> {
9     List<GradeBook> findByStudent(User student);
10     List<GradeBook> findByCourseNameAndAssessmentName(String courseName, String assessmentName);
11 }
12
```

A.51: com/example/demo/repository/ProfileChangeRequestRepository.java

```
1 package com.example.demo.repository;
2
3 import com.example.demo.entity.ProfileChangeRequest;
4 import org.springframework.data.jpa.repository.JpaRepository;
5 import java.util.List;
6
7 public interface ProfileChangeRequestRepository extends JpaRepository<ProfileChangeRequest, Long> {
8     List<ProfileChangeRequest> findByApprovedFalseAndRejectedFalse();
9     void deleteByUser(com.example.demo.entity.User user);
10 }
11
```

A.52: com/example/demo/repository/PromotionHistoryRepository.java

```
1 package com.example.demo.repository;
2
3 import com.example.demo.entity.PromotionHistory;
4 import com.example.demo.entity.StudentClass;
5 import org.springframework.data.jpa.repository.JpaRepository;
6 import java.util.List;
7
8 public interface PromotionHistoryRepository extends JpaRepository<PromotionHistory, Long> {
9     List<PromotionHistory> findByStudentClassOrderByTimestampDesc(StudentClass studentClass);
10 }
11
```

A.53: com/example/demo/repository/ResourceReactionRepository.java

```

1 package com.example.demo.repository;
2
3 import com.example.demo.entity.ResourceReaction;
4 import org.springframework.data.jpa.repository.JpaRepository;
5 import org.springframework.stereotype.Repository;
6 import java.util.List;
7 import java.util.Optional;
8
9 @Repository
10 public interface ResourceReactionRepository extends JpaRepository<ResourceReaction, Long> {
11     List<ResourceReaction> findByResourceId(Long resourceId);
12     Optional<ResourceReaction> findByResourceIdAndUsername(Long resourceId, String username);
13     void deleteByResourceId(Long resourceId);
14 }

```

A.54: com/example/demo/repository/ResourceRepository.java

```

1 package com.example.demo.repository;
2
3 import com.example.demo.entity.Resource;
4 import org.springframework.data.jpa.repository.JpaRepository;
5 import com.example.demo.entity.StudentClass;
6 import java.util.List;
7
8 public interface ResourceRepository extends JpaRepository<Resource, Long> {
9     List<Resource> findByApproved(boolean approved);
10    List<Resource> findByCourseNameAndApproved(String courseName, boolean approved);
11    List<Resource> findByStudentClassAndApproved(StudentClass studentClass, boolean approved);
12    List<Resource> findByUploadedBy(com.example.demo.entity.User uploadedBy);
13 }
14

```

A.55: com/example/demo/repository/ScheduleItemRepository.java

```

1 package com.example.demo.repository;
2
3 import com.example.demo.entity.ScheduleItem;
4 import org.springframework.data.jpa.repository.JpaRepository;
5 import com.example.demo.entity.StudentClass;
6 import java.util.List;
7
8 public interface ScheduleItemRepository extends JpaRepository<ScheduleItem, Long> {
9     List<ScheduleItem> findByDayOfWeekOrderByStartTimeAsc(String dayOfWeek);
10    List<ScheduleItem> findByStudentClass(StudentClass studentClass);
11    List<ScheduleItem> findByStudentClassAndDayOfWeekOrderByStartTimeAsc(StudentClass studentClass, String dayOfWeek);
12    List<ScheduleItem> findByUniversity(com.example.demo.entity.University university);
13 }
14

```

A.56: com/example/demo/repository/StudentClassRepository.java

```

1 package com.example.demo.repository;
2
3 import com.example.demo.entity.StudentClass;
4 import com.example.demo.entity.University;
5 import org.springframework.data.jpa.repository.JpaRepository;
6 import java.util.Optional;
7 import java.util.List;
8
9 public interface StudentClassRepository extends JpaRepository<StudentClass, Long> {
10     Optional<StudentClass> findByBatchAndDepartmentAndSection(String batch, String department, String section);
11     List<StudentClass> findByUniversity(University university);
12 }
13

```

A.57: com/example/demo/repository/SystemAdminRepository.java

```

1 package com.example.demo.repository;
2
3 import com.example.demo.entity.SystemAdmin;
4 import org.springframework.data.jpa.repository.JpaRepository;
5 import java.util.Optional;
6
7 public interface SystemAdminRepository extends JpaRepository<SystemAdmin, Long> {
8     Optional<SystemAdmin> findByUsername(String username);
9     boolean existsByUsername(String username);
10 }
11

```

A.58: com/example/demo/repository/SystemMetadataRepository.java

```

1 package com.example.demo.repository;
2
3 import com.example.demo.entity.SystemMetadata;
4 import com.example.demo.entity.University;
5 import org.springframework.data.jpa.repository.JpaRepository;
6 import java.util.List;
7
8 public interface SystemMetadataRepository extends JpaRepository<SystemMetadata, Long> {
9     List<SystemMetadata> findByType(String type);
10    List<SystemMetadata> findByTypeAndUniversity(String type, University university);
11 }
12

```

A.59: com/example/demo/repository/UniversityRepository.java

```

1 package com.example.demo.repository;
2
3 import com.example.demo.entity.University;
4 import org.springframework.data.jpa.repository.JpaRepository;
5 import java.util.Optional;
6
7 public interface UniversityRepository extends JpaRepository<University, Long> {
8     Optional<University> findByDomain(String domain);
9     Optional<University> findByName(String name);
10    boolean existsByName(String name);
11    boolean existsByDomain(String domain);
12 }
13

```

A.60: com/example/demo/repository/UserRepository.java

```

1 package com.example.demo.repository;
2
3 import com.example.demo.entity.User;
4 import com.example.demo.entity.University;
5 import com.example.demo.entity.StudentClass;
6 import org.springframework.data.jpa.repository.JpaRepository;
7 import java.util.Optional;
8
9 public interface UserRepository extends JpaRepository<User, Long> {
10    Optional<User> findByUsername(String username);
11    Optional<User> findByEmail(String email);
12    boolean existsByUsername(String username);
13    boolean existsByEmail(String email);
14    java.util.List<User> findByUniversityAndRole(University university, User.Role role);
15    java.util.List<User> findByStudentClass(StudentClass studentClass);
16 }
17

```

A.61: com/example/demo/service/CustomUserDetailsService.java

```

1 package com.example.demo.service;
2
3 import com.example.demo.entity.SystemAdmin;
4 import com.example.demo.entity.User;
5 import com.example.demo.repository.SystemAdminRepository;
6 import com.example.demo.repository.UserRepository;
7 import lombok.RequiredArgsConstructor;
8 import org.springframework.security.core.userdetails.UserDetails;
9 import org.springframework.security.core.userdetails.UserDetailsService;
10 import org.springframework.security.core.userdetails.UsernameNotFoundException;
11 import org.springframework.stereotype.Service;
12
13 import java.util.Optional;
14
15 @Service
16 @RequiredArgsConstructor
17 public class CustomUserDetailsService implements UserDetailsService {
18
19     private final UserRepository userRepository;
20     private final SystemAdminRepository systemAdminRepository;
21
22     @Override
23     public UserDetails loadUserByUsername(String username) throws UsernameNotFoundException {
24         // First check if user is a system admin (site owner / master admin)
25         Optional<SystemAdmin> sysAdmin = systemAdminRepository.findByUsername(username);
26         if (sysAdmin.isPresent()) {
27             return org.springframework.security.core.userdetails.User.builder()
28                 .username(sysAdmin.get().getUsername())
29                 .password(sysAdmin.get().getPassword())
30                 .roles("SYSTEM_ADMIN")
31                 .build();
32         }
33
34         // Then check if user is a university user
35         User user = userRepository.findByUsername(username)
36             .orElseThrow(() -> new UsernameNotFoundException("User not found with username: " + username));
37
38         return org.springframework.security.core.userdetails.User.builder()
39             .username(user.getUsername())
40             .password(user.getPassword())
41             .roles(user.getRole().name())
42             .build();
43     }
44 }
45

```

A.62: com/example/demo/service/SlotDetectionService.java

```

1 package com.example.demo.service;
2
3 import com.example.demo.entity.ClassTest;
4 import com.example.demo.entity.ScheduleItem;
5 import com.example.demo.repository.ClassTestRepository;
6 import com.example.demo.repository.ScheduleItemRepository;
7 import lombok.AllArgsConstructor;
8 import lombok.Data;
9 import lombok.NoArgsConstructor;
10 import org.springframework.stereotype.Service;
11
12 import java.time.LocalDate;
13 import java.time.LocalTime;
14 import java.util.ArrayList;
15 import java.util.Comparator;
16 import java.util.List;
17
18 @Service
19 @AllArgsConstructor
20 public class SlotDetectionService {
21
22     private final ScheduleItemRepository scheduleItemRepository;
23     private final ClassTestRepository classTestRepository;
24
25     @Data
26     @NoArgsConstructor
27     @AllArgsConstructor
28     public static class TimeInterval {
29         private LocalTime start;
30         private LocalTime end;
31     }
32
33     public List<TimeInterval> detectFreeSlots(LocalDate date, int durationMinutes, com.example.demo.entity.StudentClass
studentClass) {
34         if (studentClass == null) {
35             return new java.util.ArrayList<>();
36         }
37         // 1. Get Day of Week
38         String dayOfWeekStr = date.getDayOfWeek().name(); // e.g. "MONDAY"
39
40         // 2. Fetch routines for this day of week and student class
41         List<ScheduleItem> routines =
scheduleItemRepository.findByStudentClassAndDayOfWeekOrderByStartTimeAsc(studentClass, dayOfWeekStr);
42
43         // 3. Fetch exams scheduled on this date for this student class
44         List<ClassTest> tests = classTestRepository.findByStudentClassOrderByDateTimeAsc(studentClass);
45         List<ClassTest> testsOnDate = tests.stream()
46             .filter(t -> t.getDateTime().toLocalDate().isEqual(date))
47             .toList();
48
49         // 4. Create a list of busy intervals
50         List<TimeInterval> busyIntervals = new ArrayList<>();
51
52         for (ScheduleItem routine : routines) {
53             busyIntervals.add(new TimeInterval(routine.getStartTime(), routine.getEndTime()));
54         }
55
56         for (ClassTest test : testsOnDate) {
57             LocalTime testStart = test.getDateTime().toLocalTime();
58             LocalTime testEnd = testStart.plusMinutes(test.getDurationMinutes());
59             busyIntervals.add(new TimeInterval(testStart, testEnd));
60         }
61
62         // 5. Define Academic boundaries: 08:00 to 18:00
63         LocalTime dayStart = LocalTime.of(8, 0);
64         LocalTime dayEnd = LocalTime.of(18, 0);
65
66         // 6. Merge overlapping busy intervals
67         List<TimeInterval> mergedBusy = mergeIntervals(busyIntervals, dayStart, dayEnd);
68
69         // 7. Find gaps
70         List<TimeInterval> freeSlots = new ArrayList<>();

```

```

71     LocalTime current = dayStart;
72
73     for (TimeInterval busy : mergedBusy) {
74         if (current.isBefore(busy.getStart())) {
75             long duration = java.time.Duration.between(current, busy.getStart()).toMinutes();
76             if (duration >= durationMinutes) {
77                 freeSlots.add(new TimeInterval(current, busy.getStart()));
78             }
79         }
80         if (busy.getEnd().isAfter(current)) {
81             current = busy.getEnd();
82         }
83     }
84
85     if (current.isBefore(dayEnd)) {
86         long duration = java.time.Duration.between(current, dayEnd).toMinutes();
87         if (duration >= durationMinutes) {
88             freeSlots.add(new TimeInterval(current, dayEnd));
89         }
90     }
91
92     return freeSlots;
93 }
94
95 private List<TimeInterval> mergeIntervals(List<TimeInterval> intervals, LocalTime boundsStart, LocalTime boundsEnd) {
96     if (intervals.isEmpty()) {
97         return new ArrayList<>();
98     }
99
100    // Filter and clamp intervals to the academic bounds
101    List<TimeInterval> clamped = new ArrayList<>();
102    for (TimeInterval interval : intervals) {
103        LocalTime start = interval.getStart();
104        LocalTime end = interval.getEnd();
105
106        if (start.isBefore(boundsStart)) start = boundsStart;
107        if (end.isAfter(boundsEnd)) end = boundsEnd;
108
109        if (start.isBefore(end)) {
110            clamped.add(new TimeInterval(start, end));
111        }
112    }
113
114    if (clamped.isEmpty()) {
115        return new ArrayList<>();
116    }
117
118    // Sort by start time
119    clamped.sort(Comparator.comparing(TimeInterval::getStart));
120
121    List<TimeInterval> merged = new ArrayList<>();
122    TimeInterval active = clamped.get(0);
123
124    for (int i = 1; i < clamped.size(); i++) {
125        TimeInterval next = clamped.get(i);
126        if (!next.getStart().isAfter(active.getEnd())) {
127            // Overlap or touch, merge
128            if (next.getEnd().isAfter(active.getEnd())) {
129                active.setEnd(next.getEnd());
130            }
131        } else {
132            merged.add(active);
133            active = next;
134        }
135    }
136    merged.add(active);
137
138    return merged;
139 }
140 }
141

```

Appendix B: User Manual

System Configuration and Execution Details

Follow these steps to deploy and run learnX.help locally:

1. Pre-requisites: Ensure JDK 21 and Maven are installed on your machine.
2. Database Configuration: The system runs with an active file-based H2 database by default. To switch to PostgreSQL, configure environment variables: `SPRING_DATASOURCE_URL`, `SPRING_DATASOURCE_USERNAME`, and `SPRING_DATASOURCE_PASSWORD`.
3. Build Command: In the root folder, compile the Maven project: `mvn clean install`.
4. Execution: Boot the Spring application: `mvn spring-boot:run`.
5. Client Access: Open your browser and navigate to `http://localhost:8080`. To view database contents directly, navigate to `http://localhost:8080/h2-console` using `jdbc:h2:file:~/.learnx/learnxdb` as the JDBC URL.

Appendix C: Screenshots

Below are UI validation frames. Place your actual screenshot PNGs in the root workspace folder to automatically embed them.

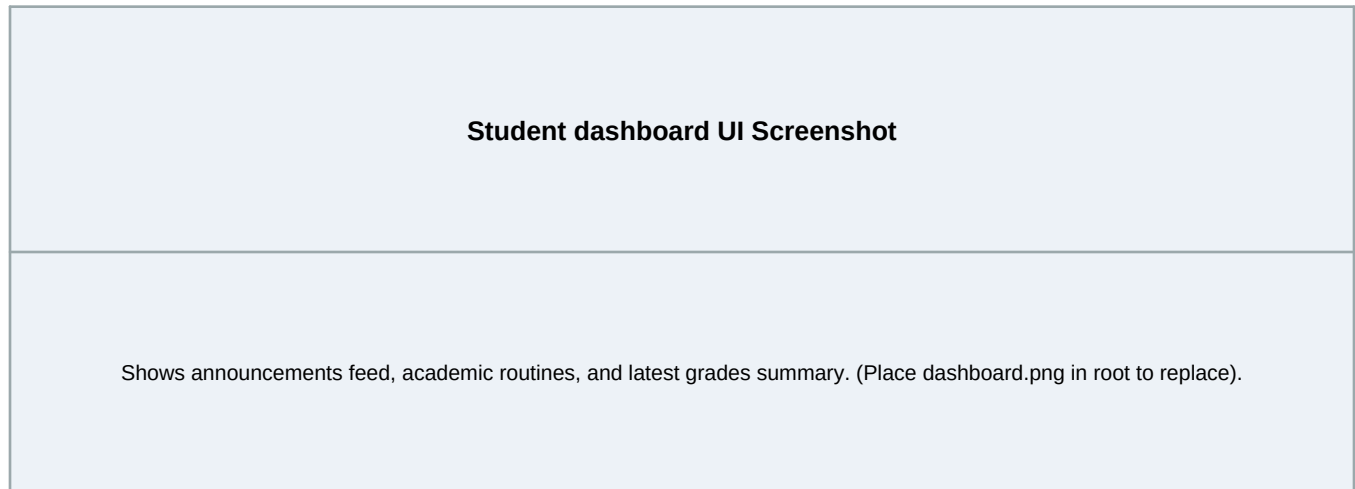


Figure 6.1: Appendix C Placeholder for Dashboard UI Screenshot

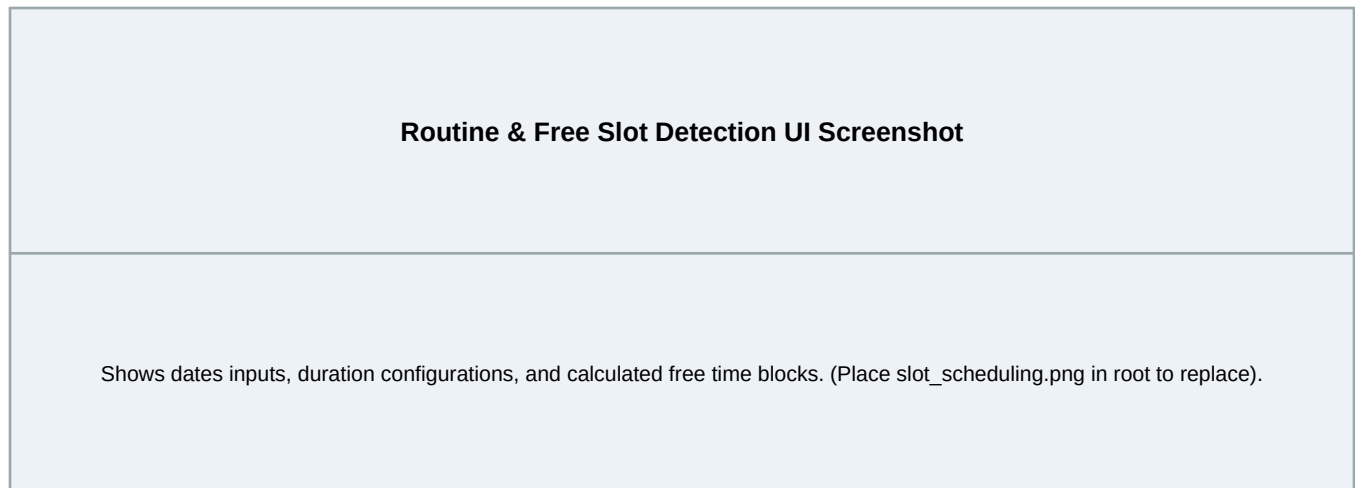


Figure 6.2: Appendix C Placeholder for Routine & Slot Scheduling UI Screenshot